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Descriptive Inorganic Chemistry

written by

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Target of the course:

By focusing on the systematic disposition of the elements in the periodic table, learn to compare and predict the properties of the elements in their uncombined state, as well as their tendency to form particular type of compounds.

PERIODIC TABLE OF THE ELEMENTS

http://www.ktf-split.hr/periodni/en/

GROUP / PERIOD

PERIOD	1 IA	2 IIA	3 IIIB	4 IVB	5 VB	6 VIB	7 VIIB	8 VIIIB	9 VIIIB	10 VIIIB	11 IB	12 IIB	13 IIIA	14 IVA	15 VA	16 VIA	17 VIIA	18 VIIIA
1	1 1.0079 **H** HYDROGEN																	2 4.0026 **He** HELIUM
2	3 6.941 **Li** LITHIUM	4 9.0122 **Be** BERYLLIUM											5 10.811 **B** BORON	6 12.011 **C** CARBON	7 14.007 **N** NITROGEN	8 15.999 **O** OXYGEN	9 18.998 **F** FLUORINE	10 20.180 **Ne** NEON
3	11 22.990 **Na** SODIUM	12 24.305 **Mg** MAGNESIUM											13 26.982 **Al** ALUMINIUM	14 28.086 **Si** SILICON	15 30.974 **P** PHOSPHORUS	16 32.065 **S** SULPHUR	17 35.453 **Cl** CHLORINE	18 39.948 **Ar** ARGON
4	19 39.098 **K** POTASSIUM	20 40.078 **Ca** CALCIUM	21 44.956 **Sc** SCANDIUM	22 47.867 **Ti** TITANIUM	23 50.942 **V** VANADIUM	24 51.996 **Cr** CHROMIUM	25 54.938 **Mn** MANGANESE	26 55.845 **Fe** IRON	27 58.933 **Co** COBALT	28 58.693 **Ni** NICKEL	29 63.546 **Cu** COPPER	30 65.39 **Zn** ZINC	31 69.723 **Ga** GALLIUM	32 72.64 **Ge** GERMANIUM	33 74.922 **As** ARSENIC	34 78.96 **Se** SELENIUM	35 79.904 **Br** BROMINE	36 83.80 **Kr** KRYPTON
5	37 85.468 **Rb** RUBIDIUM	38 87.62 **Sr** STRONTIUM	39 88.906 **Y** YTTRIUM	40 91.224 **Zr** ZIRCONIUM	41 92.906 **Nb** NIOBIUM	42 95.94 **Mo** MOLYBDENUM	43 (98) **Tc** TECHNETIUM	44 101.07 **Ru** RUTHENIUM	45 102.91 **Rh** RHODIUM	46 106.42 **Pd** PALLADIUM	47 107.87 **Ag** SILVER	48 112.41 **Cd** CADMIUM	49 114.82 **In** INDIUM	50 118.71 **Sn** TIN	51 121.76 **Sb** ANTIMONY	52 127.60 **Te** TELLURIUM	53 126.90 **I** IODINE	54 131.29 **Xe** XENON
6	55 132.91 **Cs** CAESIUM	56 137.33 **Ba** BARIUM	57-71 **La-Lu** Lanthanide	72 178.49 **Hf** HAFNIUM	73 180.95 **Ta** TANTALUM	74 183.84 **W** TUNGSTEN	75 186.21 **Re** RHENIUM	76 190.23 **Os** OSMIUM	77 192.22 **Ir** IRIDIUM	78 195.08 **Pt** PLATINUM	79 196.97 **Au** GOLD	80 200.59 **Hg** MERCURY	81 204.38 **Tl** THALLIUM	82 207.2 **Pb** LEAD	83 208.98 **Bi** BISMUTH	84 (209) **Po** POLONIUM	85 (210) **At** ASTATINE	86 (222) **Rn** RADON
7	87 (223) **Fr** FRANCIUM	88 (226) **Ra** RADIUM	89-103 **Ac-Lr** Actinide	104 (261) **Rf** RUTHERFORDIUM	105 (262) **Db** DUBNIUM	106 (266) **Sg** SEABORGIUM	107 (264) **Bh** BOHRIUM	108 (277) **Hs** HASSIUM	109 (268) **Mt** MEITNERIUM	110 (281) **Uun** UNUNNILIUM	111 (272) **Uuu** UNUNUNIUM	112 (285) **Uub** UNUNBIUM		114 (289) **Uuq** UNUNQUADIUM				

Key: RELATIVE ATOMIC MASS (1) · GROUP IUPAC 13 · GROUP CAS IIIA · ATOMIC NUMBER 5 · 10.811 · SYMBOL **B** · ELEMENT NAME BORON

- Metal
- Semimetal
- Nonmetal
- 1 Alkali metal
- 2 Alkaline earth metal
- Transition metals
- Lanthanide
- Actinide
- 16 Chalcogens element
- 17 Halogens element
- 18 Noble gas

STANDARD STATE (25 °C; 101 kPa)

- **Ne** - gas
- **Ga** - liquid
- **Fe** - solid
- **Tc** - synthetic

LANTHANIDE

57 138.91 **La** LANTHANUM	58 140.12 **Ce** CERIUM	59 140.91 **Pr** PRASEODYMIUM	60 144.24 **Nd** NEODYMIUM	61 (145) **Pm** PROMETHIUM	62 150.36 **Sm** SAMARIUM	63 151.96 **Eu** EUROPIUM	64 157.25 **Gd** GADOLINIUM	65 158.93 **Tb** TERBIUM	66 162.50 **Dy** DYSPROSIUM	67 164.93 **Ho** HOLMIUM	68 167.26 **Er** ERBIUM	69 168.93 **Tm** THULIUM	70 173.04 **Yb** YTTERBIUM	71 174.97 **Lu** LUTETIUM

ACTINIDE

89 (227) **Ac** ACTINIUM	90 232.04 **Th** THORIUM	91 231.04 **Pa** PROTACTINIUM	92 238.03 **U** URANIUM	93 (237) **Np** NEPTUNIUM	94 (244) **Pu** PLUTONIUM	95 (243) **Am** AMERICIUM	96 (247) **Cm** CURIUM	97 (247) **Bk** BERKELIUM	98 (251) **Cf** CALIFORNIUM	99 (252) **Es** EINSTEINIUM	100 (257) **Fm** FERMIUM	101 (258) **Md** MENDELEVIUM	102 (259) **No** NOBELIUM	103 (262) **Lr** LAWRENCIUM

(1) Pure Appl. Chem., **73**, No. 4, 667-683 (2001)
Relative atomic mass is shown with five significant figures. For elements have no stable nuclides, the value enclosed in brackets indicates the mass number of the longest-lived isotope of the element.

However three such elements (Th, Pa, and U) do have a characteristic terrestrial isotopic composition, and for these an atomic weight is tabulated.

Editor: Aditya Vardhan (adivar@nettlinx.com)

Lecture 1

The electronic structure of the atom

Chapter 1

(pages 1-17)

The structure of the periodic Table

Chapter 2

(pages 19-40)

Content:

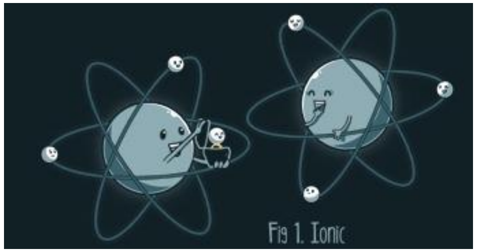
- to give a quick review of the quantum model of the electronic structure of the atom;

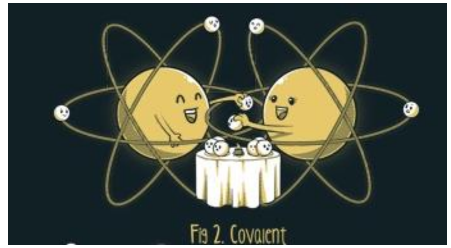
- to introduce Slater's rules;

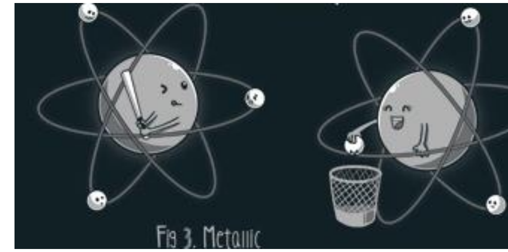
- to use the concept of affective nuclear charge to predict trends in the periodic table;

- Electronegativity and bond triangle

What types of bond do we know?







Ionic bond

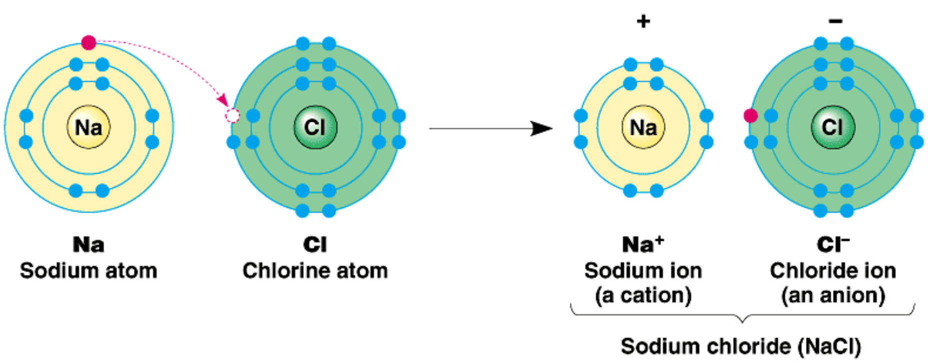
Covalent bond

Metallic bond

polar bonds

Coordination bond

What type of bond is involved in sodium chloride (NaCl)?



Why Na^+ and Cl^- ?

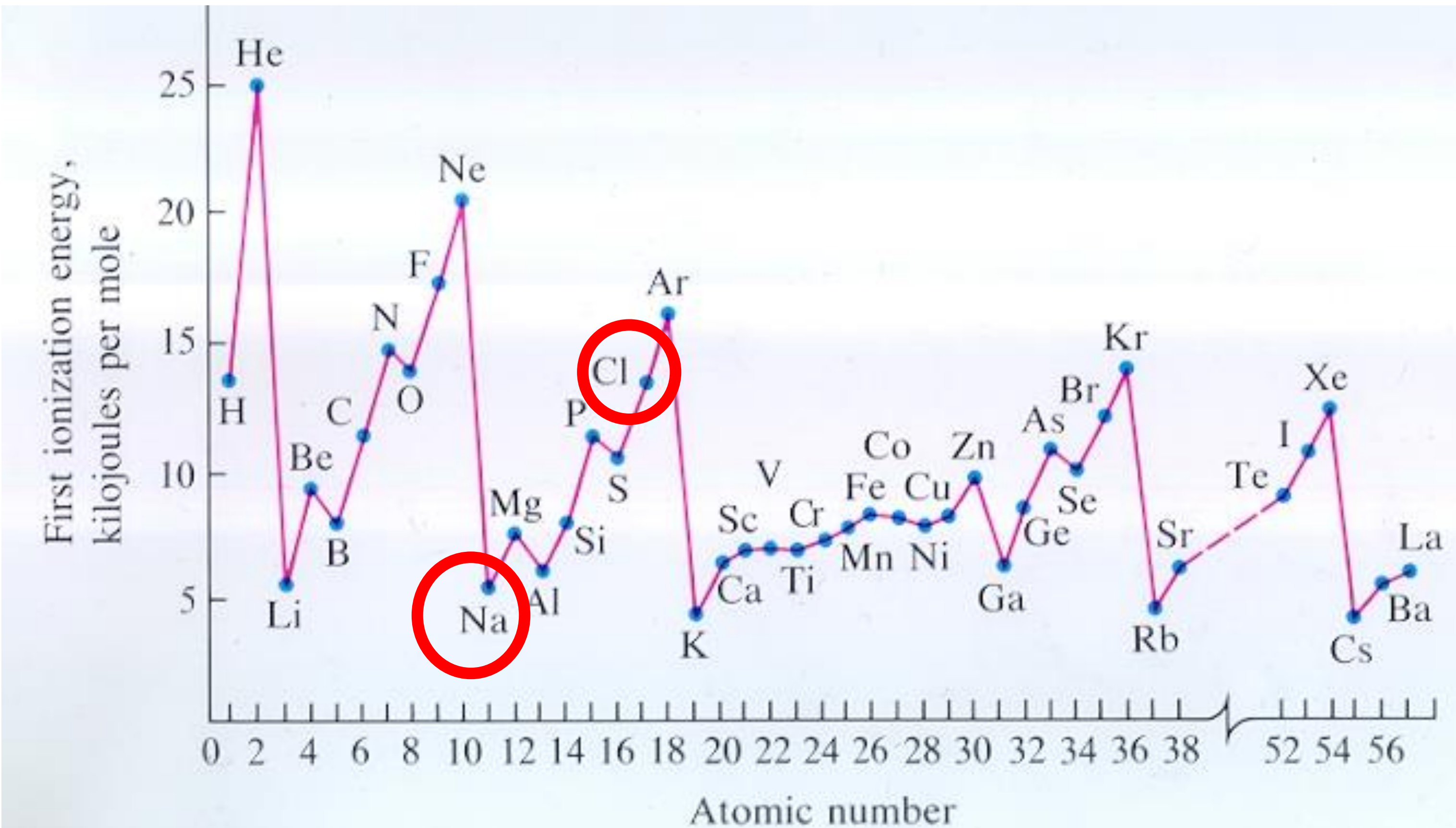
Why not Cl^+ and Na^- ?

(a) First ionization energy

(b) Electron affinity

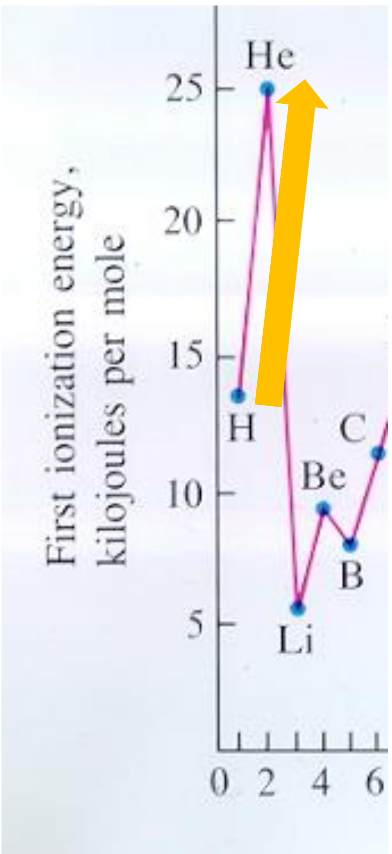
First ionization energy

$A(g) \rightarrow A^+(g) + e^- \qquad \Delta H > 0$



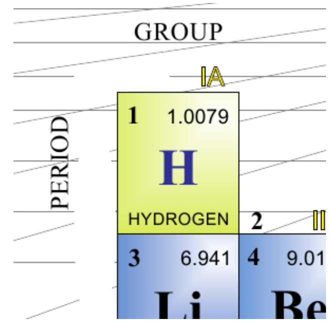
First ionization energy

$$A(g) \rightarrow A^{+}(g) + e^{-} \qquad \Delta H > 0$$

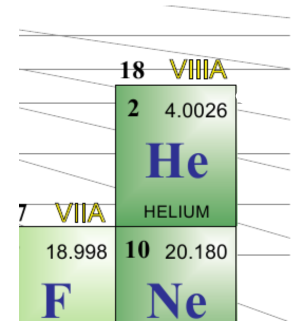


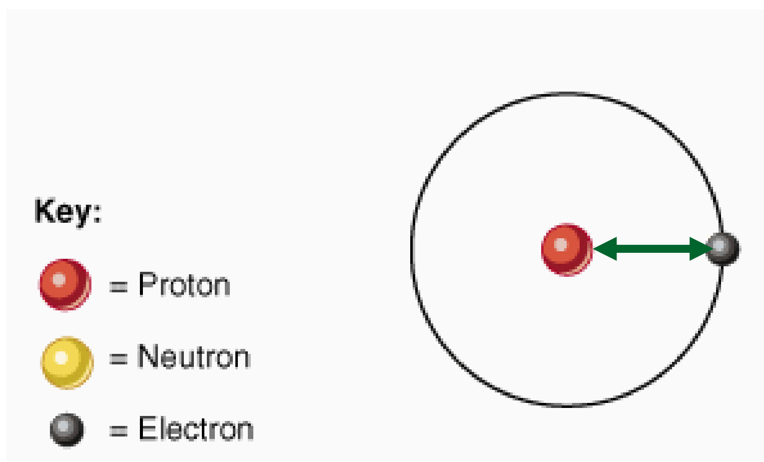
$$E = K\frac{Q_{+}Q_{-}}{r}$$

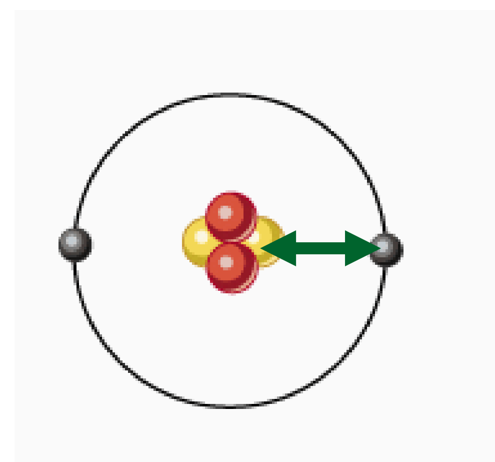
Hydrogen



Helium



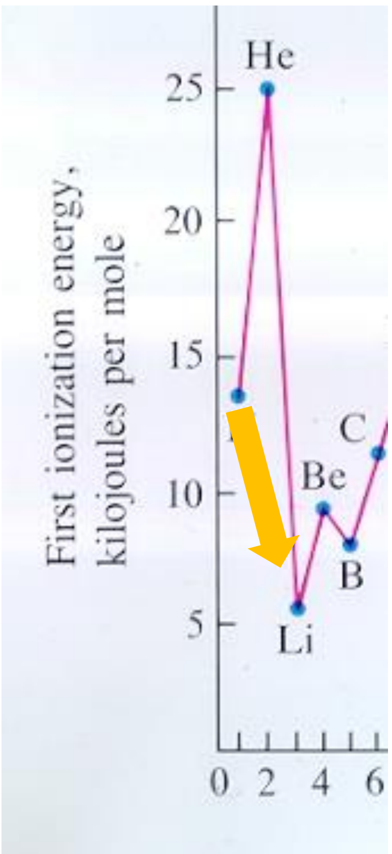




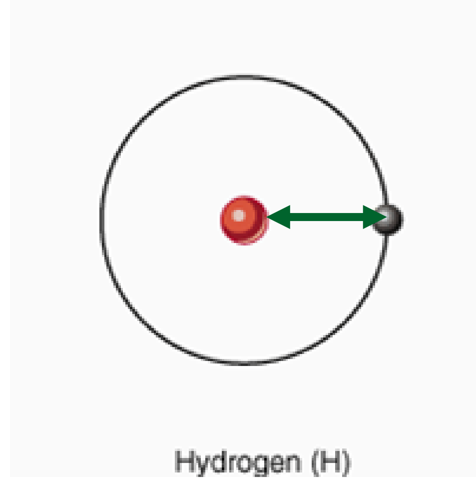
Electron configuration: **H : 1s^1** **He : 1s^2**

First ionization energy

$$A(g) \rightarrow A^{+}(g) + e^{-} \qquad \Delta H > 0$$

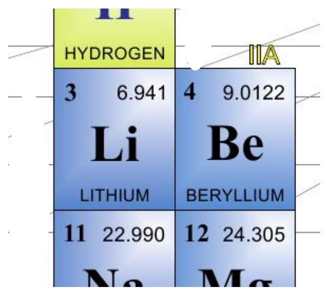


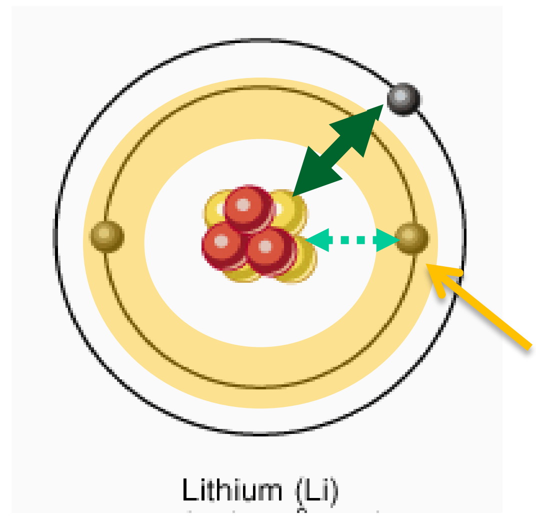
$$E = K\frac{Q_{+}Q_{-}}{r}$$



No shielding electrons

Lithium





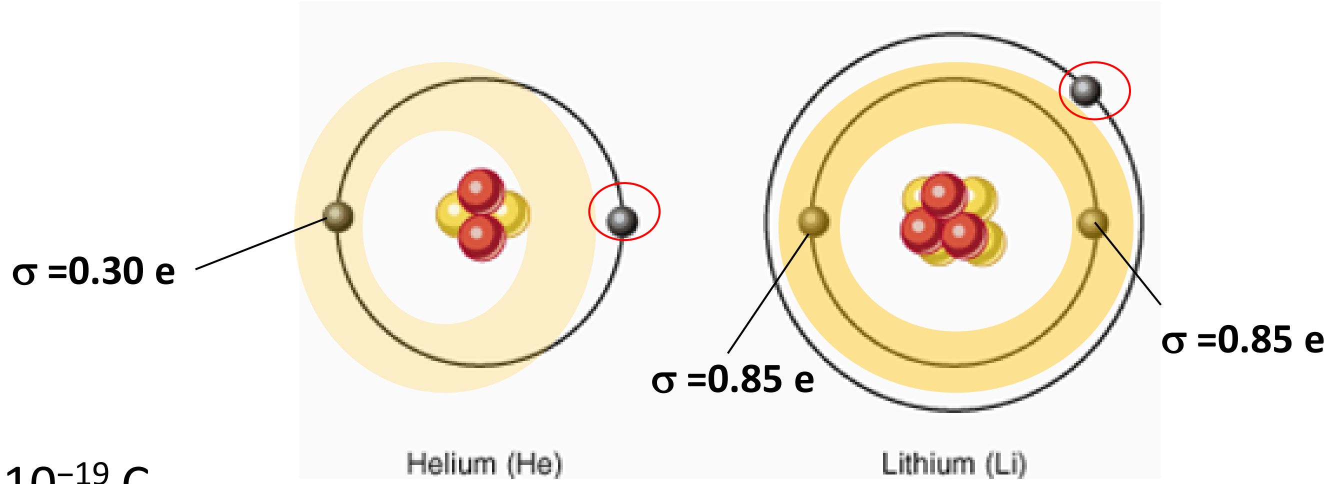
(a) Longer distance

(b) Inner electrons shield nucleolus charge

Electron configuration: $1s^2\ 2s^1$

How can we calculate the effective nuclear charge?

J. K. Slater (1900-1976) proposed an empirical constant screening constant (σ) that represent the cumulative extent to which the inner electrons of an atom shield from nuclear charge.



Unit = 1e = 1.602×10^{-19} C

Thus, Slater's cab be used in the equation:

$Z_{eff} = Z - \sigma$

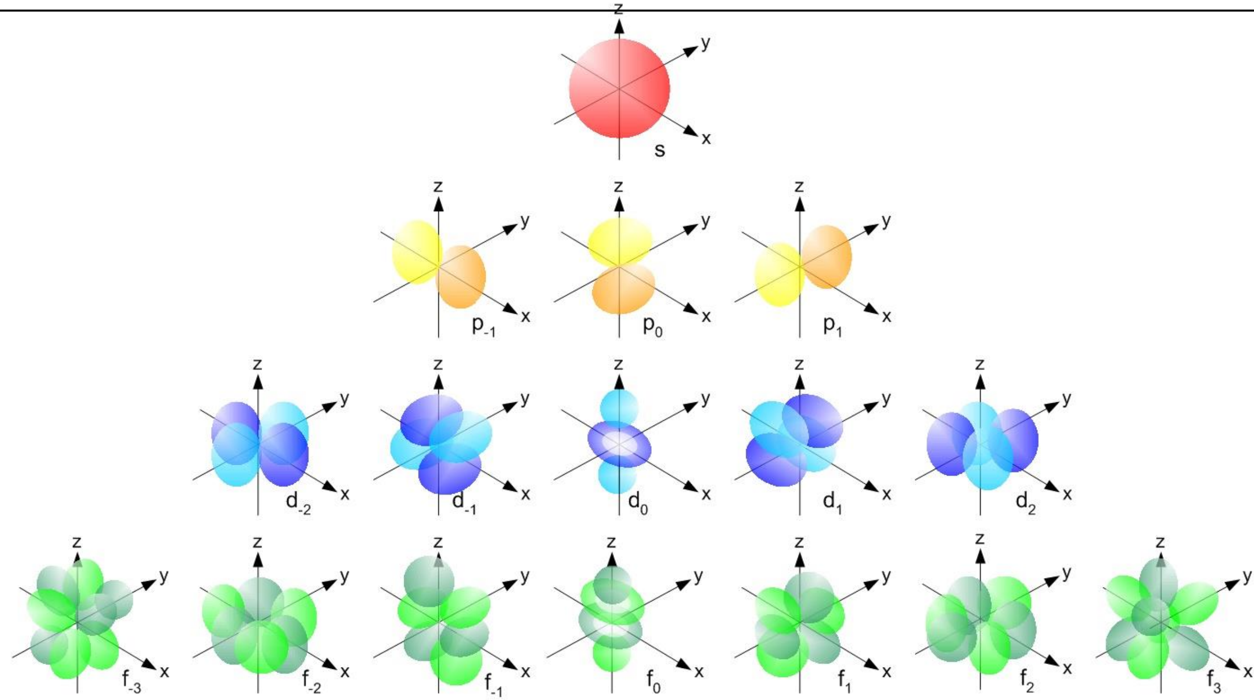
Z_{eff} is the effective nuclear charge

Z is the atomic number

σ is the Slater's shield constant

Helium: $Z_{eff} = 2 - 0.30 = 1.70$ *Lithium:* $Z_{eff} = 3 - (0.85 \times 2) = 1.30$

Quantum Number	Allowed Values	Name and Meaning
n	n = 1, 2, 3, ……	Principal quantum number: orbital energy and size.
l	**l** = (n-1), (n-2), …., 0	Azimuthal (or angular) quantum number: orbital shape
m_l	m_l = -**l** , …, 0, …, **l**	Magnetic quantum number: orbital orientation
m_s	$m_s = 1/2, -1/2$	Electron spin quantum number



Slater's Rules: how to calculate the shielding constant

1. All electrons in orbitals of greater principal quantum number contribute 0.

2. Each electron in the same principal quantum number contributes 0.35, except for the [1s] group, where the other electron contributes only 0.30.

3. If the reference is a *n*s or *n*p electron, we count 0.85 for each electron with principal quantum number (*n–1*), and 1.00 for each electron with principal quantum number (*n–2*) or less.

4. If the reference is a *n*d or *n*f electron, we count 1.00 and for electrons with principal quantum number *n* and a smaller azimuthal quantum number *l,* and for from each electron with principal quantum number (n–1) or less.

	s_i for Electrons in			
Electron Group	*All Higher Groups*	*Same Group*	*Groups with* $n' = n - 1$	*Groups with* $n' < n - 1$
$(1s)$	0	0.30		
(ns,p)	0	0.35	0.85	1.00
$(nd), (nf)$	0	0.35	1.00	1.00

Slater's Rules

How to use them:

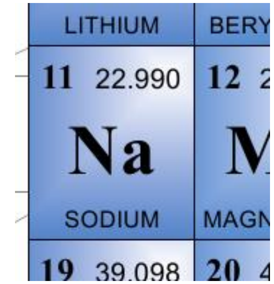
Step 1: Write the electron configuration, separating the different sub-shells:
e.g. (1s) (2s, 2p) (3s, 3p) (3d) (4s, 4p) (4d) (4f) (5s, 5p) . . .

Step 2: Identify the electron of interest, and ignore all electrons in higher groups, because, these do not shield electrons in lower groups

Step 3: Use the Slater's Rules. Be aware that rules for *s* or *p* electron are different from those for a *d* or *f* electron.

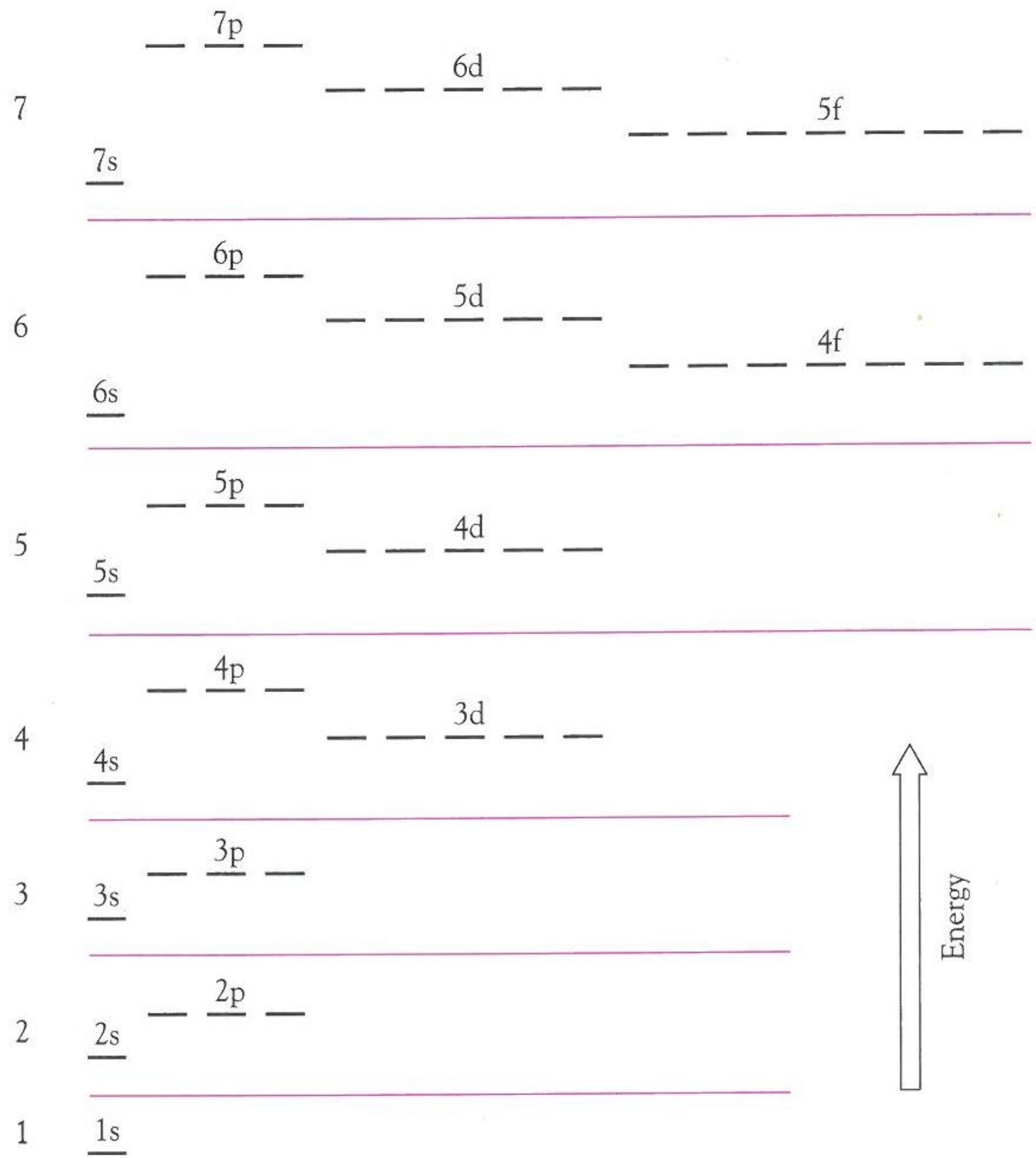
	s_i for Electrons in			
Electron Group	*All Higher Groups*	*Same Group*	*Groups with* $n' = n - 1$	*Groups with* $n' < n - 1$
(1*s*)	0	0.30		
(*ns,p*)	0	0.35	0.85	1.00
(*nd*), (*nf*)	0	0.35	1.00	1.00

Exercise A

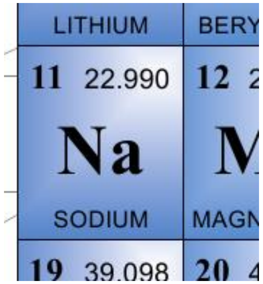


Calculate the effective nuclear charge on the outer shell electrons of Sodium.

Step 1: Write the electron configuration and separate the electrons in the the different sub-shells:


7p
6d
7
5f
7s
6p
5d
6
4f
6s
5p
5
4d
5s
4p
4
3d
4s
3p
3
3s
2p
2
2s
1
1s
Energy

Exercise A



Calculate the effective nuclear charge on the outer shell electrons of Sodium.

Step 1: Write the electron configuration and separate the electrons in the the different sub-shells:

$1s^2\ 2s^2\ 2p^6\ 3s^1$ $(1s^2)\ (2s^2\ 2p^6)\ 3s^1$

Step 2: Identify the electron of interest: $(1s^2)\ (2s^2\ 2p^6)\ 3s^1$

Step 3: Use the Slater's Rules:

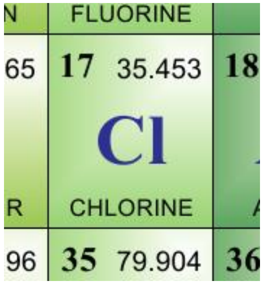
$\sigma = 0 \times 0.35 + 8 \times 0.85 + 2 \times 1 = 8.80$

$Z_{eff} = Z - \sigma = 11 - 8.80 = 2.20$

Electron Group	s_i for Electrons in: All Higher Groups	Same Group	Groups with $n' = n - 1$	Groups with $n' < n - 1$
$(1s)$	0	0.30		
n = 3 (ns,p)	0	0.35	0.85	1.00
(nd), (nf)	0	0.35	1.00	1.00

Exercise B

7 minutes



Calculate the effective nuclear charge on the outer shell electrons of Chlorine.

Exercise C

10 minutes

Using Slater's rules, calculate the effective nuclear charge on one of the 3*d* electrons compared with one of the 4*s* electrons for an atom of manganese. What is the electronic configuration of an hypothetical Mn^{1+} ion?

Exercise D

7 minutes

Use the Slater's rules to calculate the effective nuclear charge on one of the *4f* electrons and one of the 6*s* electrons in a neutral Europium atom (Z=63)

Atomic radii

What do we mean by the "size" of an atom?

The concept of "size" is somewhat ambiguous when applied to the scale of atoms and molecules.

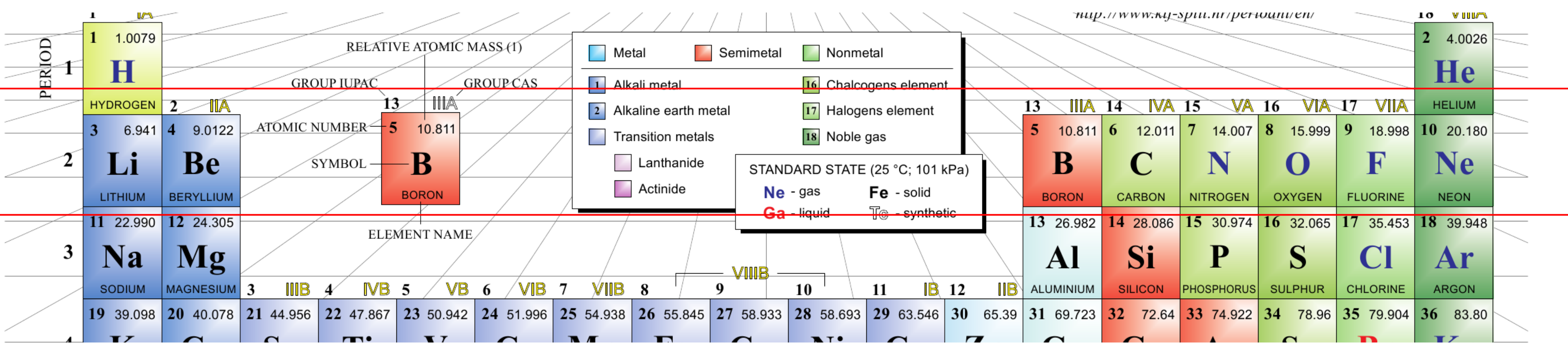
The reason for this is apparent when you recall that an atom has no definite boundary:
there is a finite (but very small) probability of finding the electron of a hydrogen atom, for example, 1 cm, or even 1 km from the nucleus.

It is not possible to specify a definite value for the radius of an isolated atom;

Exercise 2.23 page 40

.... For each of the elements (of the second period), calculate the effective nuclear charge of each of the 1*s*, 2*s*, 2*p* orbitals according to the Slater's rules....

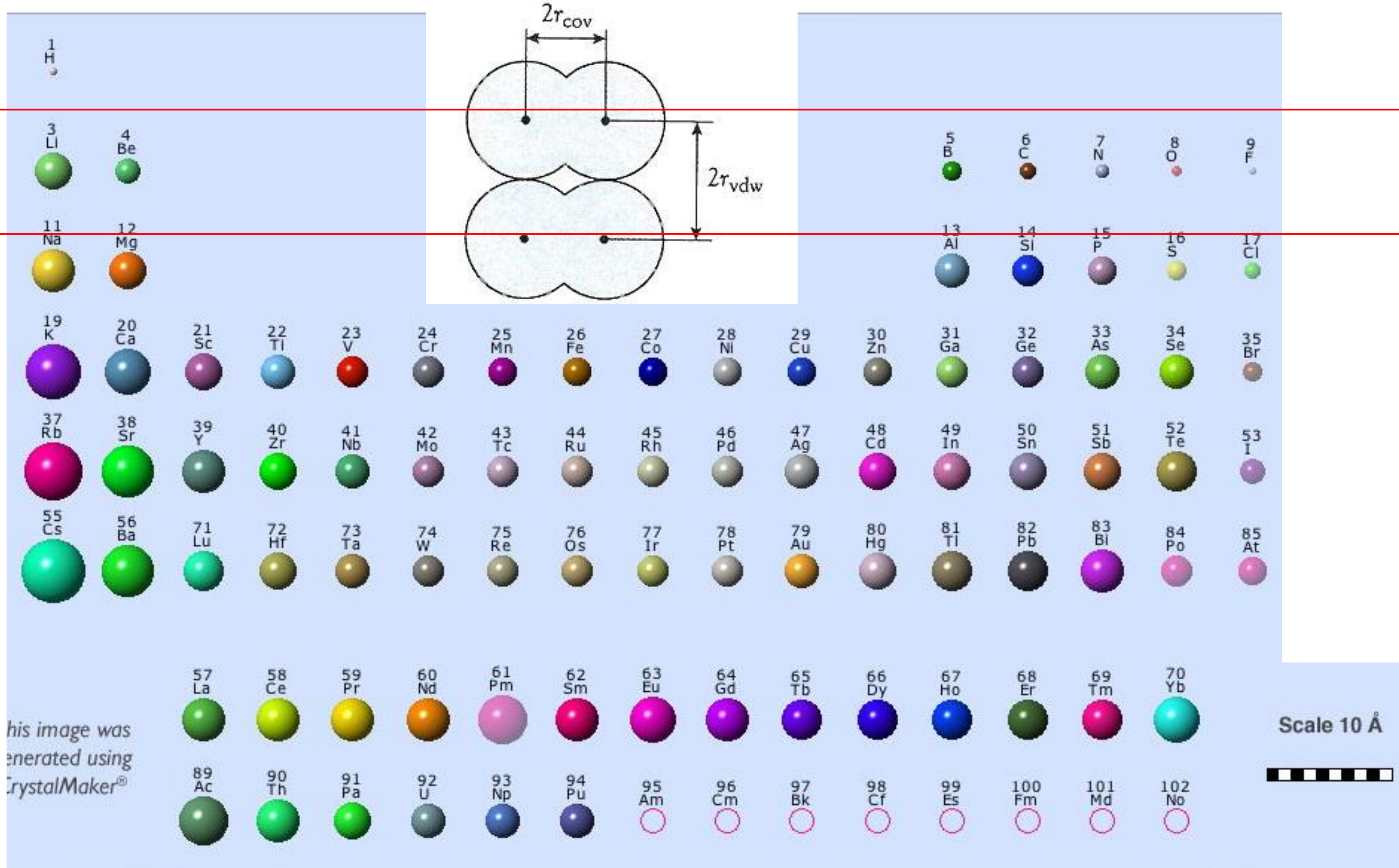
Element	Li	Be	B	C	N	O	F	Ne
Z	3	4	5	6	7	8	9	10
1s	2.65	3.65	4.65	5.65	6.65	7.65	8.65	9.65
2s	**1.30**	**1.95**	2.60	3.25	3.90	4.55	5.20	5.85
2p	-	-	**2.60**	**3.25**	**3.90**	**4.55**	**5.20**	**5.85**



Covalent radius

Increasing

Increasing



Electronegativity

Pauling

Electronegativity is the power of an atom in a molecule to attract electrons for itself.

If two atoms, A and B, had the same electronegativity, the strenght of the A-B bond would be equal to the average of A-A and B-B bond energy:

Geometric average $$D(A\text{-}B) = \sqrt{D(A-A) \times D(B-B)}$$

D = dissociation energy

5 minutes

Exercise E

If H and F had the same electronegativity, what would be the dissociatio energy for the HCl molecule?

D (Cl_2) = 242 KJ mol^{-1}

D (H_2) = 432 KJ mol^{-1}

Electronegativity

0.7 — 4

Pauling scale

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
H 2.1																	He --
Li 1.0	Be 1.5											B 2.0	C 2.5	N 3.0	O 3.5	F 4.0	Ne --
Na 0.9	Mg 1.2											Al 1.5	Si 1.8	P 2.1	S 2.5	Cl 3.0	Ar --
K 0.8	Ca 1.0	Sc 1.3	Ti 1.5	V 1.6	Cr 1.6	Mn 1.5	Fe 1.8	Co 1.8	Ni 1.8	Cu 1.9	Zn 1.6	Ga 1.6	Ge 1.8	As 2.0	Se 2.4	Br 2.8	Kr 3.0
Rb 0.8	Sr 1.0	Y 1.2	Zr 1.4	Nb 1.6	Mo 1.8	Tc 1.9	Ru 2.2	Rh 2.2	Pd 2.2	Ag 1.9	Cd 1.7	In 1.7	Sn 1.8	Sb 1.9	Te 2.1	I 2.5	Xe 2.6
Cs 0.7	Ba 0.9	La 1.1	Hf 1.3	Ta 1.5	W 1.7	Re 1.9	Os 2.2	Ir 2.2	Pt 2.2	Au 2.4	Hg 1.9	Tl 1.8	Pb 1.9	Bi 1.9	Po 2.0	At 2.2	Rn --
Fr 0.7	Ra 0.9	Ac 1.1	Rf --	Db --	Sg --	Bh --	Hs --	Mt --	Uun --	Uuu --	Uub --	113 --	Uuq --	115 --	116 --	117 --	118 --

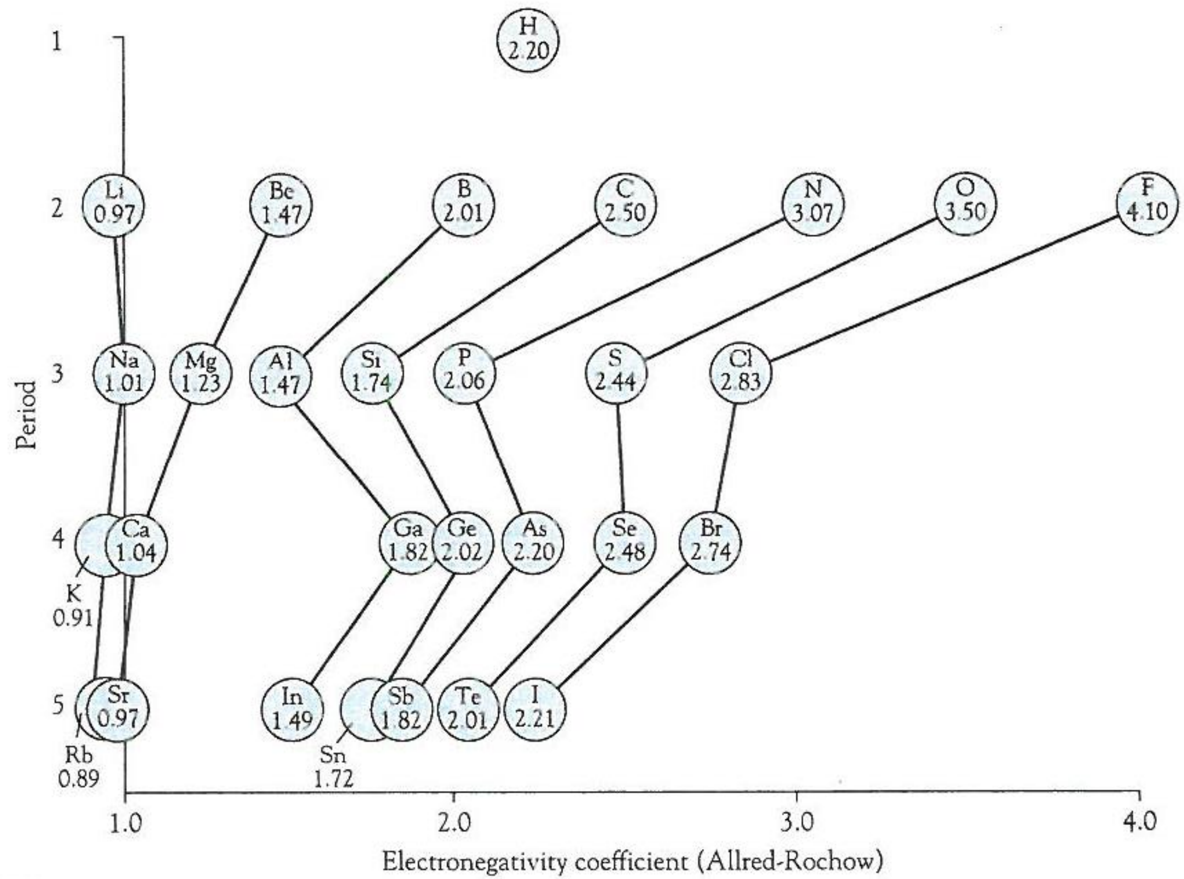
Ce 1.1	Pr 1.1	Nd 1.1	Pm 1.2	Sm 1.2	Eu 1.1	Gd 1.2	Tb 1.2	Dy 1.2	Ho 1.2	Er 1.2	Tm 1.2	Yb 1.2	Lu 1.3
Th 1.3	Pa 1.5	U 1.7	Np 1.3	Pu 1.3	Am 1.3	Cm 1.3	Bk 1.3	Cf 1.3	Es 1.3	Fm 1.3	Md 1.3	No 1.5	Lr --

There are various Electronegativity scales

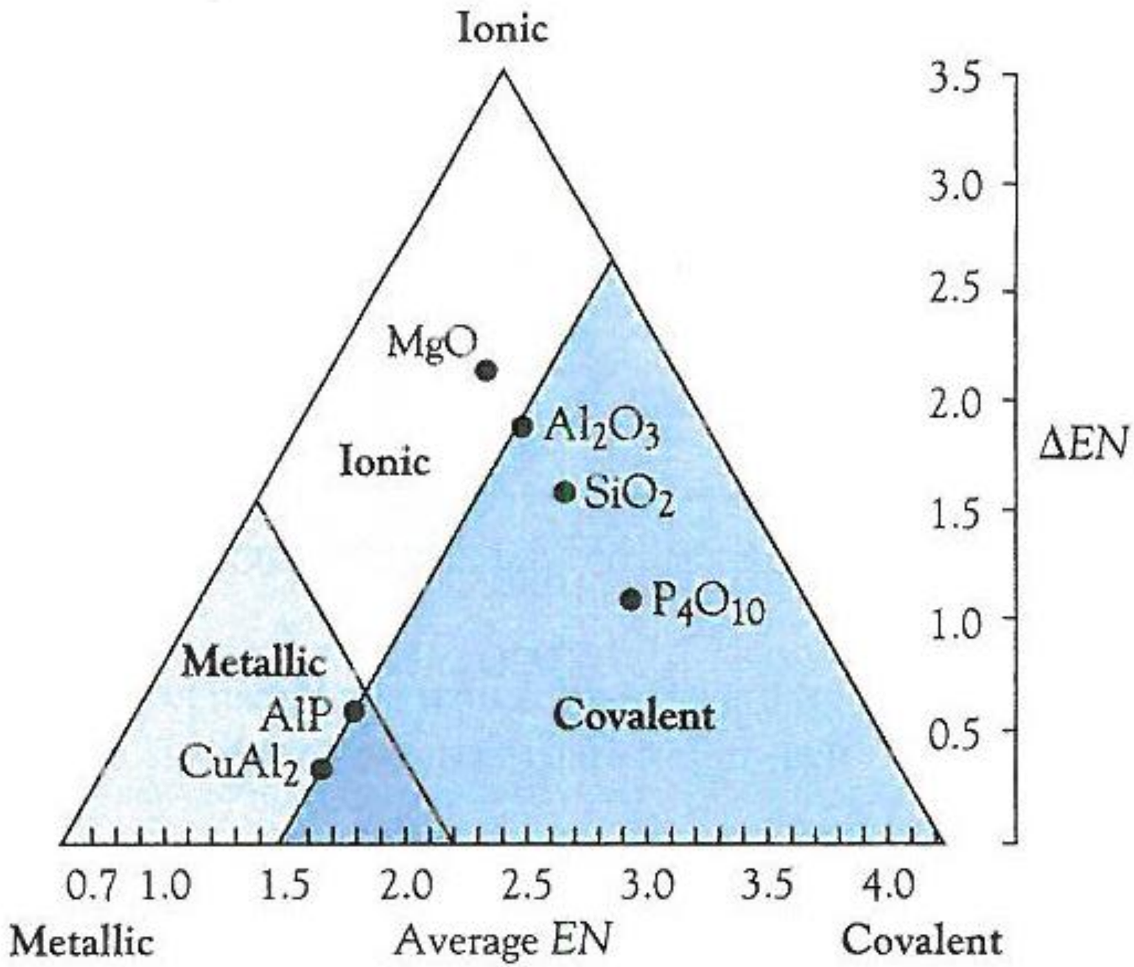
Allred-Rochow

An atom will attract the electrons

$$\chi^{AR} = 3590\frac{Z_{eff}}{r^2{}_{cov}} + 0.744 \text{ (radius expressed in pm)}$$



Bond continuum

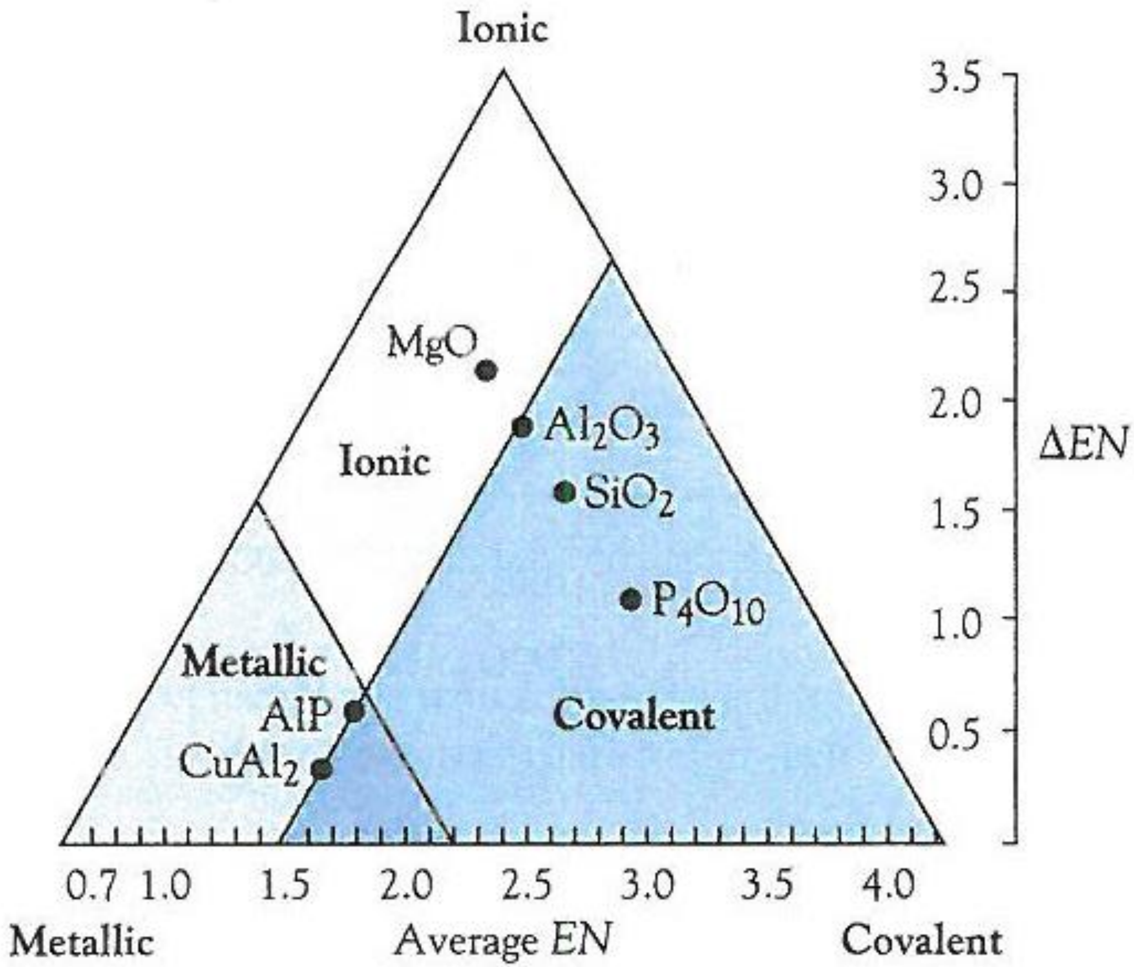


Lecture 2

Covalent bonding

Chapter 3

Bond triangle



Content

The covalent bond

- Lewis structure
- VSEPR model
- Valence Bond Theory
- Molecular Orbital Theory

Exercise 1

Determine: (a) the bond order, (b) the presence of lone pairs, (c) the geometry, and (d) the hybridization for ammonia (NH_3) and the nitrate ion (NO_3^-)

1) Lewis structures

Rules:

1. Count the total number of **valence electrons**.

2. Identify the **central atom**: the least electronegative, except H

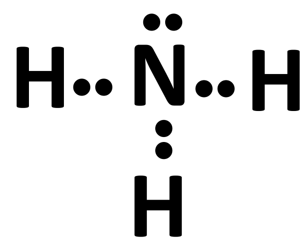
3. Place an **electron pair** between the central atom and each of the surrounding atoms and add **lone pairs** to the surrounding atoms.

4. If the octet rule is not respected and there are "**leftover electrons",** add to them to the lone pairs to the central atom. If the octet rule is not respected and there NO MORE electrons are left, construct double and triple bonds using lone pairs from the surrounding atoms

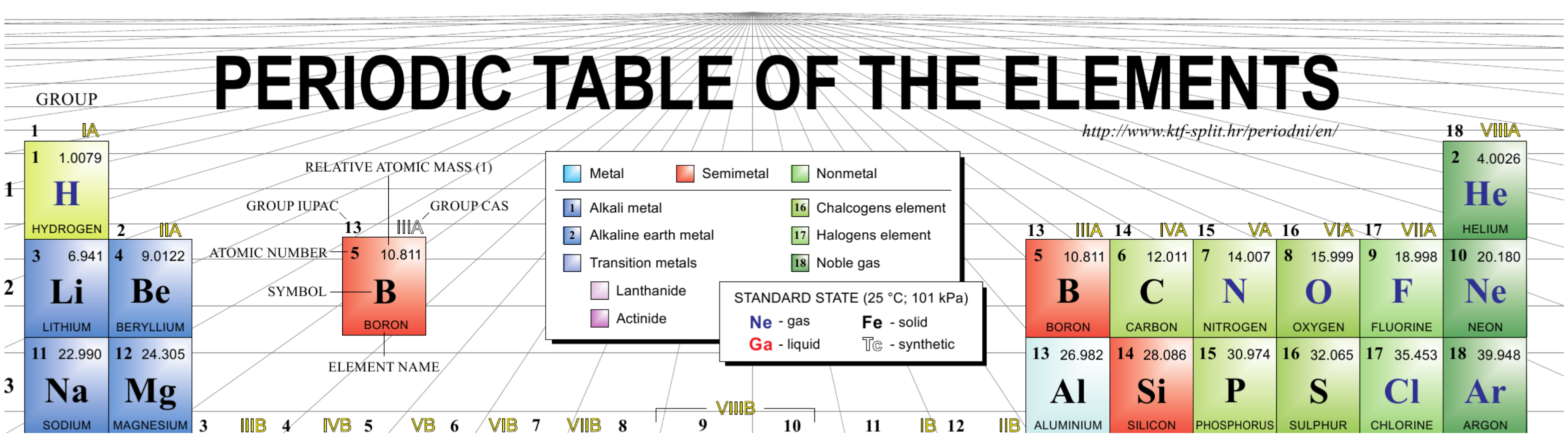
Exercise 1

Determine: (a) the bond order, (b) the presence of lone pairs, (c) the geometry, and (d) the hybridization for ammonia (NH_3) and the nitrate ion (NO_3^-)

1	N	$1 \times 5 = 5\ e^-$
3	H	$3 \times 1 = 3\ e^-$
Total		$8\ e^-$



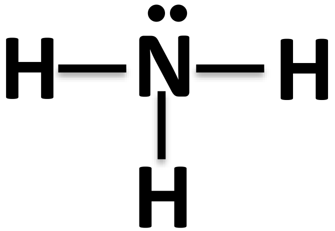
$8\ e^-$



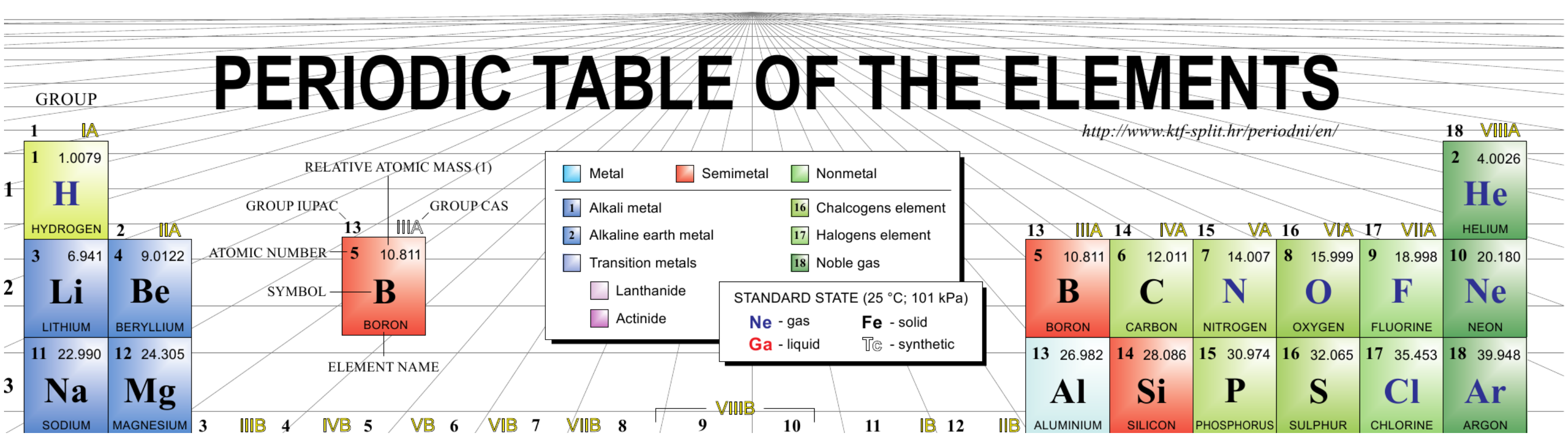
Exercise 1

Determine: (a) the bond order, (b) the presence of lone pairs, (c) the geometry, and (d) the hybridization for ammonia (NH_3) and the nitrate ion (NO_3^-)

1	N	$1 \times 5 = 5\ e^-$
3	H	$3 \times 1 = 3\ e^-$
Total		$8\ e^-$



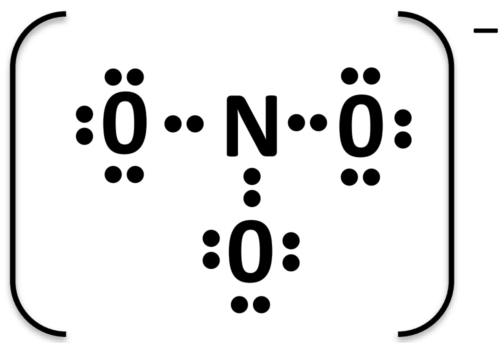
Single bonds, i.e.
Bond order = 1



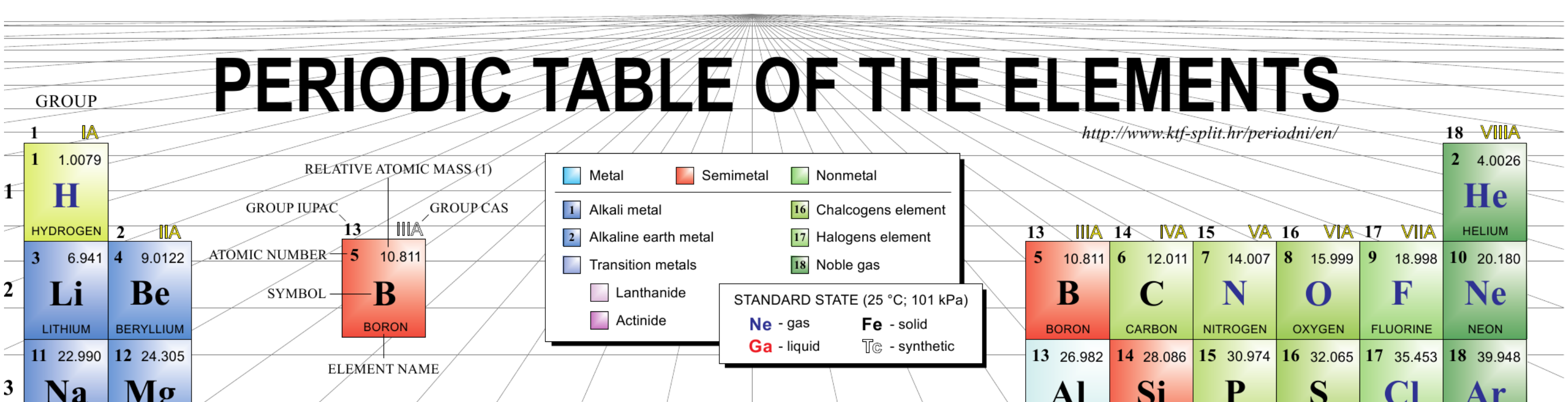
Exercise 1

Determine: (a) the bond order, (b) the presence of lone pairs, (c) the geometry, and (d) the hybridization for ammonia (NH_3) and the nitrate ion (NO_3^-)

1	**N**	**$1 \times 5 = 5\ e^-$**
3	**O**	**$3 \times 6 = 18\ e^-$**
1	**e^-**	**$1 \times 1 = 1\ e^-$**
Total		**24 e^-**



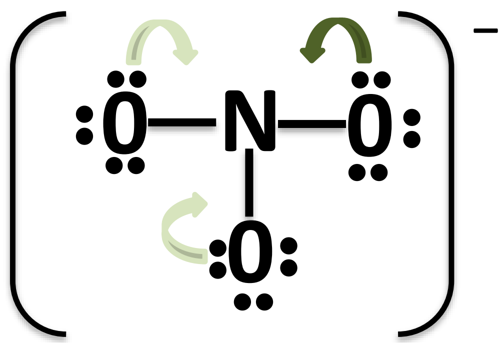
24 e^-



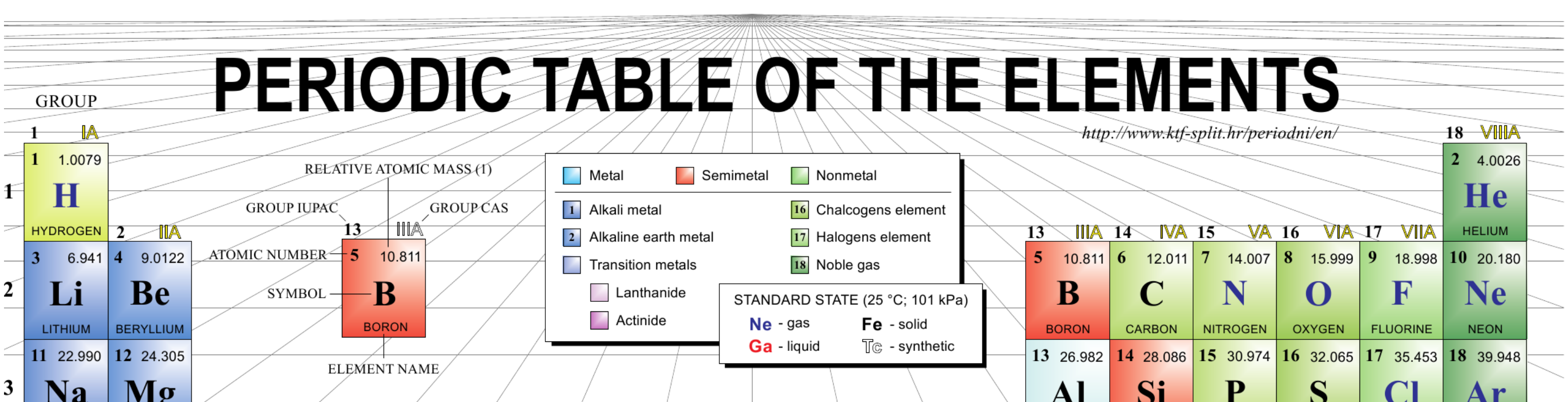
Exercise 1

Determine: (a) the bond order, (b) the presence of lone pairs, (c) the geometry, and (d) the hybridization for ammonia (NH_3) and the nitrate ion (NO_3^-)

1	**N**	**$1 \times 5 = 5\ e^-$**
3	**O**	**$3 \times 6 = 18\ e^-$**
1	**e^-**	**$1 \times 1 = 1\ e^-$**
Total		**$24\ e^-$**

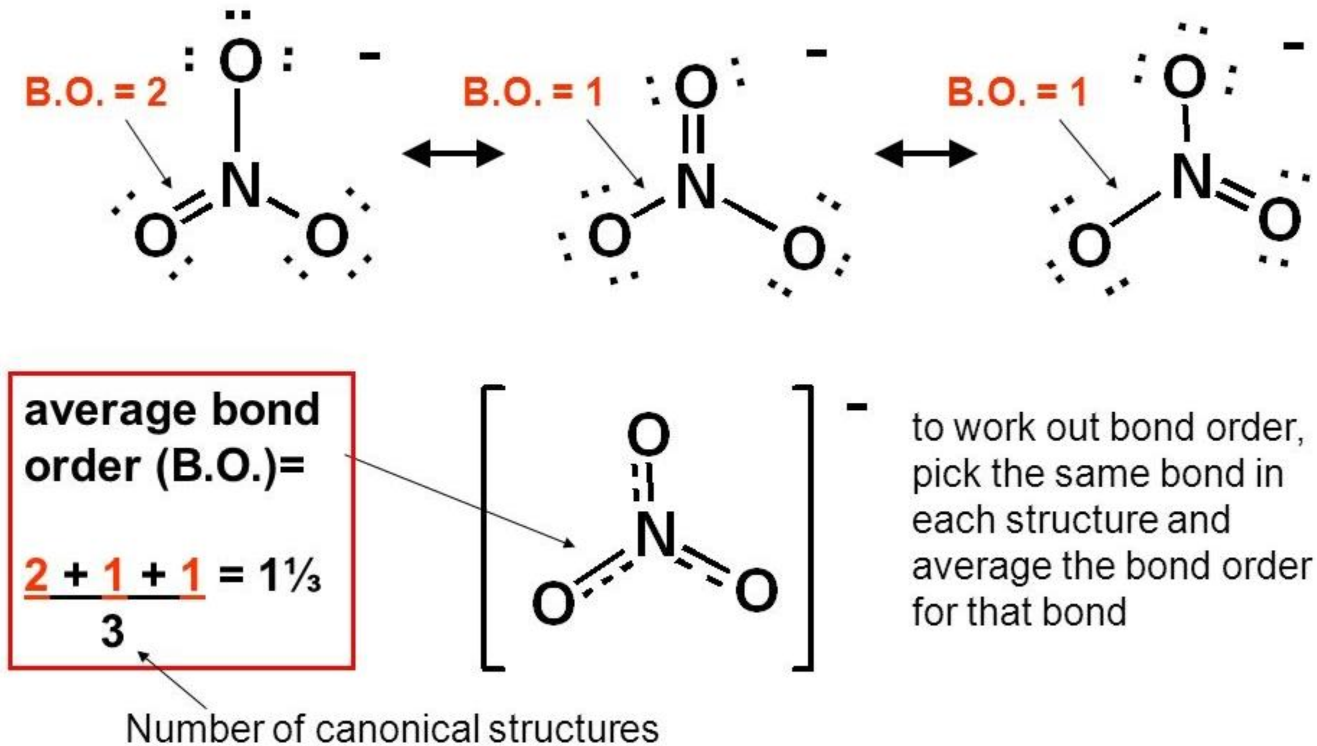


$24\ e^-$



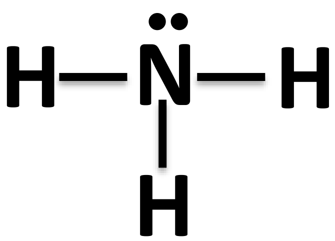
Exercise 1

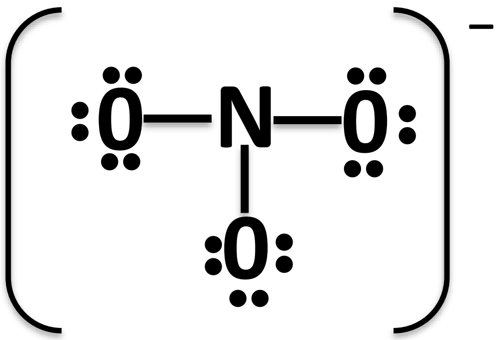
Determine: (a) the bond order, (b) the presence of lone pairs, (c) the geometry, and (d) the hybridization for ammonia (NH_3) and the nitrate ion (NO_3^-)



Exercise 1

Determine: (a) the bond order, (b) the presence of lone pairs, (c) the geometry, and (d) the hybridization for ammonia (NH_3) and the nitrate ion (NO_3^-)

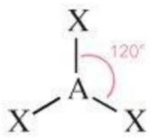
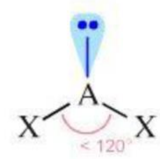
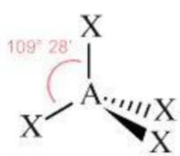
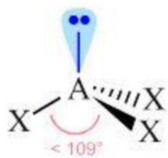
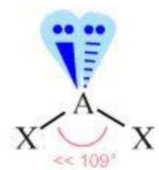
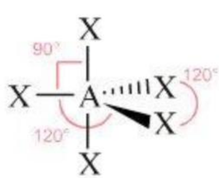
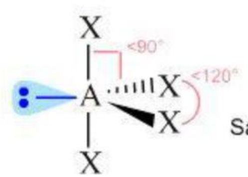
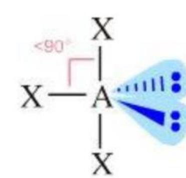




2) Valence-Shell Electron-Pair Repulsion (VSEPR)

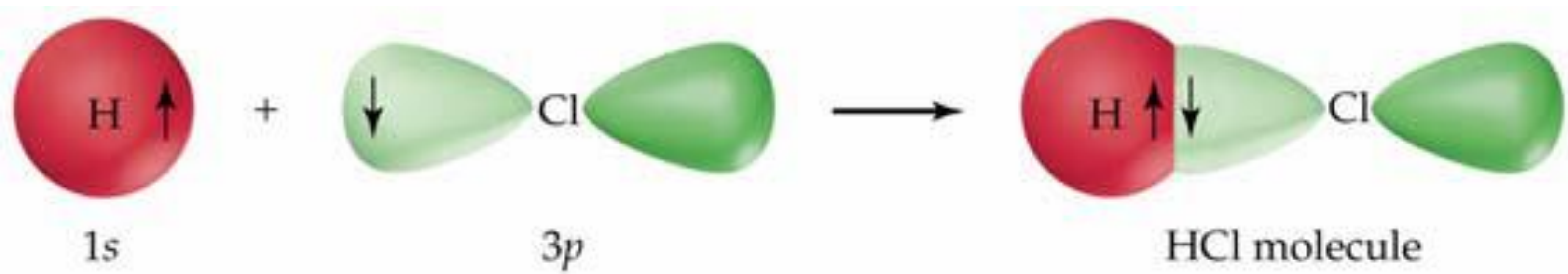
-The approximate shape of a molecule can be predicted from the Lewis dot structure using this model, which depicts electrons in bonds and lone pairs as "**electron groups" or "charge clouds" that repel one another and stay as far apart as possible.**

- Every group of electrons, whether in a **bond or a lone pair**, counts as a charge cloud.

Bonding pairs + Lone pairs	0 Lone pairs	1 Lone pair	2 Lone pairs	
2	 Linear AX_2			
3	nitrate ion  AX_3 Trigonal planar	 AX_2E_1 Bent or Angular		
4	 AX_4 Tetrahedral	ammonia  AX_3E_1 Trigonal pyramidal	 AX_2E_2 Bent or Angular	
5	 AX_5 Trigonal bipyramidal	 AX_4E_1 Sawhorse or Seesaw	 AX_3E_2 T-shape	

3) Valence Bond Theory

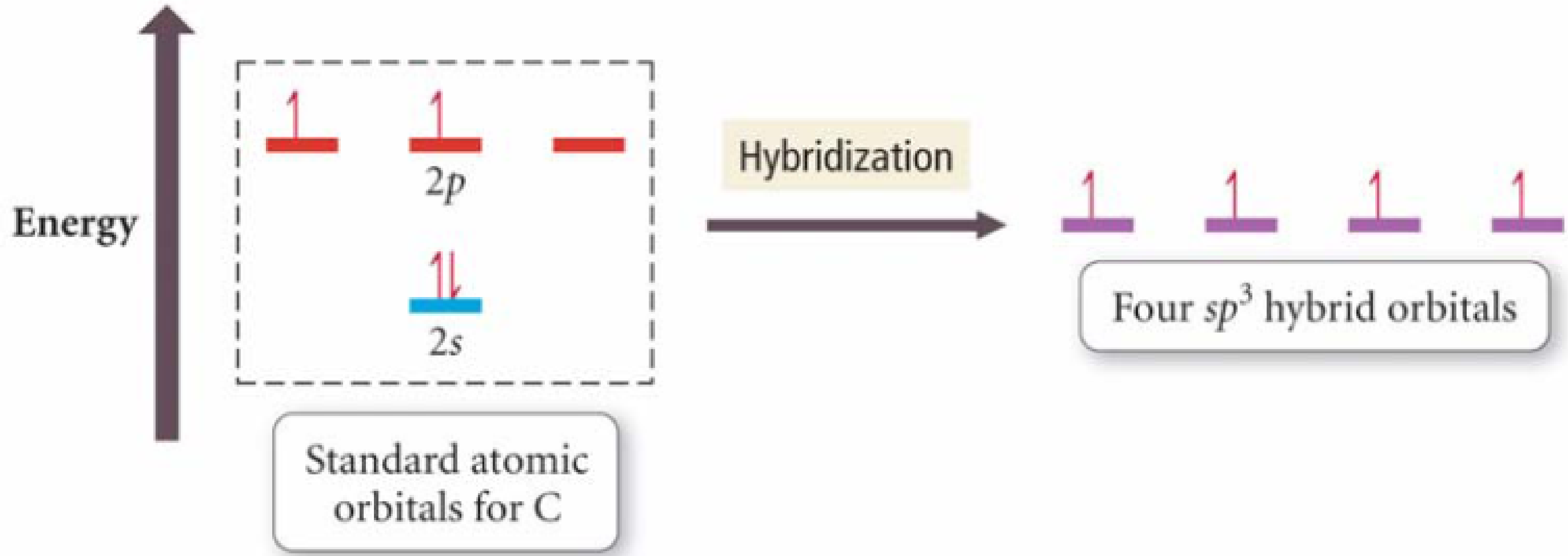
- A covalent bond is formed by pairing the unpaired electrons in neighboring atoms;
- The spins of the unpaired electrons must be antiparallel;



- To provide enough unpaired electrons in each atom for the maximum bond formation, electrons can be excited to fill empty orbitals during bond formation;
- The shape of the molecule results from the directions in which the orbitals of the central atom point.

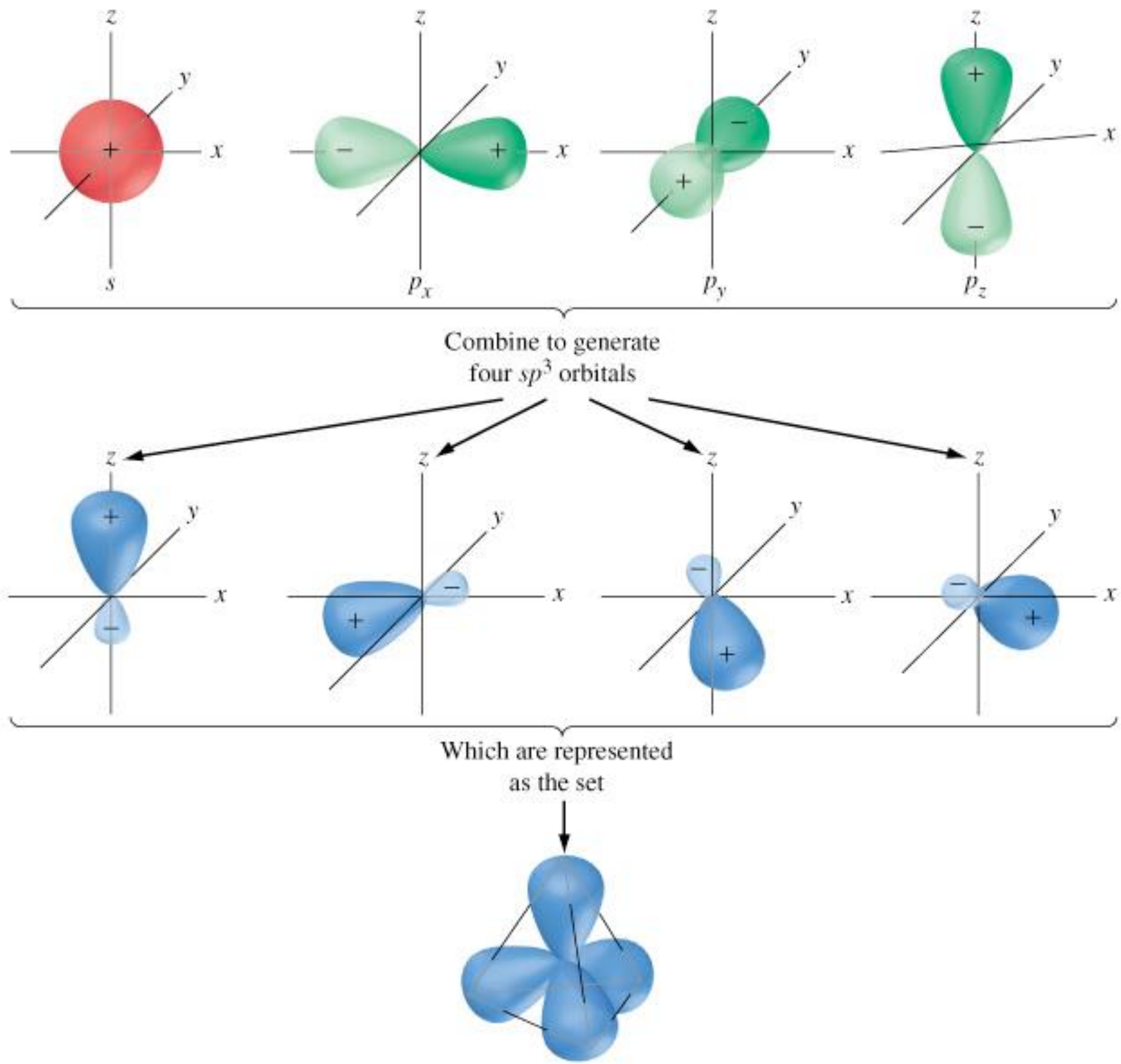
sp^3 Hybridization

- When the *s* and all three *p* orbitals combine, the resulting hybrid orbitals are ***sp*^{3} hybrid orbitals**.

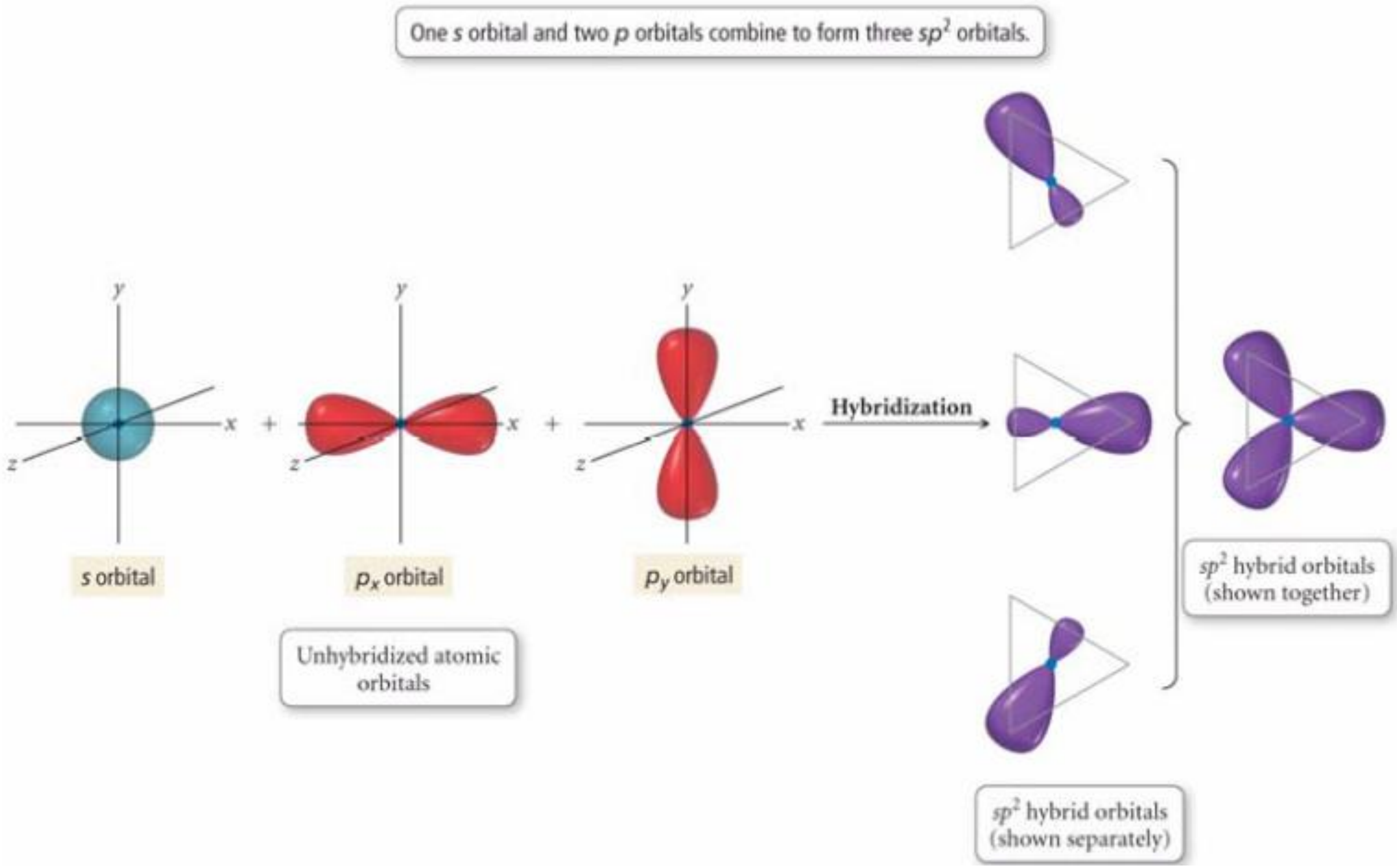


- These 4 orbitals point to the corners of a tetrahedron (109.5°).
- sp^3 orbitals are used to describe the C atoms in CH_4 and C_2H_6, the N in NH_3, and the O in H_2O.

sp^3 Hybridization

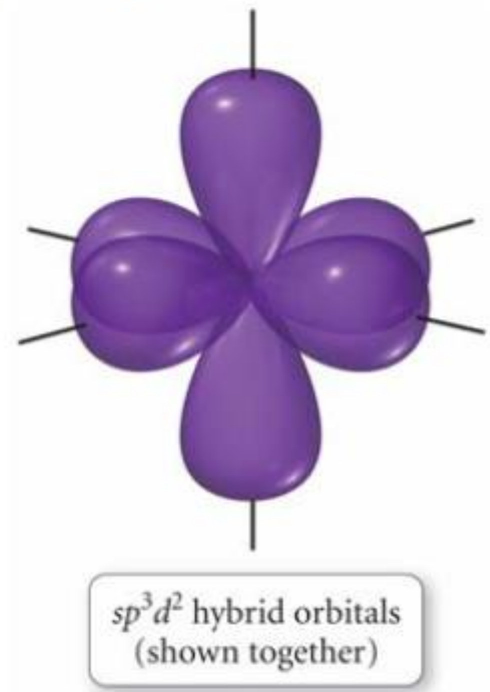


sp^2 Hybridization

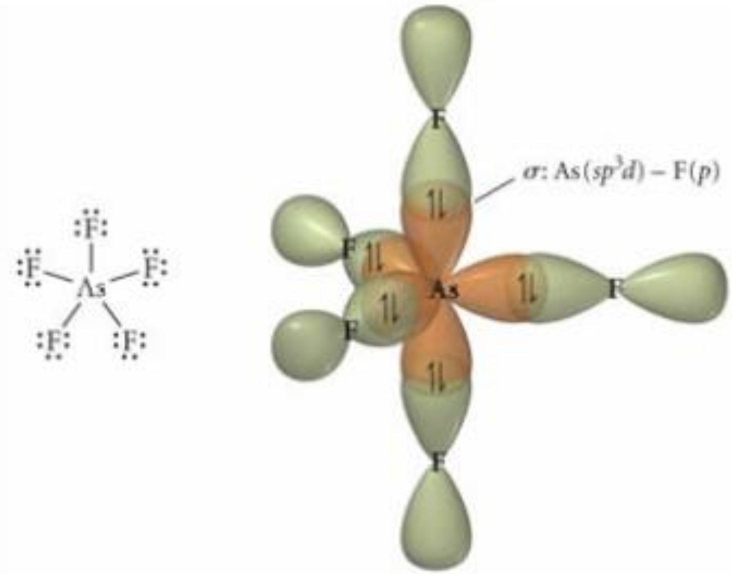


sp^3d and sp^3d^2 Hybridization

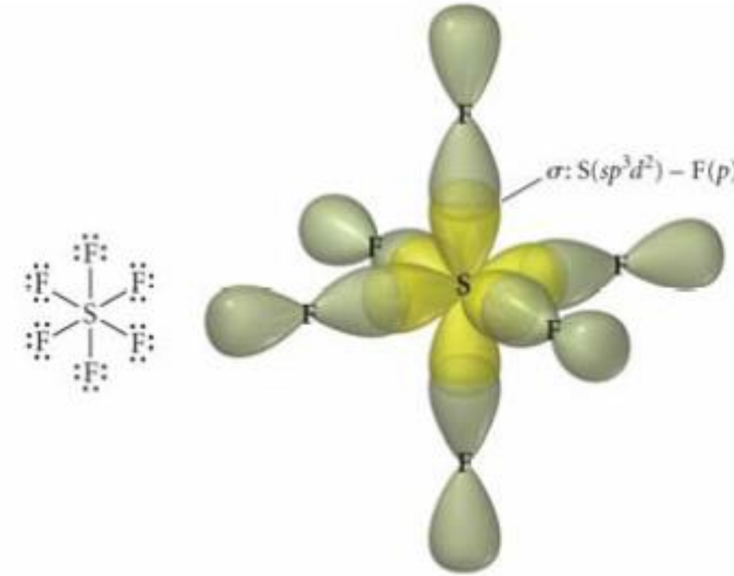
- The *s* and the three *p* orbitals can also combine witl d orbitals in molecules with expanded octets:
  - $s + p + p + p + d =$ **sp^3d hybrid orbitals**
  - $s + p + p + p + d + d =$ **sp^3d^2 hybrid orbitals**



sp^3d Orbitals in AsF_5; sp^3d^2 Orbitals in SF_6



sp^3d hybrid orbitals in AsF_5



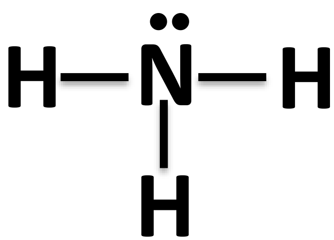
sp^3d^2 hybrid orbitals in SF_6

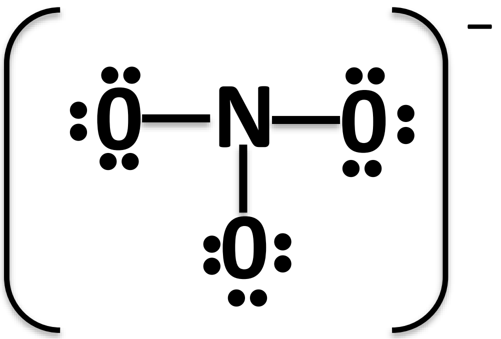
Hybrid Orbitals

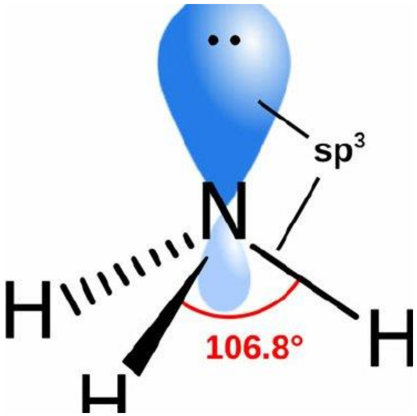
2 charge clouds	→	2 *sp* orbitals	→	linear (180°)
3 charge clouds	→	3 sp^2 orbitals	→	trigonal planar (120°)
4 charge clouds	→	4 sp^3 orbitals	→	tetrahedral (109.5°)
5 charge clouds	→	5 sp^3d orbitals	→	trigonal bipyramidal (90°, 120°)
6 charge clouds	→	6 sp^3d^2 orbitals	→	octahedral (90°)

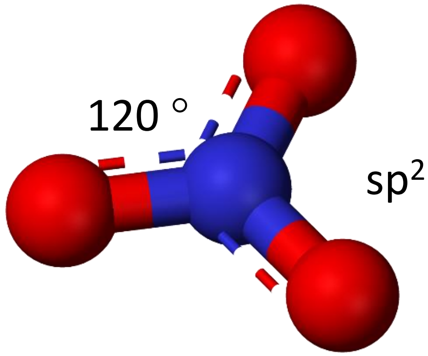
Exercise 1

Determine: (a) the bond order, (b) the presence of lone pairs, (c) the geometry, and (d) the hybridization for ammonia (NH_3) and the nitrate ion (NO_3^-)





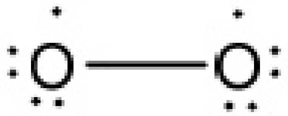


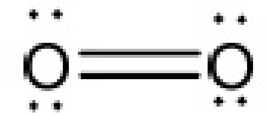


Exercise 2

Determine the bond order, the presence of lone pairs, and the geometry for an oxygen molecule **O_2**

2	**O**	**$2 \times 6 = 12\ e^-$**





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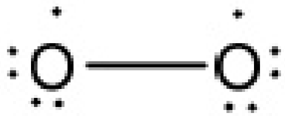
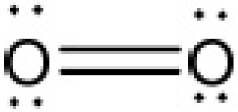
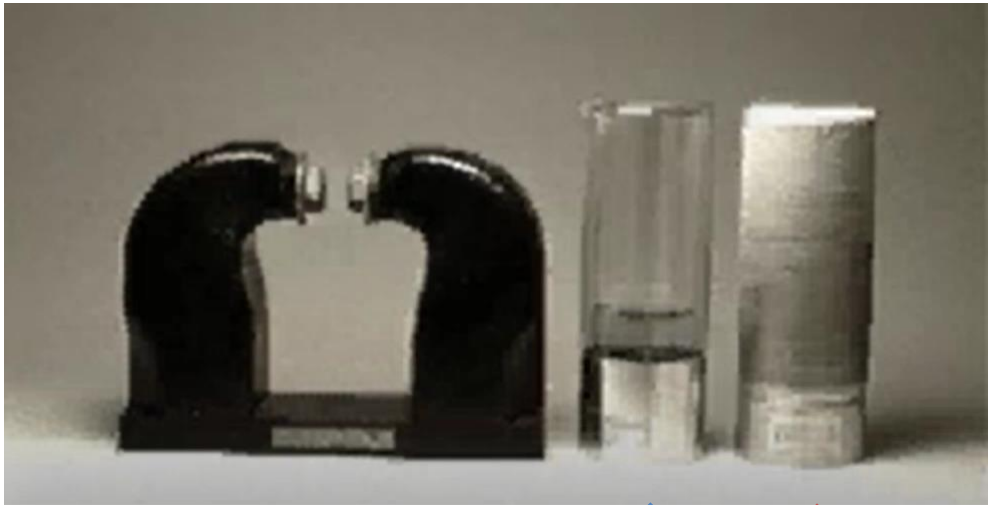
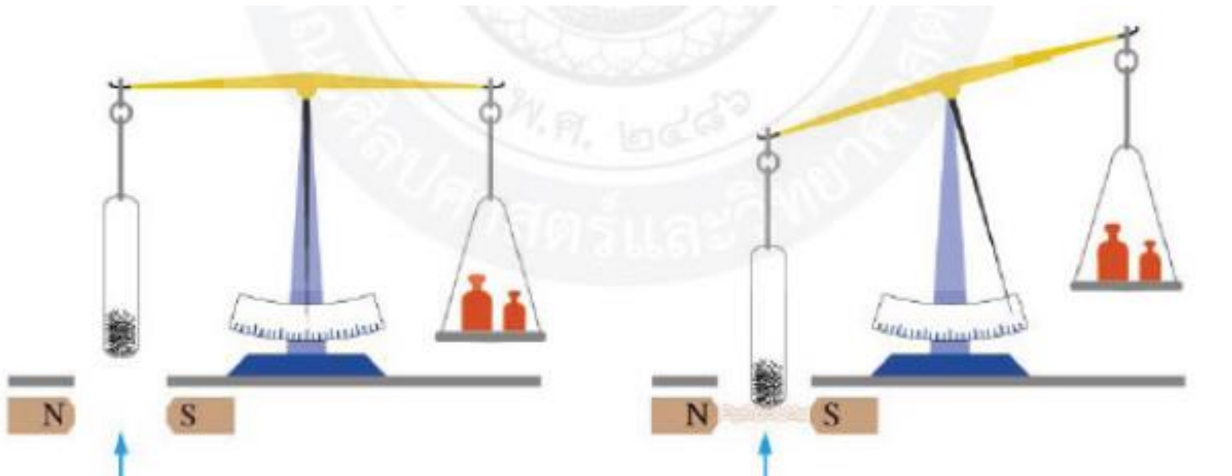






Unpaired electrons produce **paramagnetism**. The material is *attracted to the field* and it is only induced when an external field is applied.
When the field is removed all electrons are randomized by thermal motion. This is why paramagnetism decreases as temperature increases.



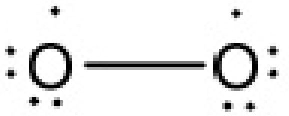
(a) No magnetic field (b) Magnetic field turned on

Property	Effect of external field	Specific susceptibility (χ) at 20 °C, (cgs units) g^{-1}	Temperature dependence of χ	Field dependence of χ
Diamagnetism	Weak repulsion	-1×10^{-6}	None	None
Paramagnetism	Moderate attraction	100×10^{-6}	$1/T$	None

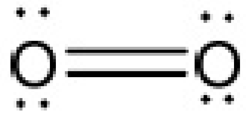
Exercise 2

Determine the bond order, the presence of lone pairs, and the geometry for an oxygen molecule **O_2**

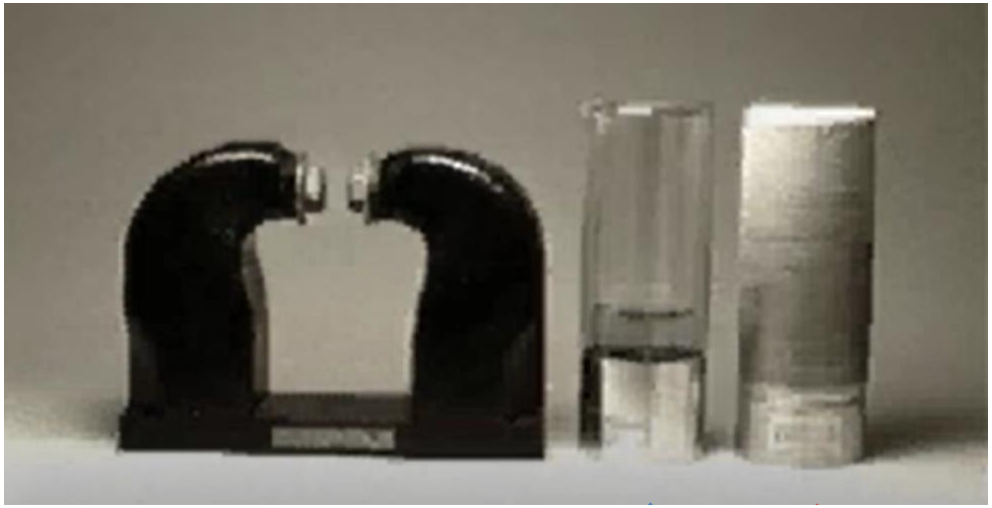
2	O	$2 \times 6 = 12\ e^-$



Paramagnetic



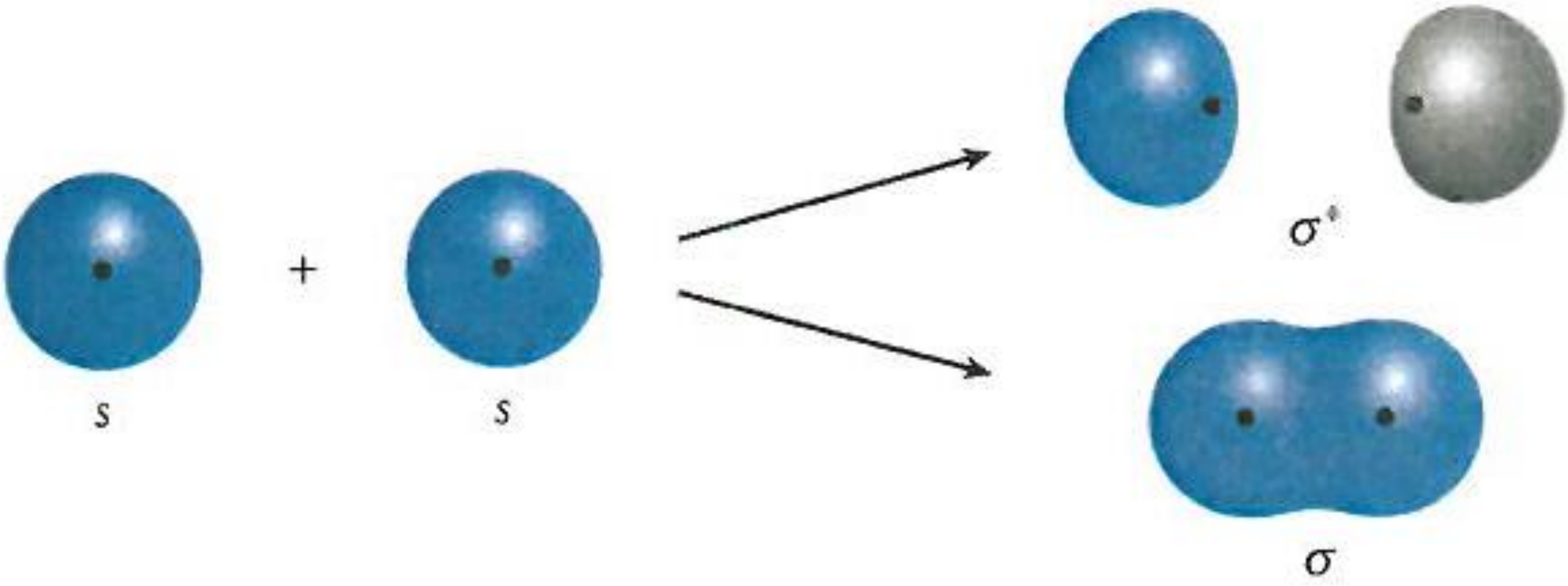
Bond order = 2



4) Molecular orbitals theory

Assumptions:

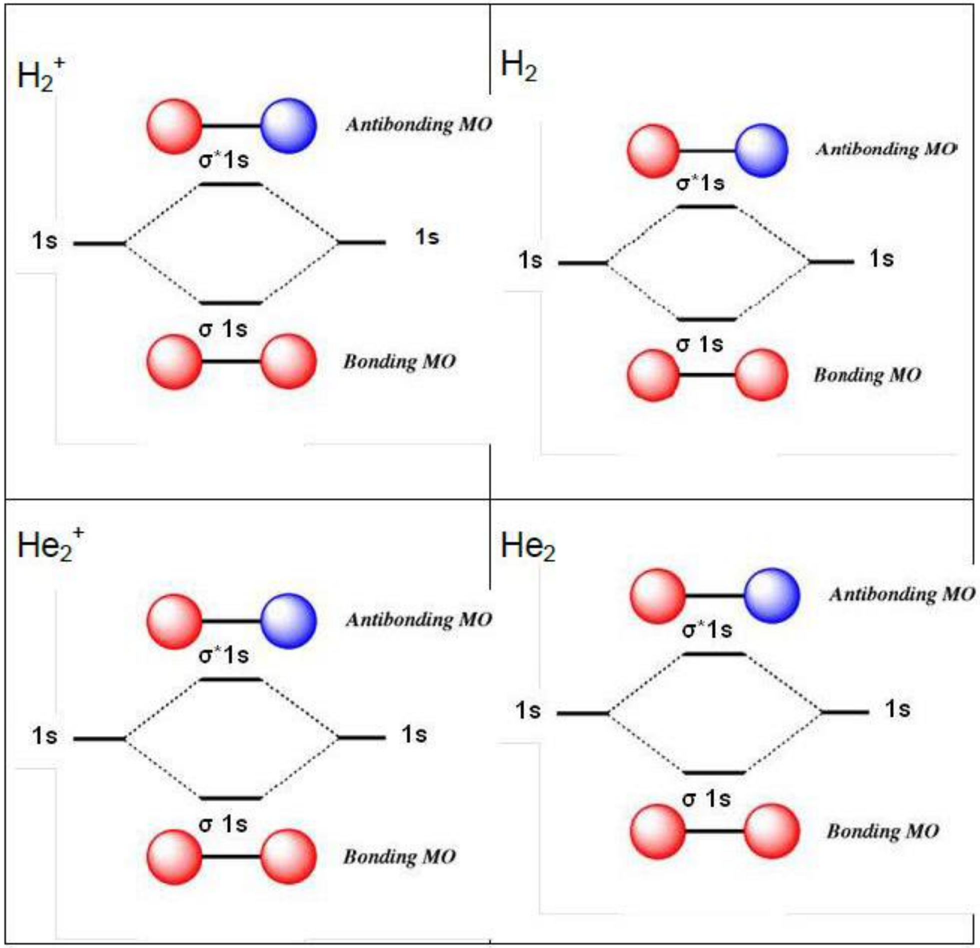
- Molecular orbitals are formed from the **overlap of atomic orbitals**.
- Only atomic orbitals of about the **same energy** interact to a significant degree.
- When two atomic orbitals overlap, they interact in two extreme ways to form two molecular orbitals, a **bonding** molecular orbital and an **antibonding** molecular orbital.
- Molecular orbitals are **filled** by following Pauli’s exclusion principle, aufbau rule and Hund’s principle.



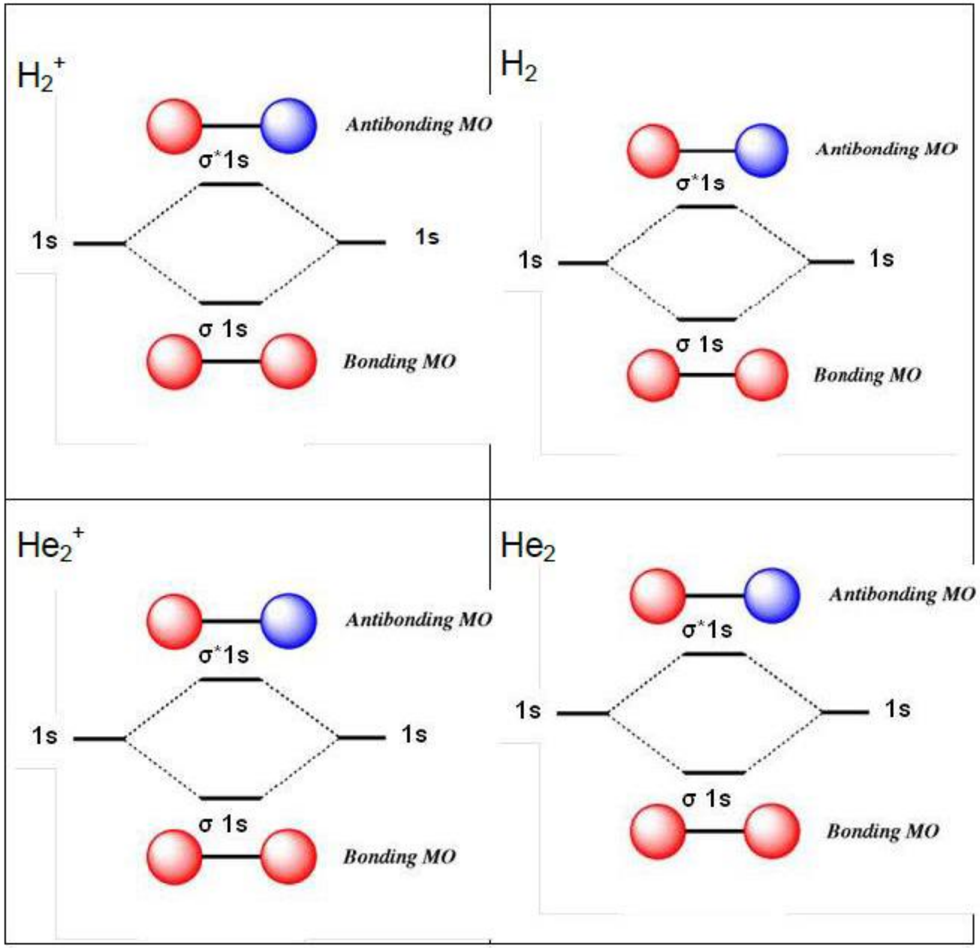
Exercise 3

Draw the molecular orbital diagram for the following molecules:

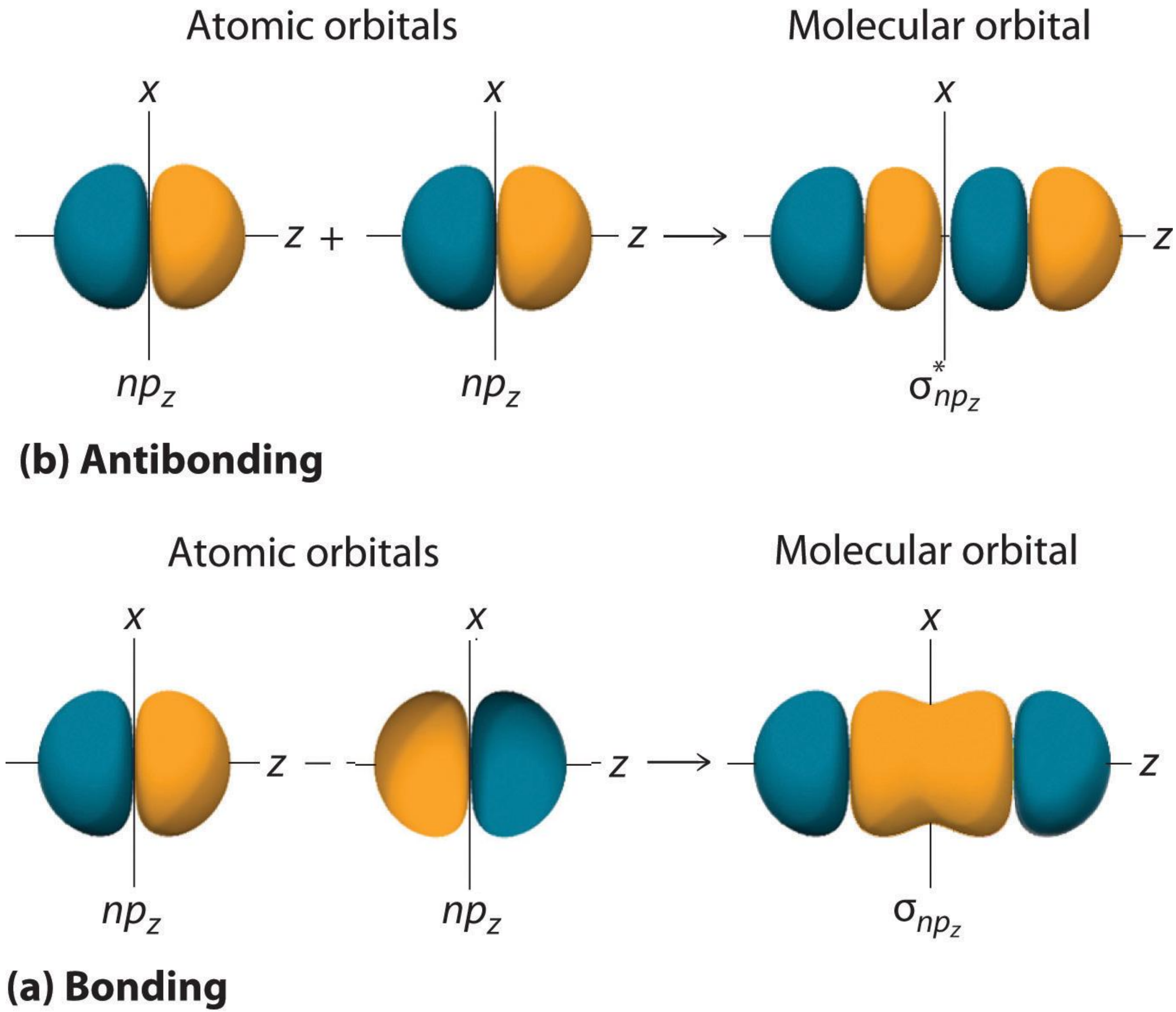
H_2, H_2^+, He_2^+, He_2

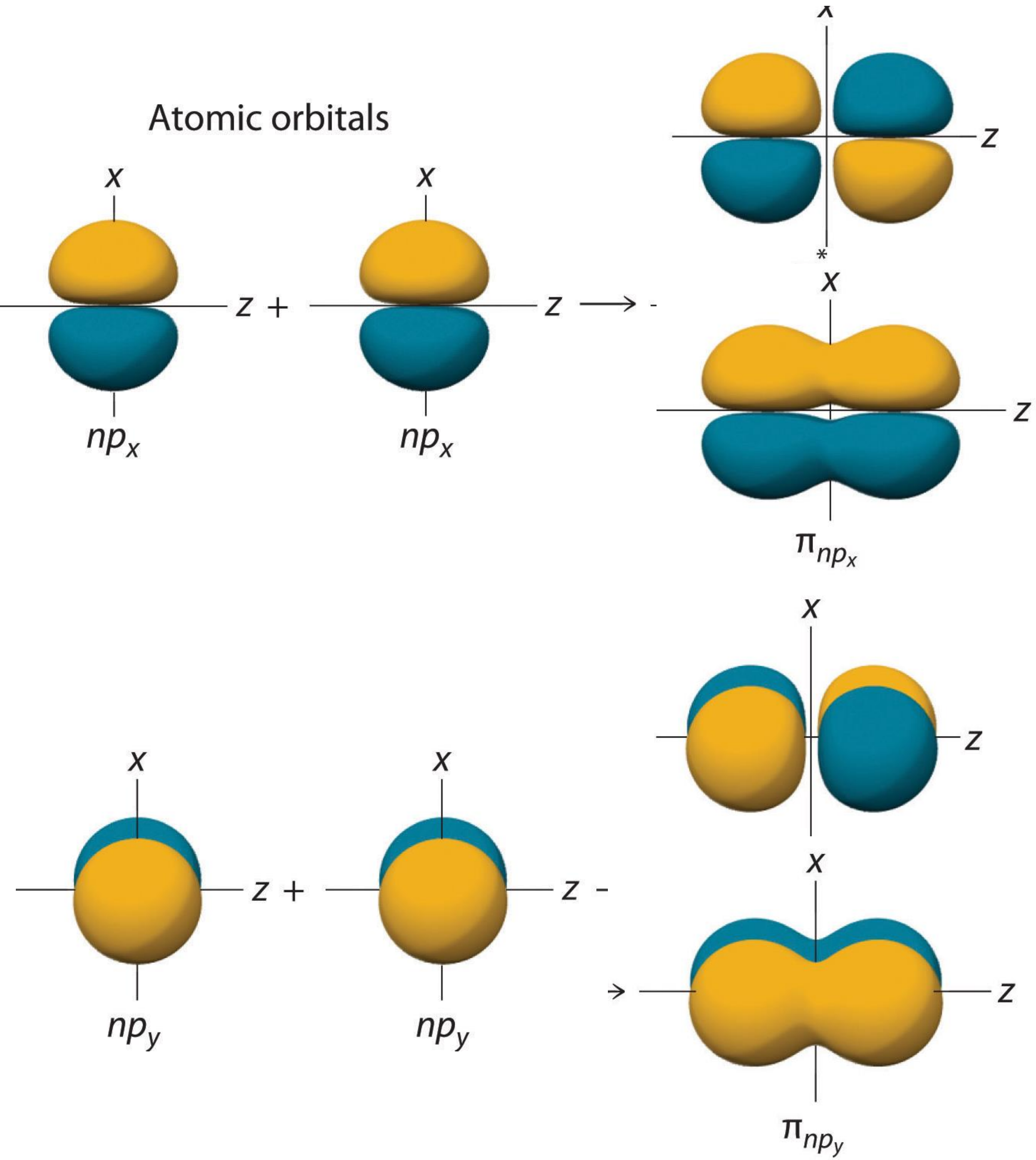


$$\text{Bond order} = \frac{e^- \text{ in bonding MO} - e^- \text{ in antibonding MO}}{2}$$


H2+
Antibonding MO
σ*1s
1s
1s
σ 1s
Bonding MO
H2
Antibonding MO
σ*1s
1s
1s
σ 1s
Bonding MO
He2+
Antibonding MO
σ*1s
1s
1s
σ 1s
Bonding MO
He2
Antibonding MO
σ*1s
1s
1s
σ 1s
Bonding MO

MO orbitals p orbitals



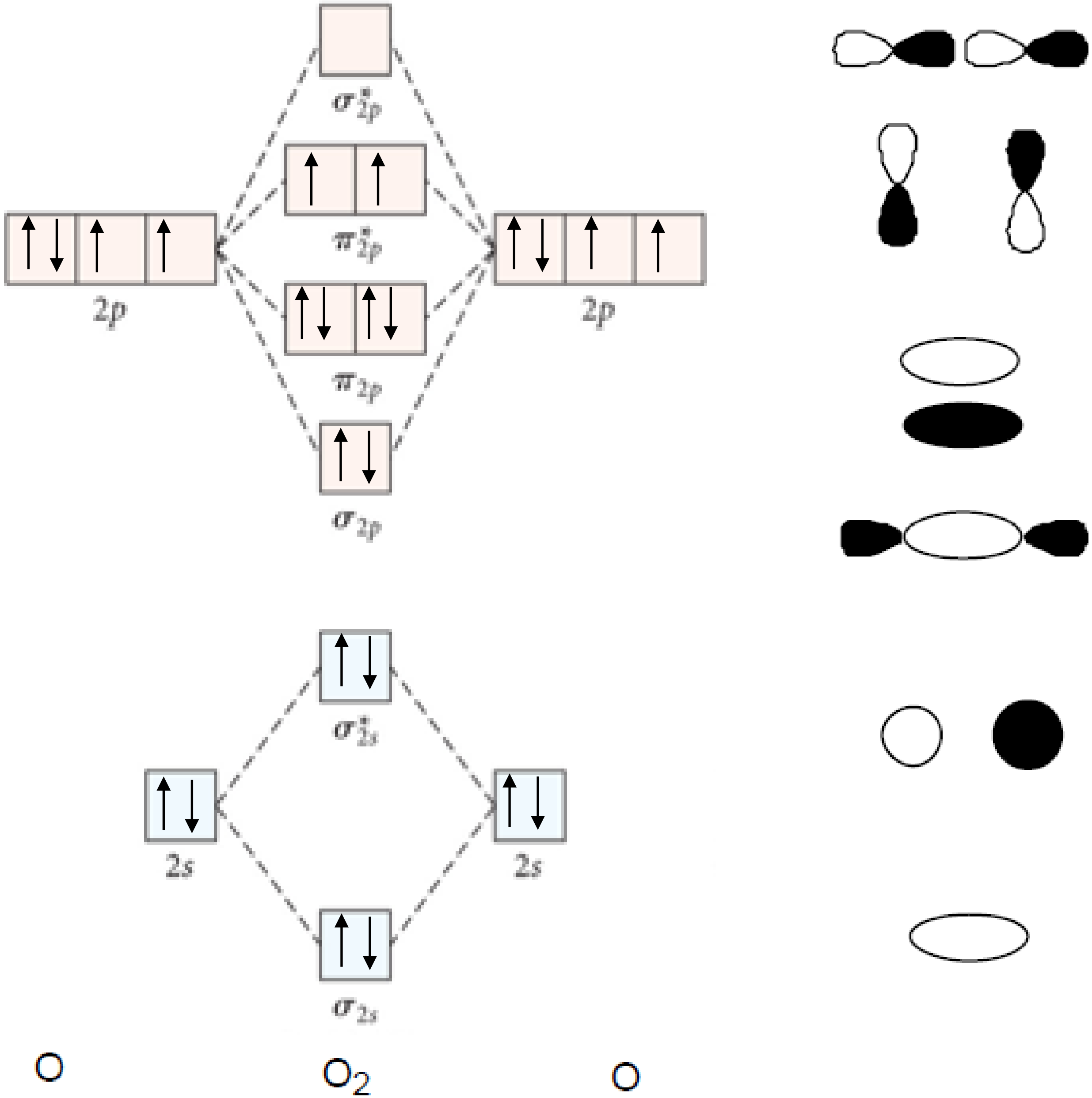


antibonding

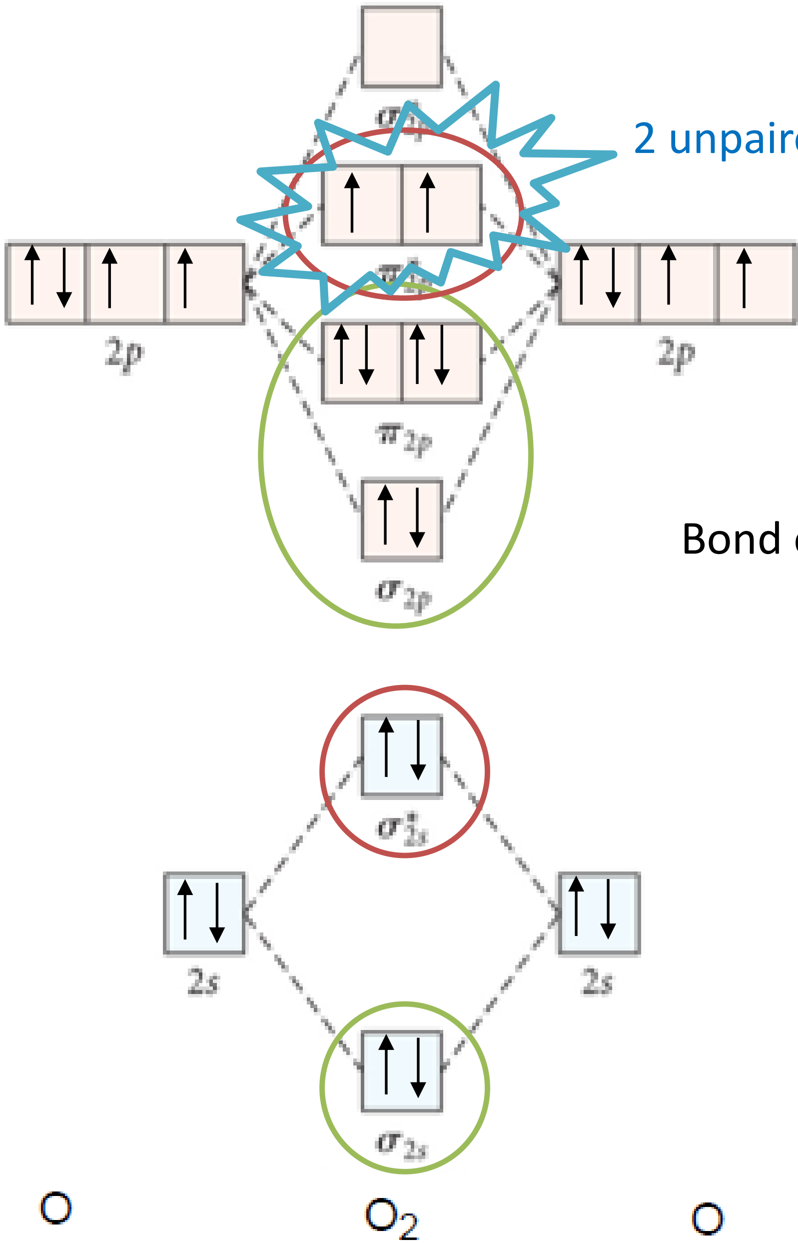
bonding

antibonding

bonding



σ^*_{2p}
π^*_{2p}
2p
2p
π_{2p}
σ_{2p}
σ^*_{2s}
2s
2s
σ_{2s}
O
O_2
O



2 unpaired e^- : paramagnetic !!!!

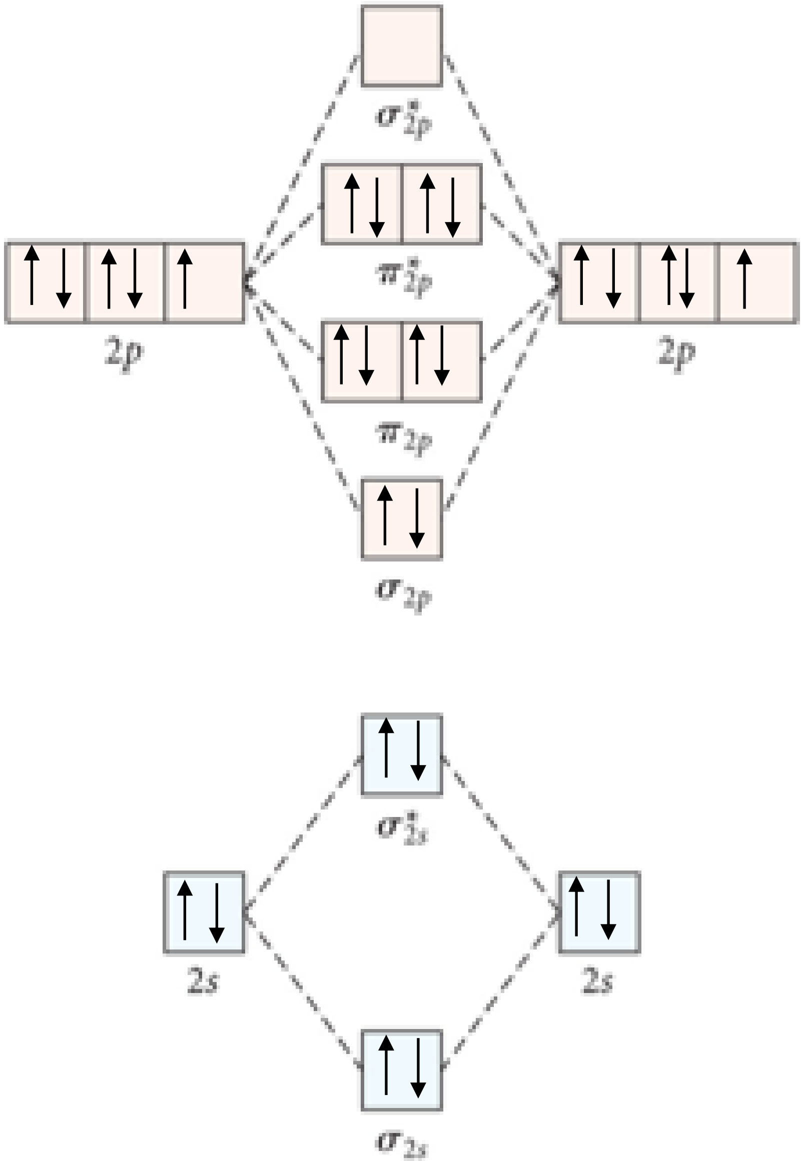
$$\text{Bond order} = \frac{e^- \text{ in bonding MO} - e^- \text{ in antibonding MO}}{2}$$

Exercise 4

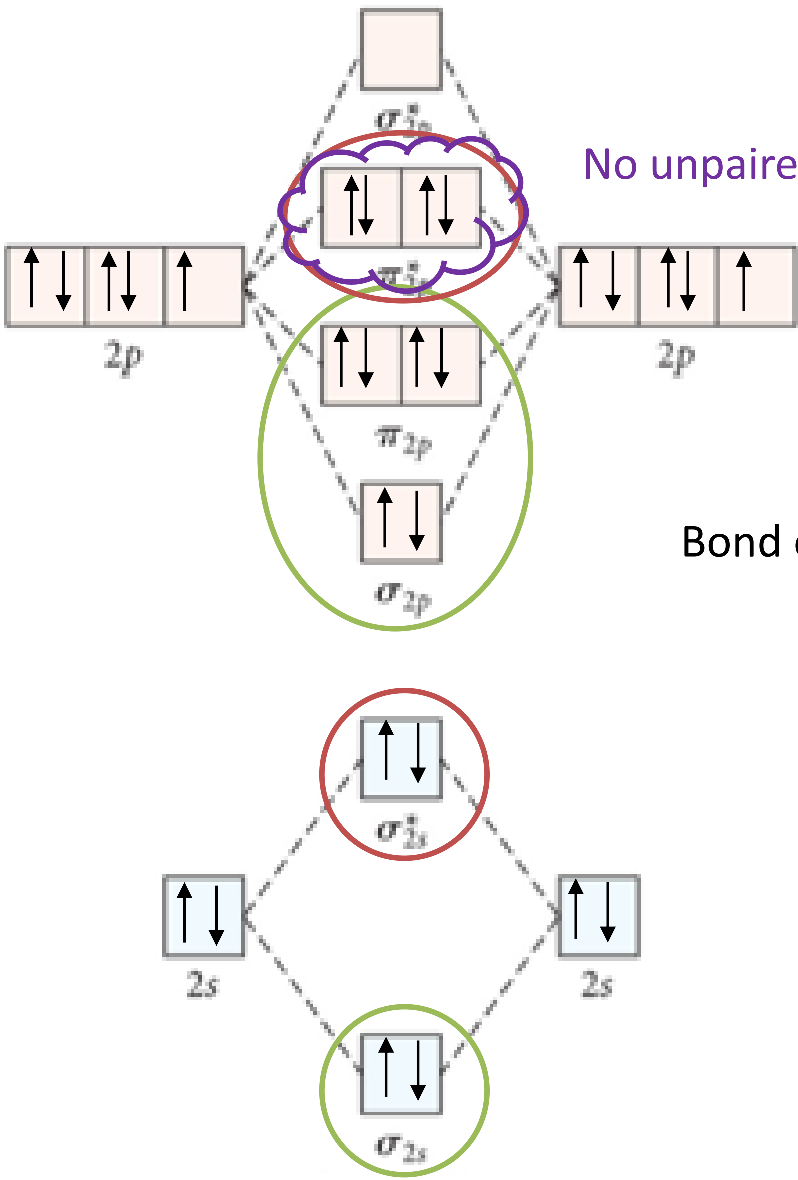
Draw the molecular orbital diagram for F_2

Predict the bond order and if this molecule is paramagnetic or diamagnetic

Exercise 4

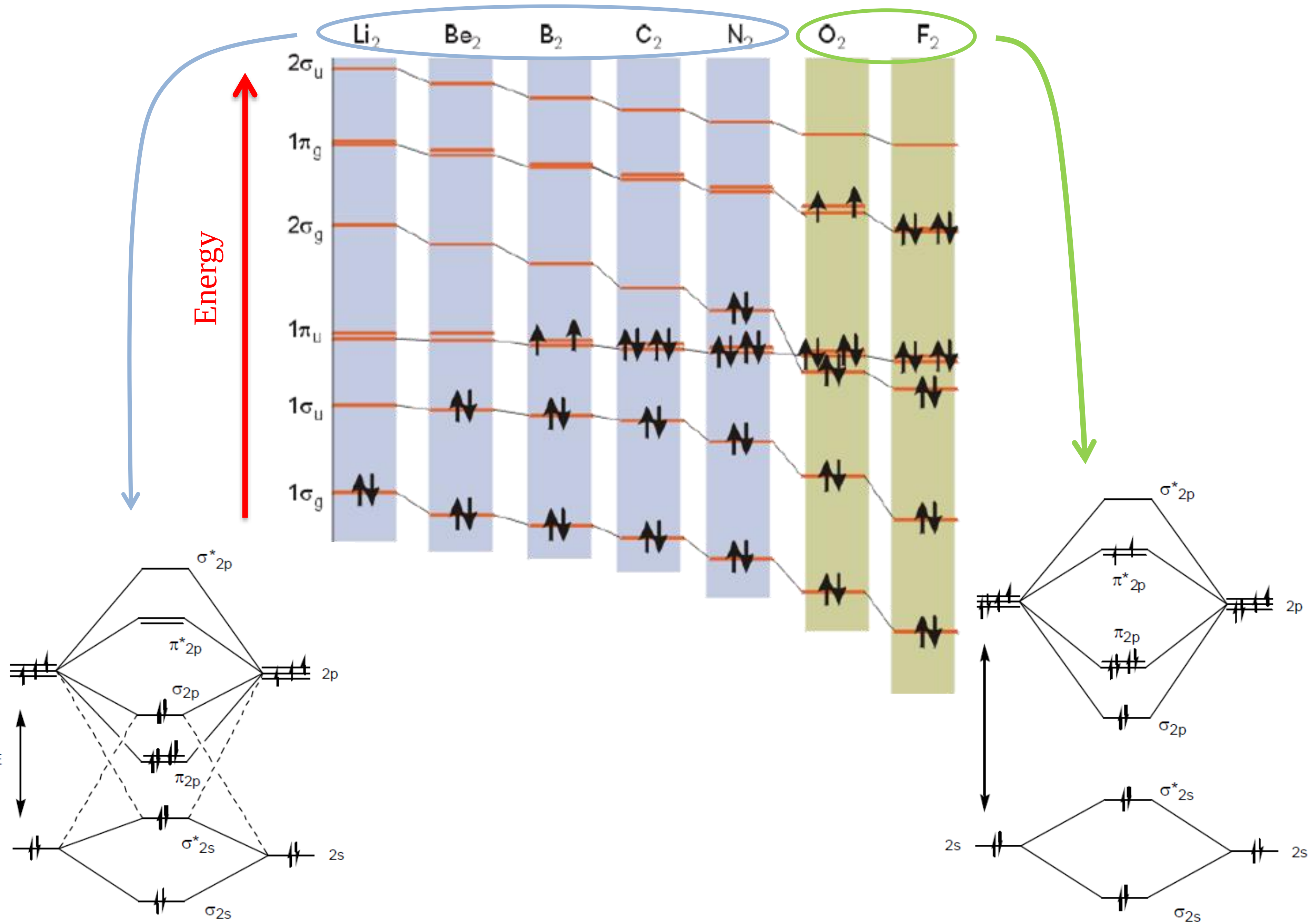


Exercise 4

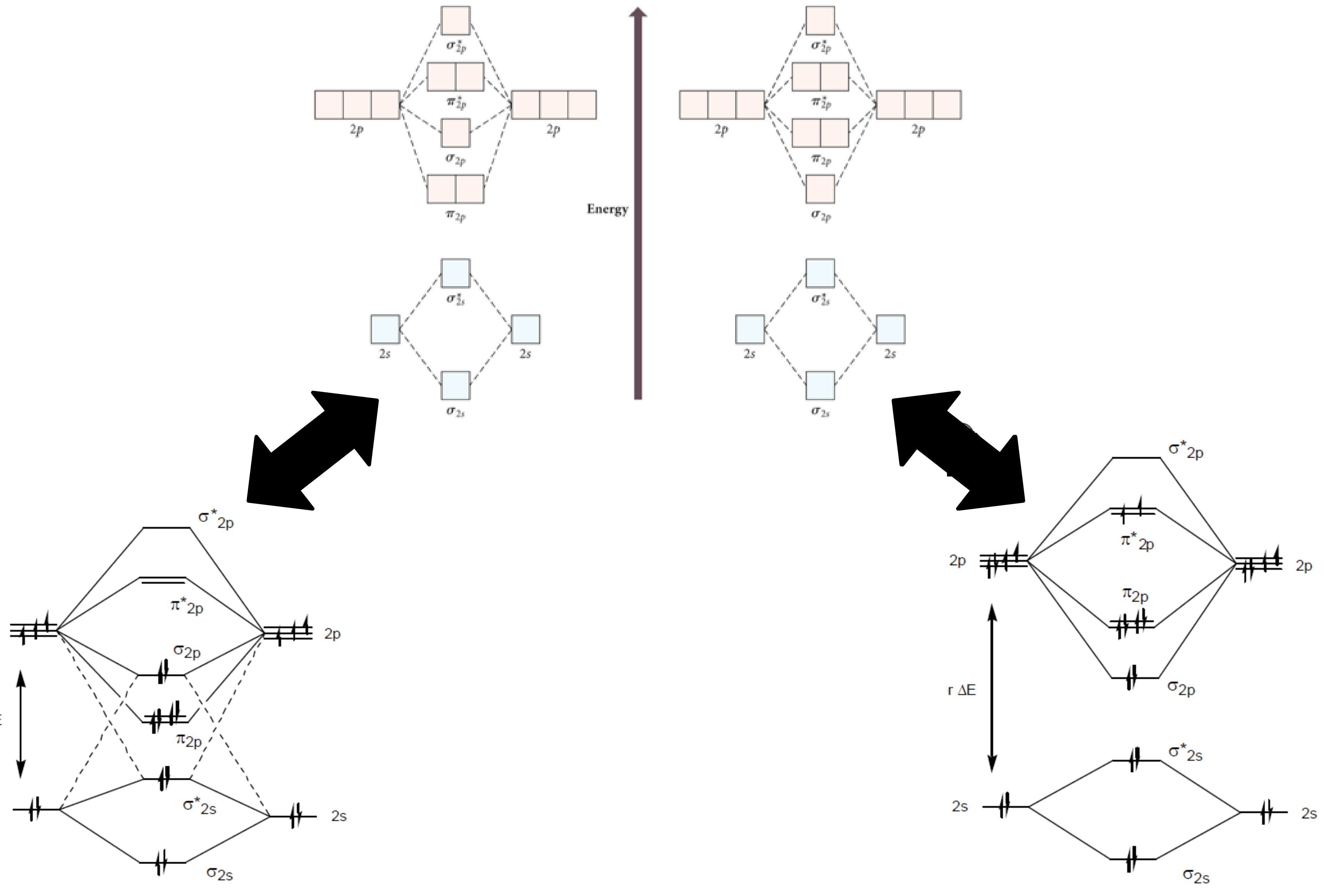


No unpaired e^- : diamagnetic !!!!

$$\text{Bond order} = \frac{e^- \text{ in bonding MO} - e^- \text{ in antibonding MO}}{2}$$



Li2
Be2
B2
C2
N2
O2
F2
2σu
1πg
2σg
1πu
1σu
1σg
Energy
σ*2p
π*2p
σ2p
π2p
σ*2s
σ2s
2p
2s
σ*2p
π*2p
π2p
σ2p
σ*2s
σ2s
2p
2s



σ^*_{2p}
π^*_{2p}
σ_{2p}
π_{2p}
2p
2p
σ^*_{2s}
2s
2s
σ_{2s}
Energy
σ^*_{2p}
π^*_{2p}
π_{2p}
σ_{2p}
2p
2p
σ^*_{2s}
2s
2s
σ_{2s}
σ^*_{2p}
π^*_{2p}
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π_{2p}
σ^*_{2s}
2s
2s
σ_{2s}
σ^*_{2p}
π^*_{2p}
2p
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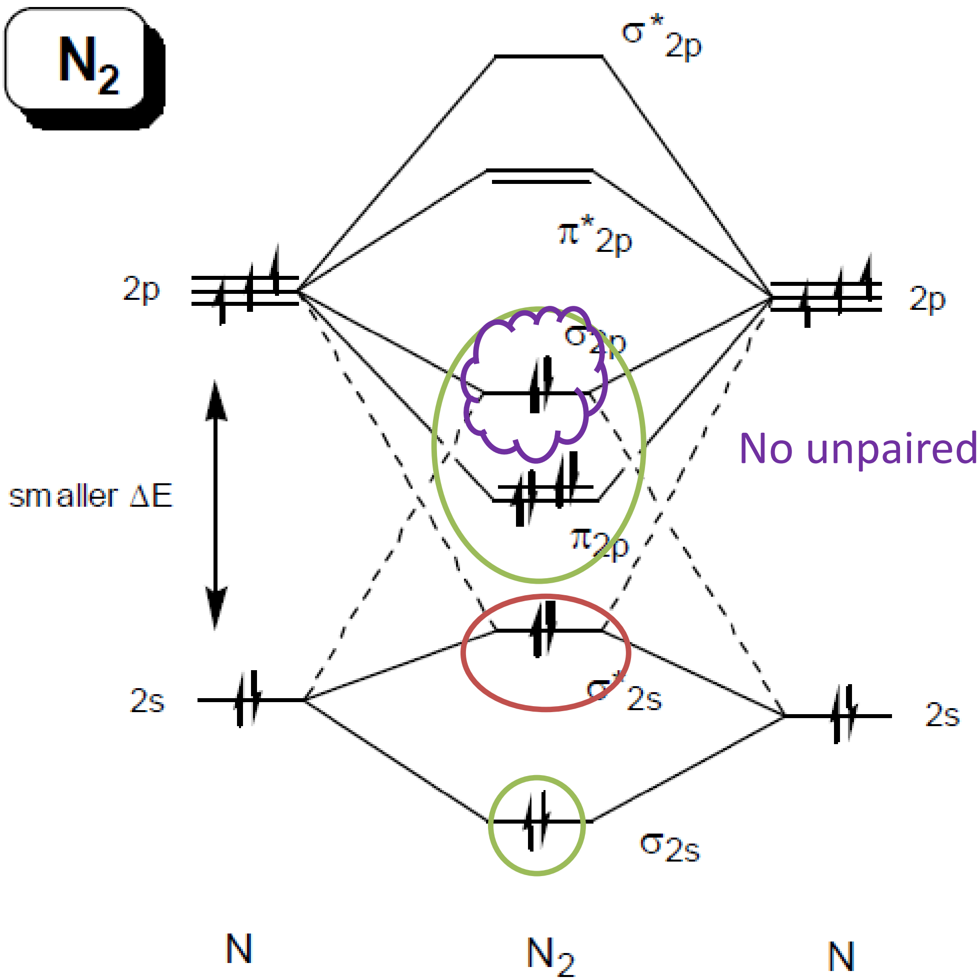
π_{2p}
r ΔE
σ_{2p}
σ^*_{2s}
2s
2s
σ_{2s}



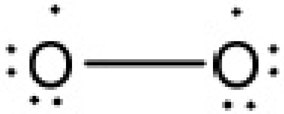
No unpaired e⁻ : diamagnetic !!!!

$$\text{B.O.} = \frac{8 - 2}{2} = 3$$ i.e. tripple bond

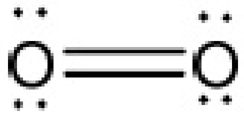
Exercise 2

Determine the bond order, the presence of lone pairs, and the geometry for an oxygen molecule **O_2**

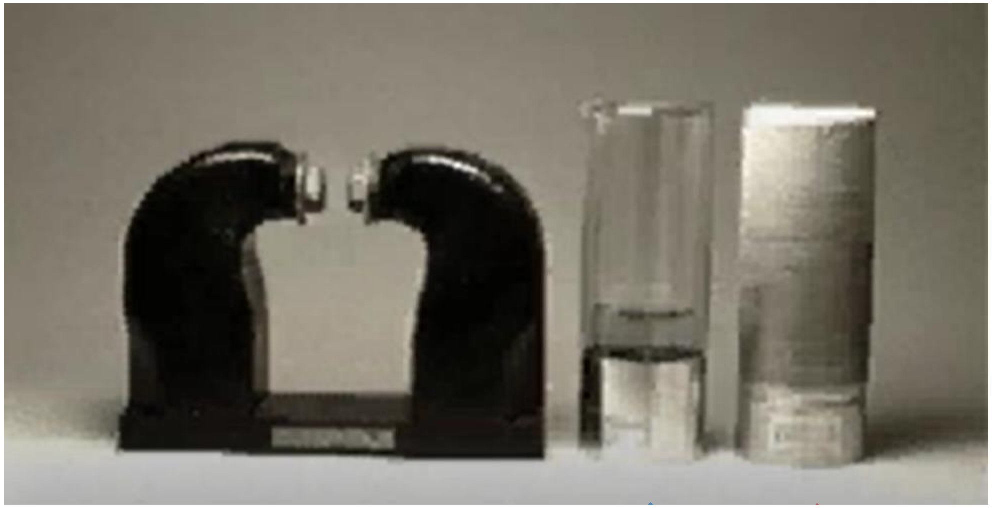
2	**O**	**$2 \times 6 = 12\ e^-$**



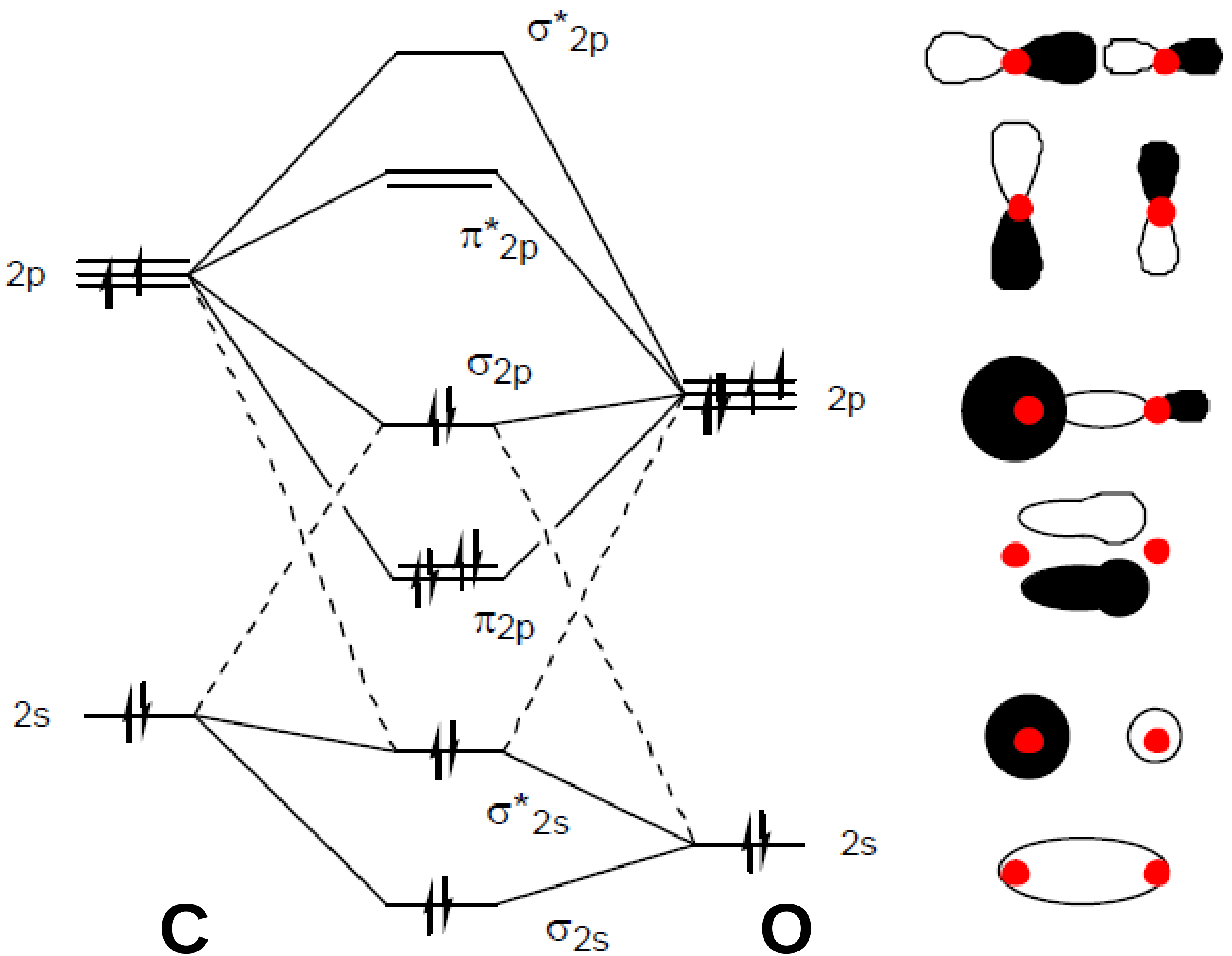
Paramagnetic



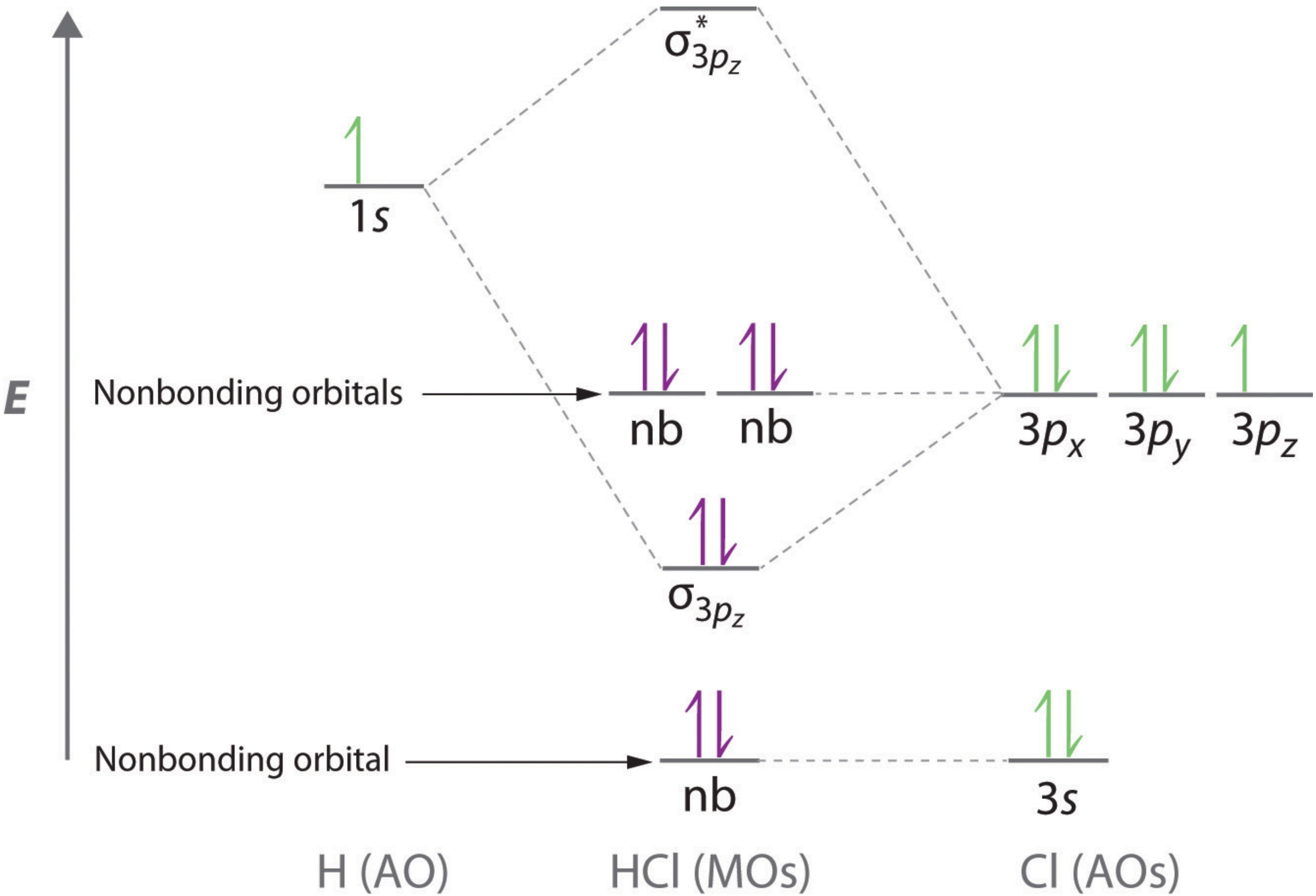
Bond order = 2



CO



HCl



E
Nonbonding orbitals
Nonbonding orbital
1s
σ*3pz
nb
nb
σ3pz
nb
3px
3py
3pz
3s
H (AO)
HCl (MOs)
Cl (AOs)

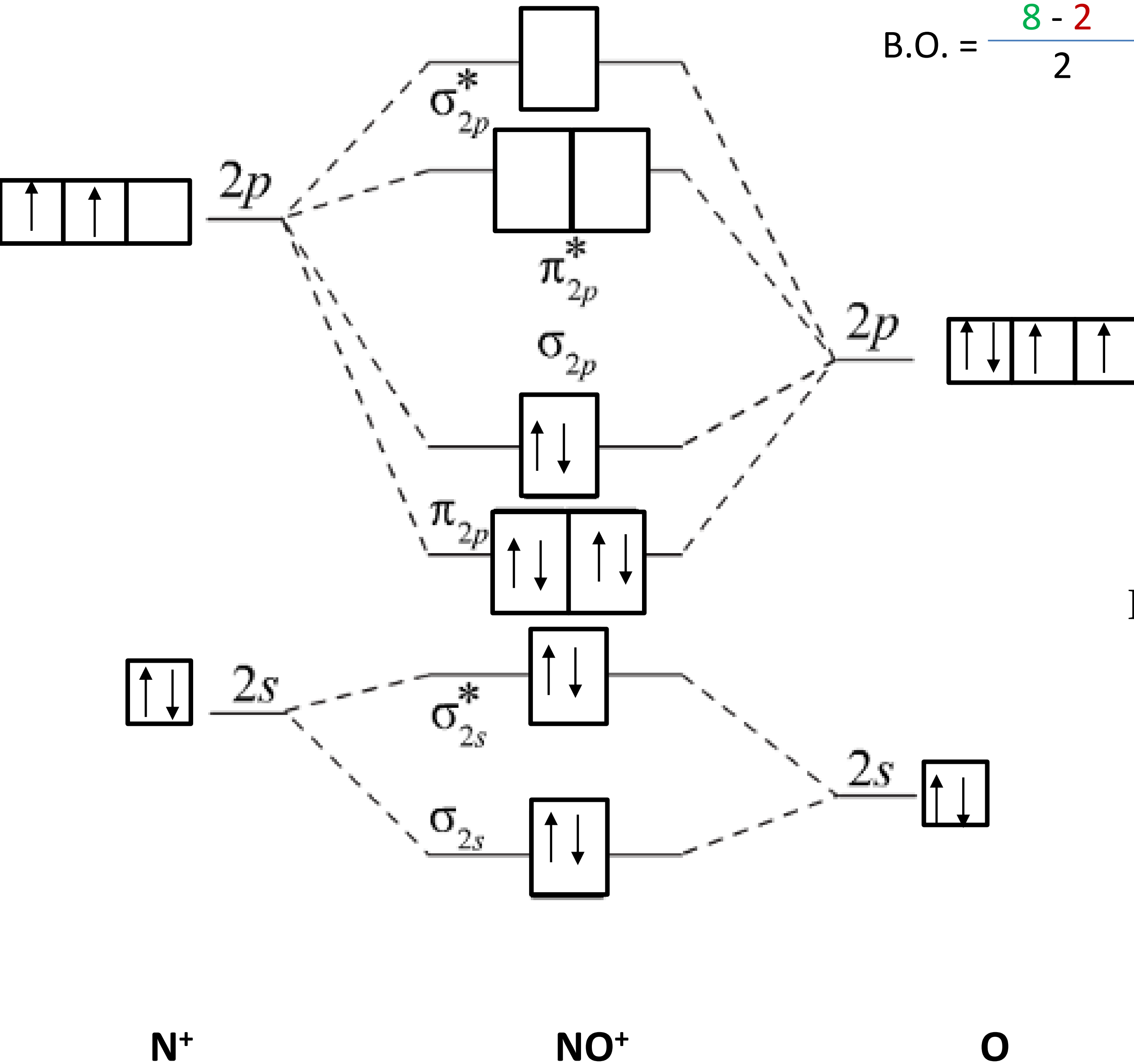
Exercise 5

Using molecular orbital diagrams, predict which of the following gas-phase reaction is the more favored and give your reasoning. You can assume that all the species in this exercise have MO structure similar to that of carbon monoxide.

$NO + CN \rightarrow NO^{+} + CN^{-}$

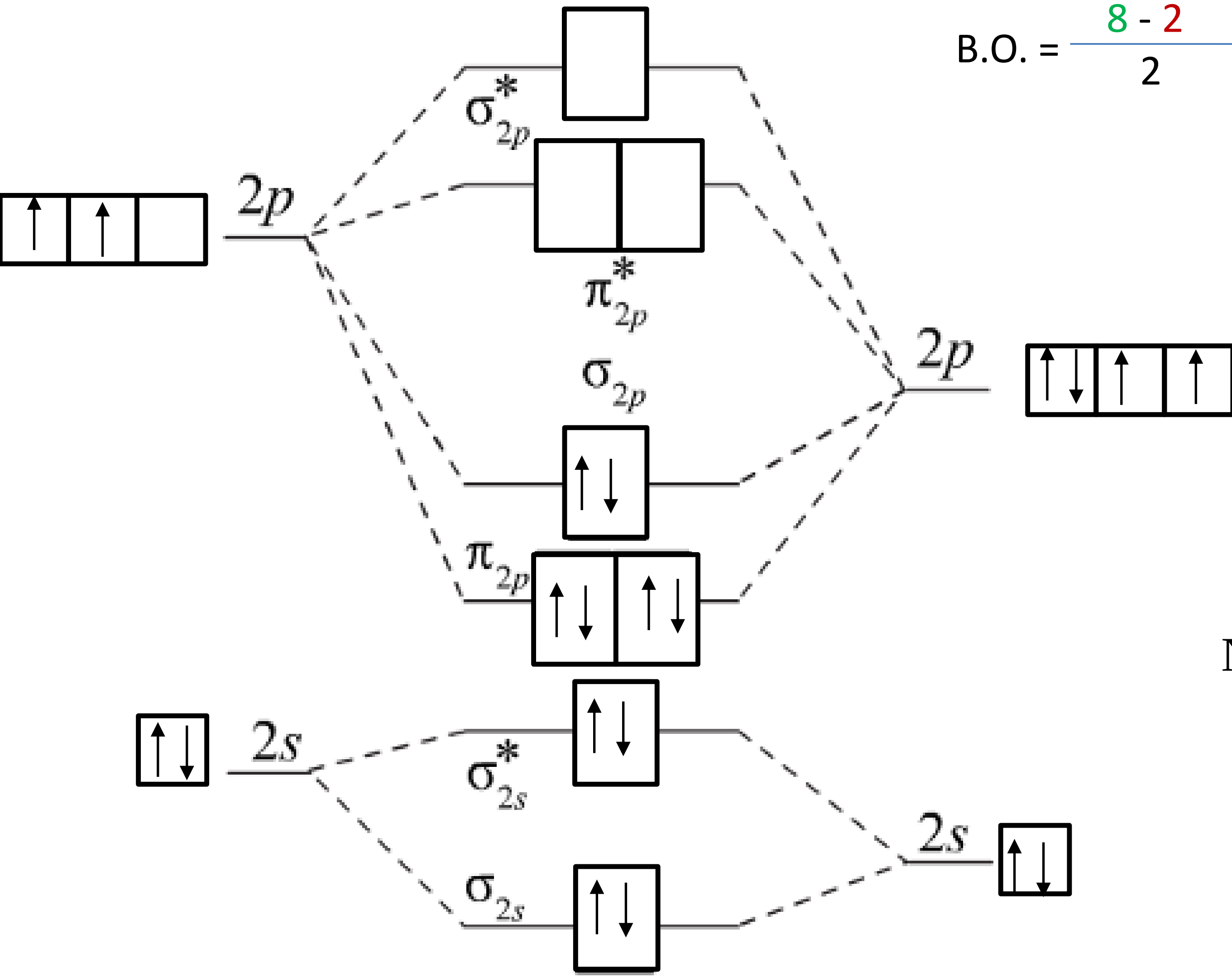
$NO + CN \rightarrow NO^{-} + CN^{+}$

$$\text{B.O.} = \frac{8 - 2}{2} = 3$$ i.e. tripple bond



$NO + CN \rightarrow NO^+ + CN^-$

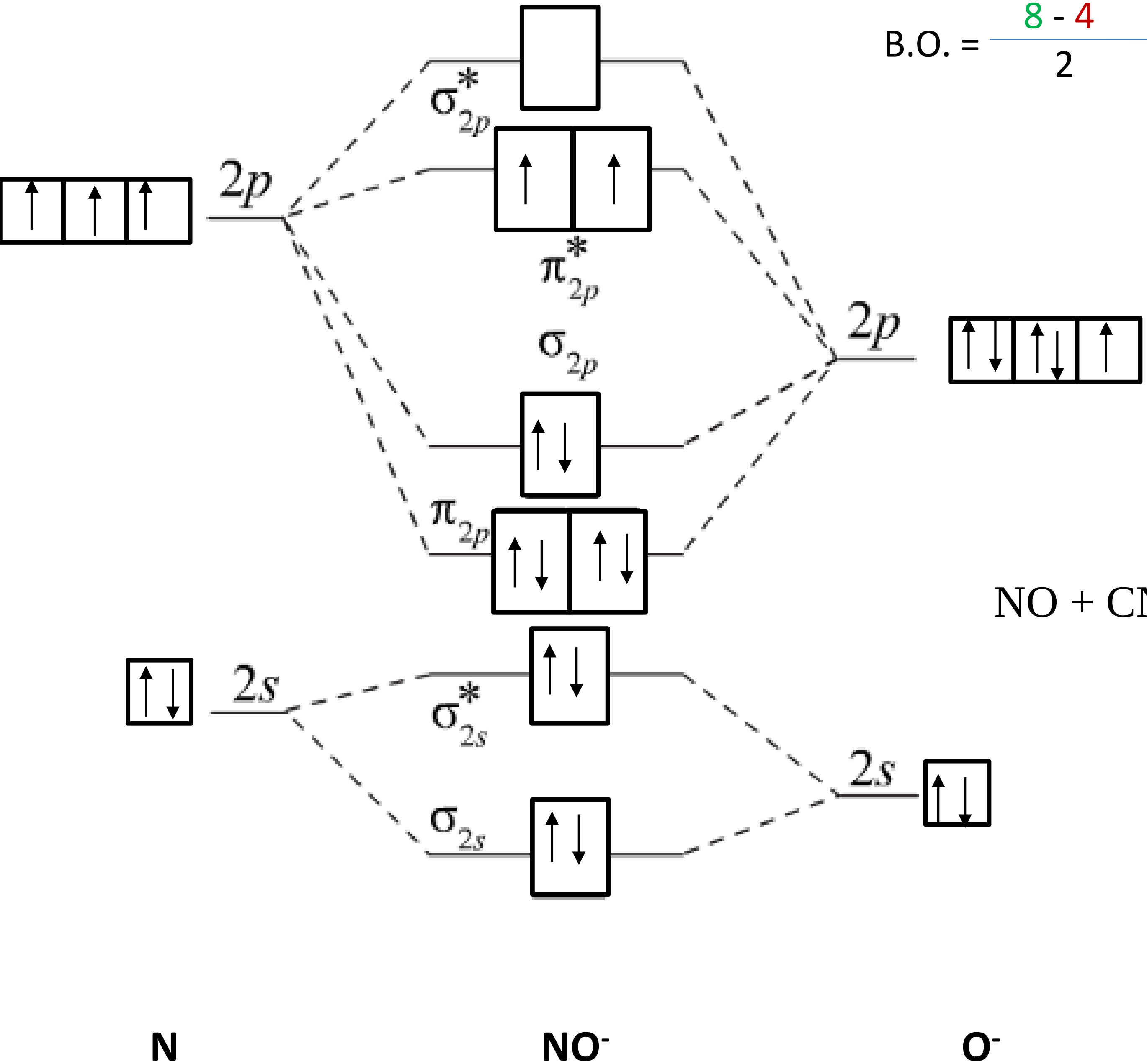
$$\text{B.O.} = \frac{8 - 2}{2} = 3 \quad \text{i.e. tripple bond}$$



$NO + CN \rightarrow NO^+ + CN^-$

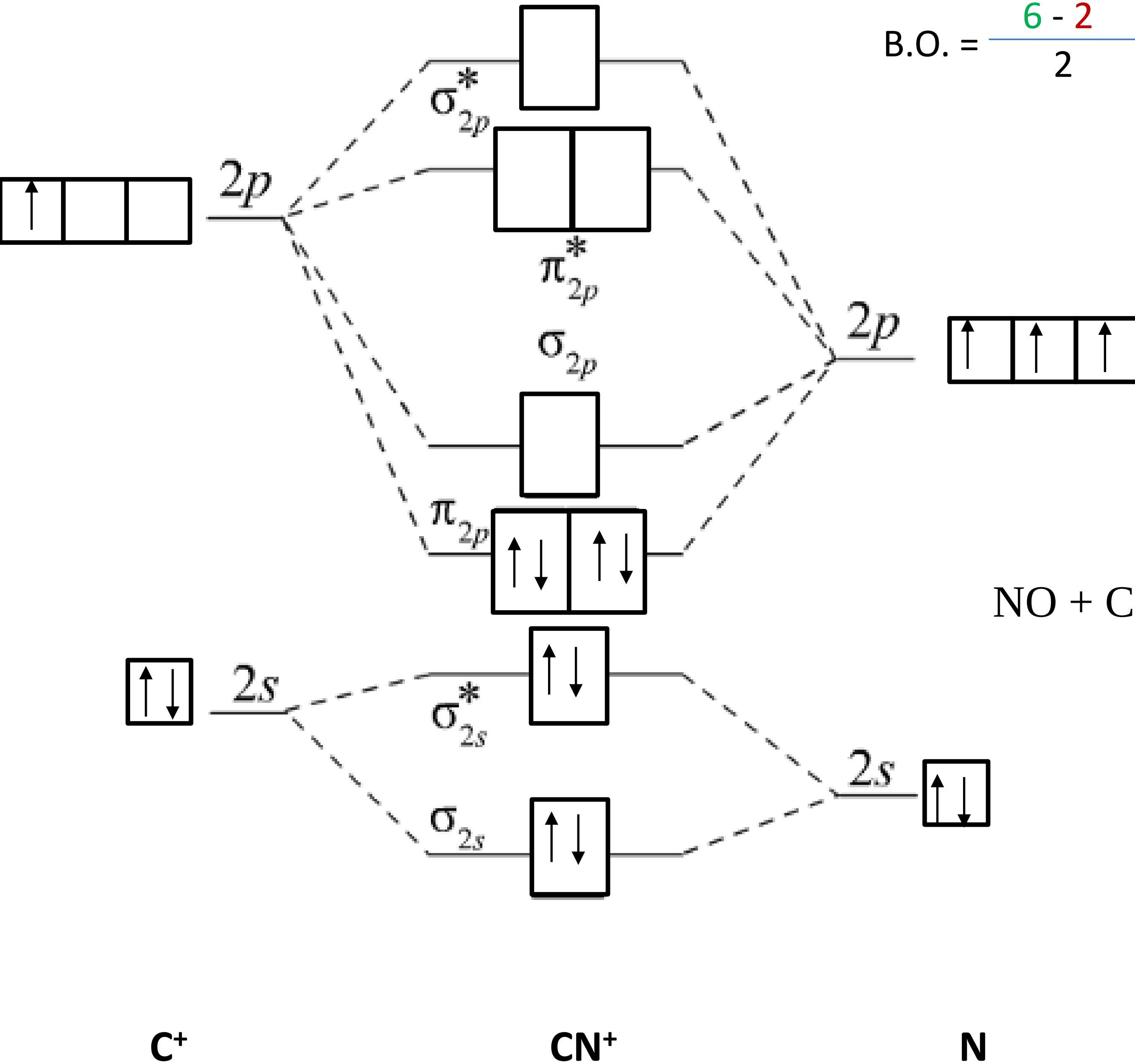
C **CN^-** **N^-**

$$\text{B.O.} = \frac{8 - 4}{2} = 2 \quad \text{i.e. double bond}$$



$NO + CN \rightarrow NO^- + CN^+$

$$B.O. = \frac{6 - 2}{2} = 2 \quad \text{i.e. double bond}$$



$$NO + CN \rightarrow NO^{-} + CN^{+}$$

Exercise 5

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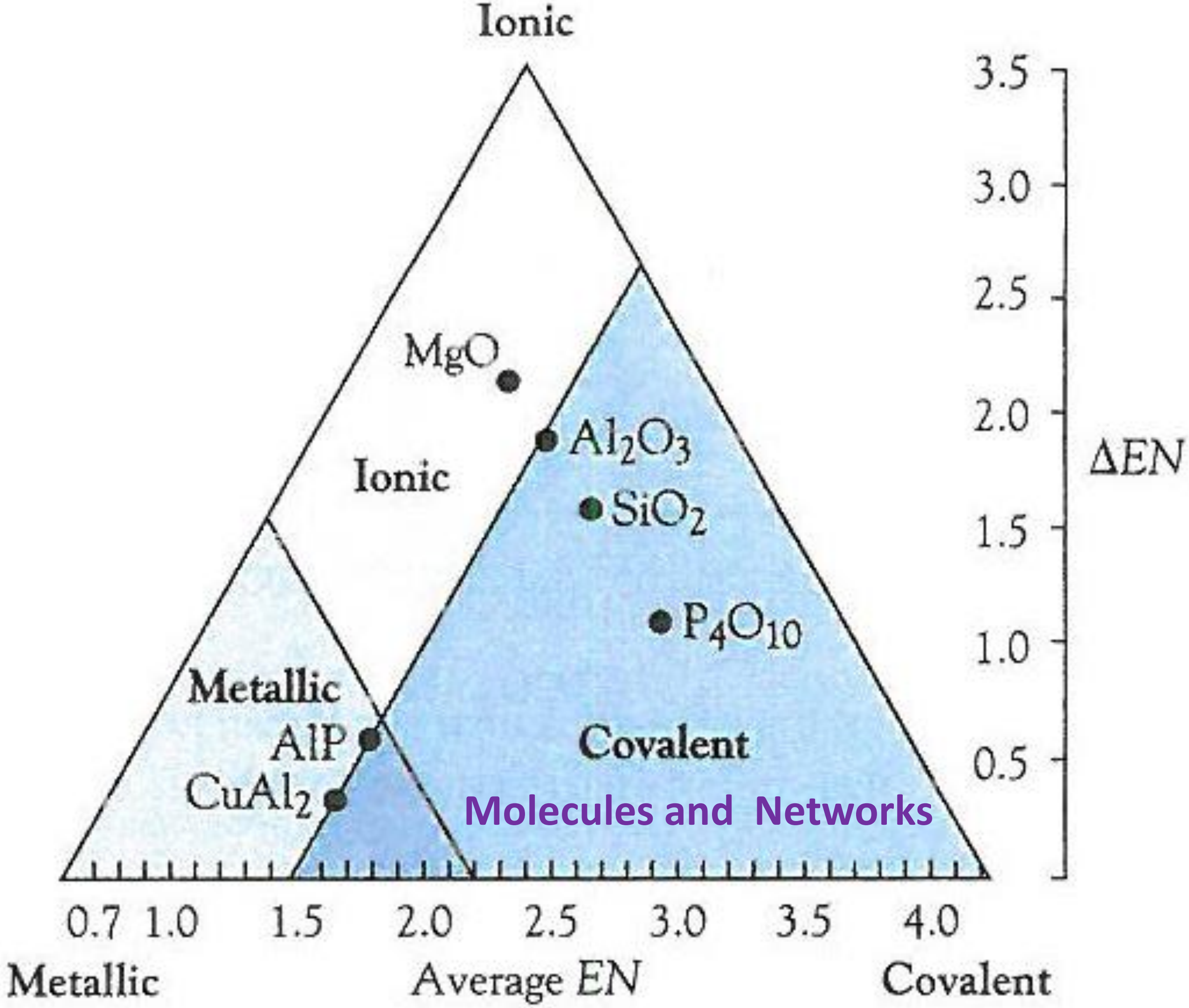
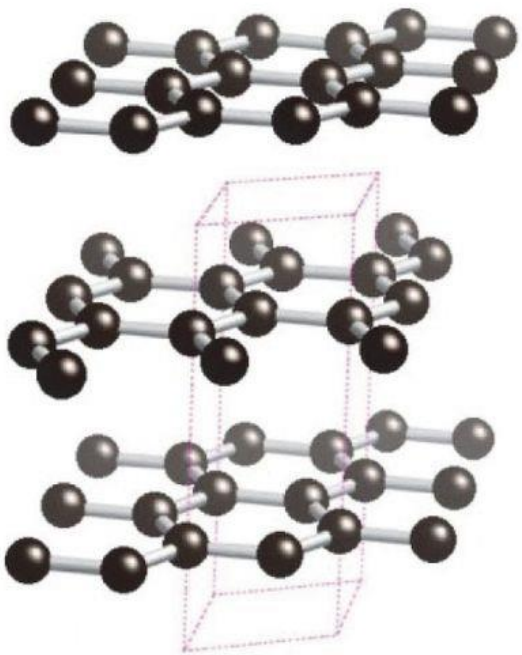
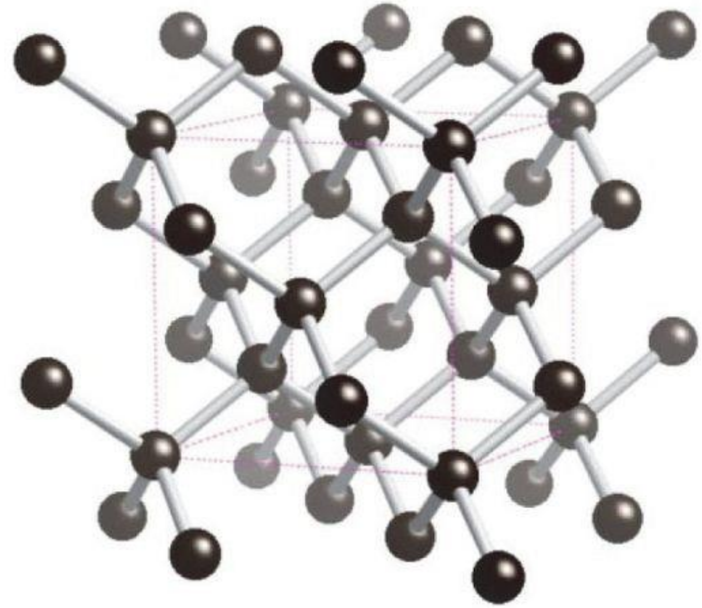

Ionic
3.5
3.0
2.5
2.0
1.5
1.0
0.5
ΔEN
MgO
Al2O3
Ionic
SiO2
P4O10
Metallic
AlP
CuAl2
Covalent
Molecules and Networks
0.7 1.0 1.5 2.0 2.5 3.0 3.5 4.0
Metallic
Average EN
Covalent

Covalent networks

Carbon

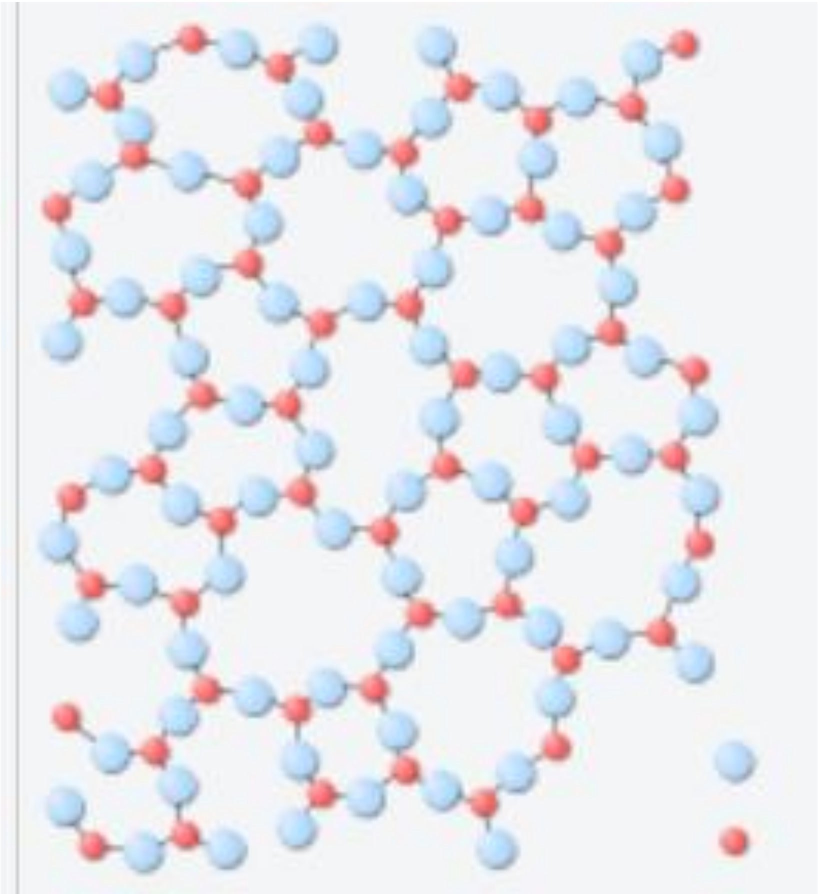


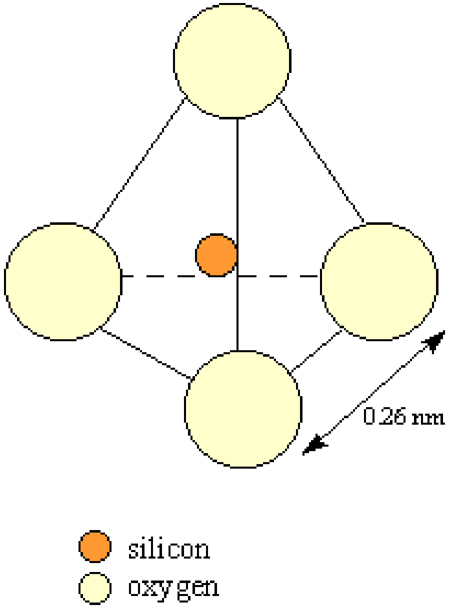
graphite

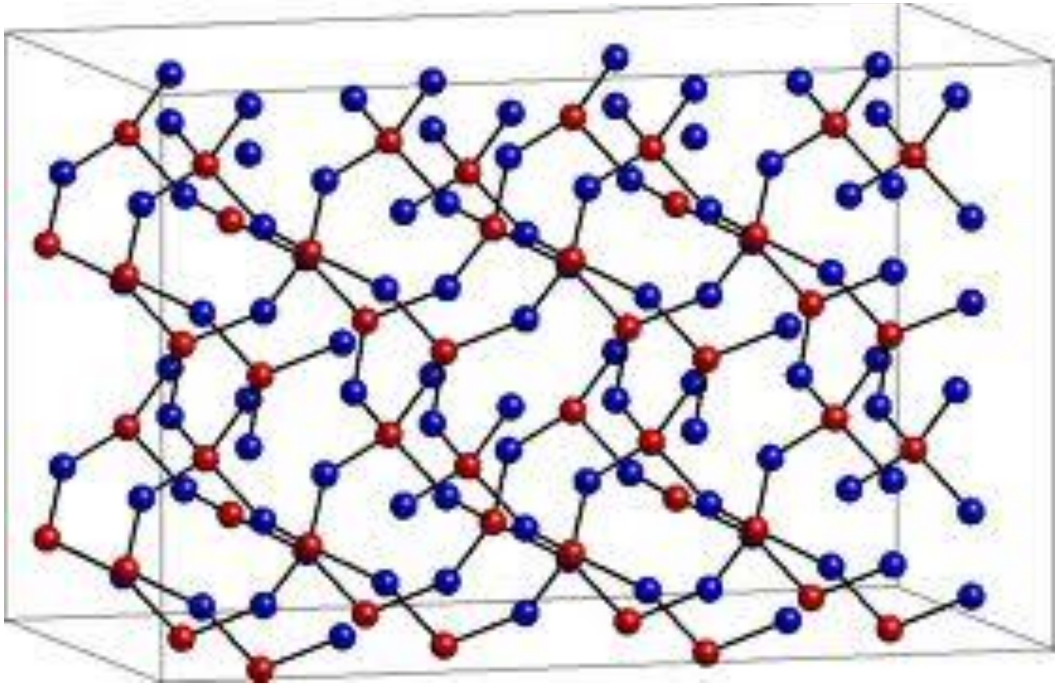


diamond

Silica (SiO_2)



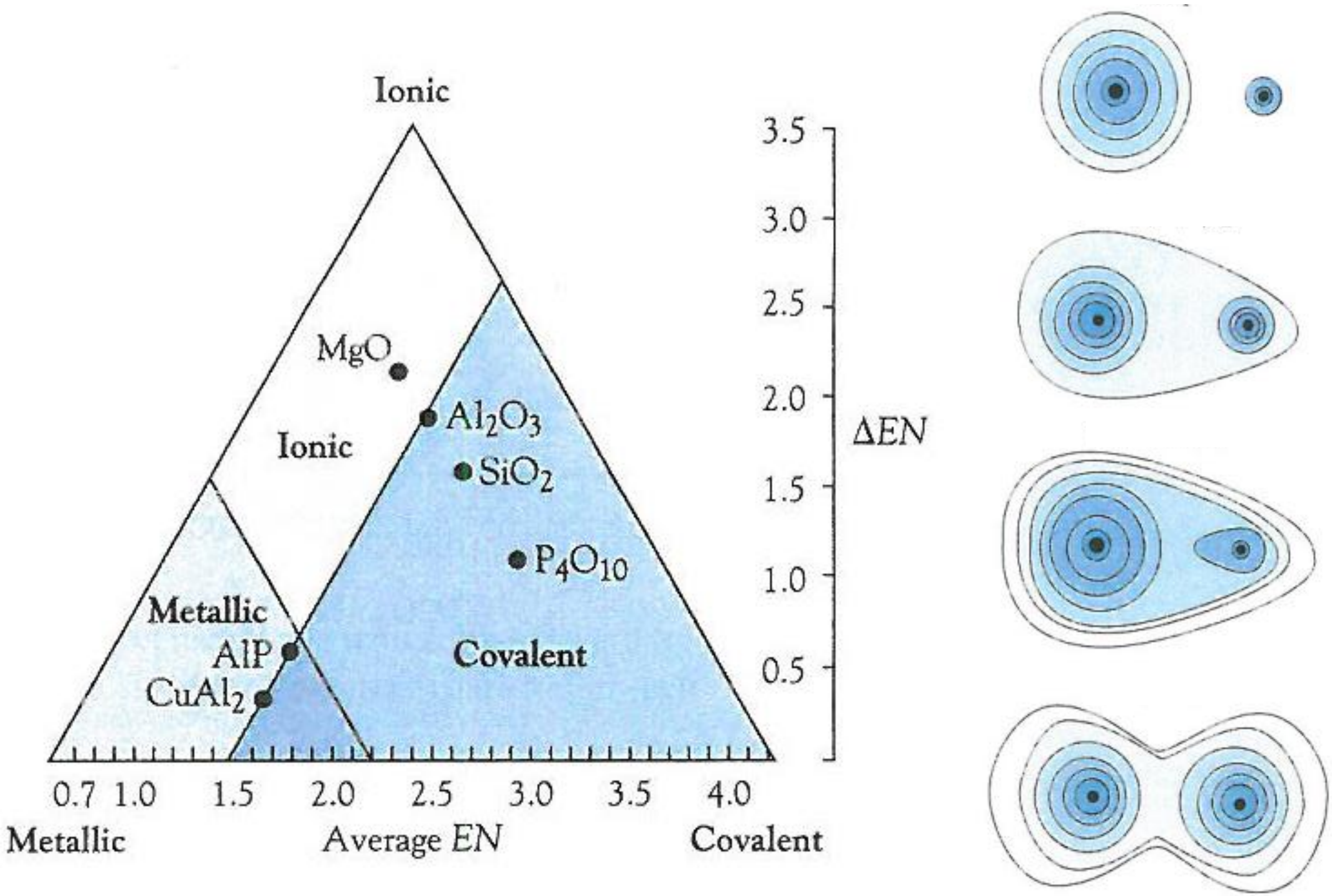




Lecture 3

Metallic bonding

Chapter 4



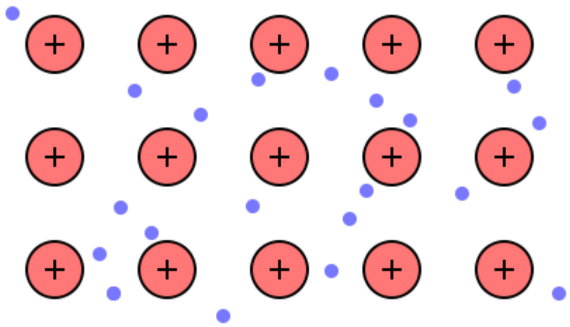
Ionic
MgO
Al_2O_3
Ionic
SiO_2
P_4O_{10}
Metallic
AlP
$CuAl_2$
Covalent
0.7 1.0 1.5 2.0 2.5 3.0 3.5 4.0
Metallic
Average EN
Covalent
3.5
3.0
2.5
2.0
1.5
1.0
0.5
ΔEN

Metallic bond

What are the properties of metals?

Bonding in metals

Electron sea



Band theory

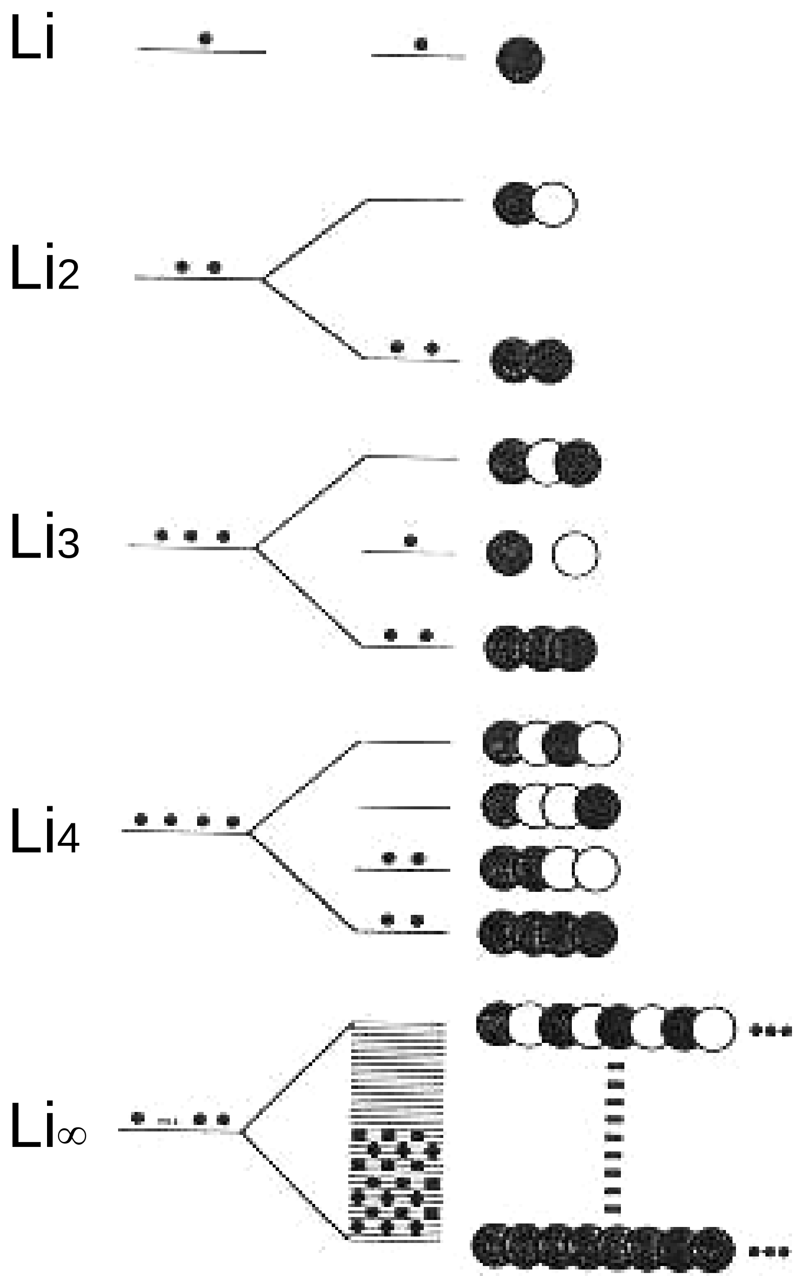
This theory can give an explanation of the characteristic properties of metals.

1) The orbitals of neighbouring atoms overlap and interact with each other.

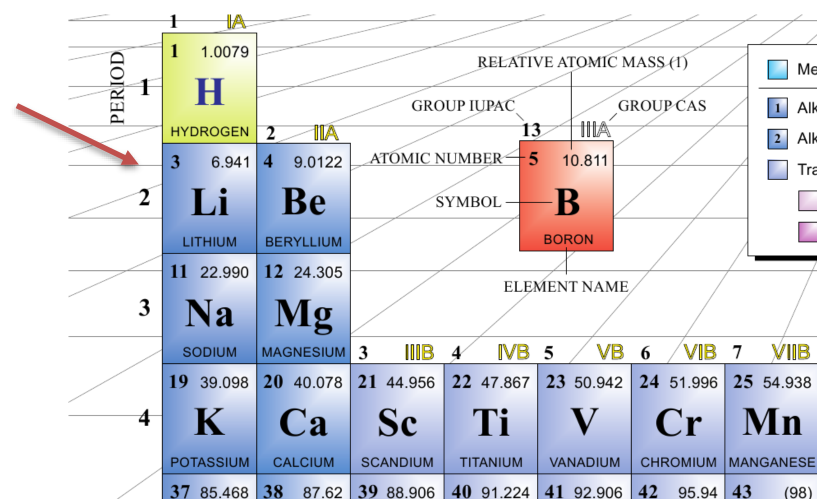
2) The interaction forms continuous energy bands.

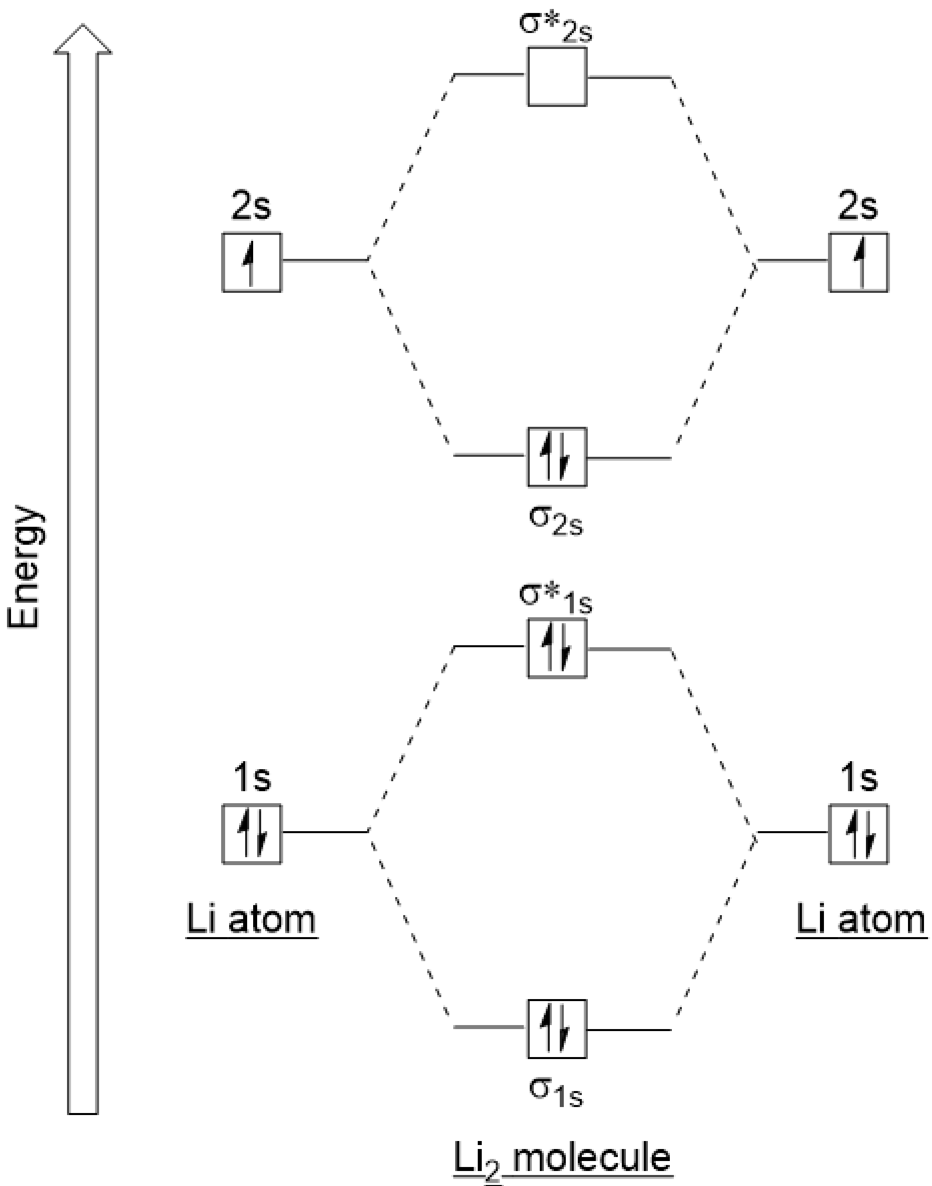
3) The electrons in these bands are completely delocalised.

Band theory



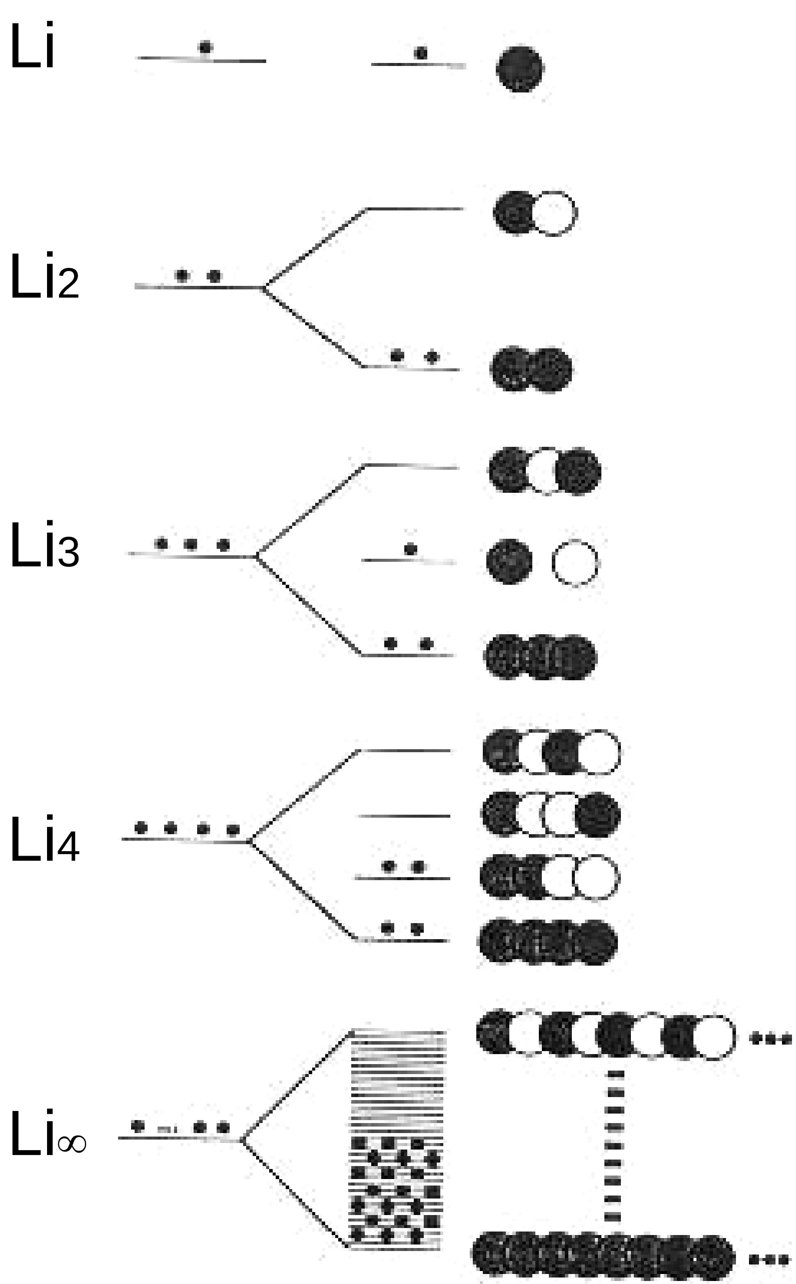
When a very large number of Li atoms are combined, a “*2s* band" is formed.





σ*2s
2s
2s
σ2s
Energy
σ*1s
1s
1s
Li atom
Li atom
σ1s
Li2 molecule

Band theory



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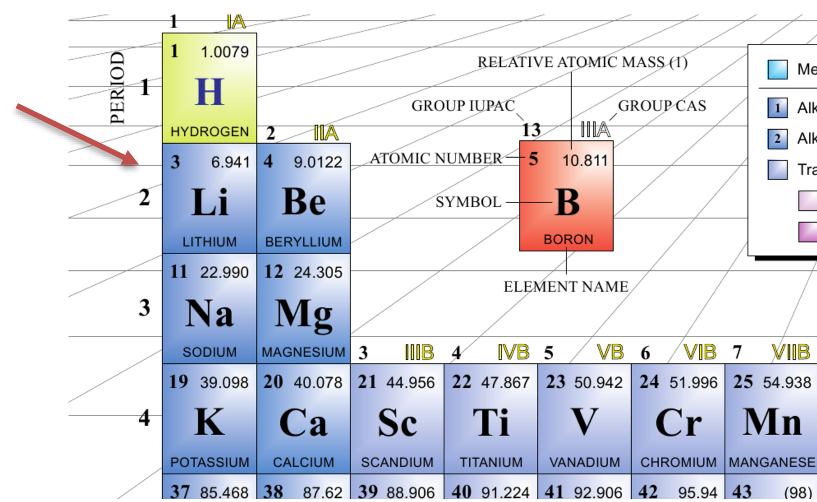
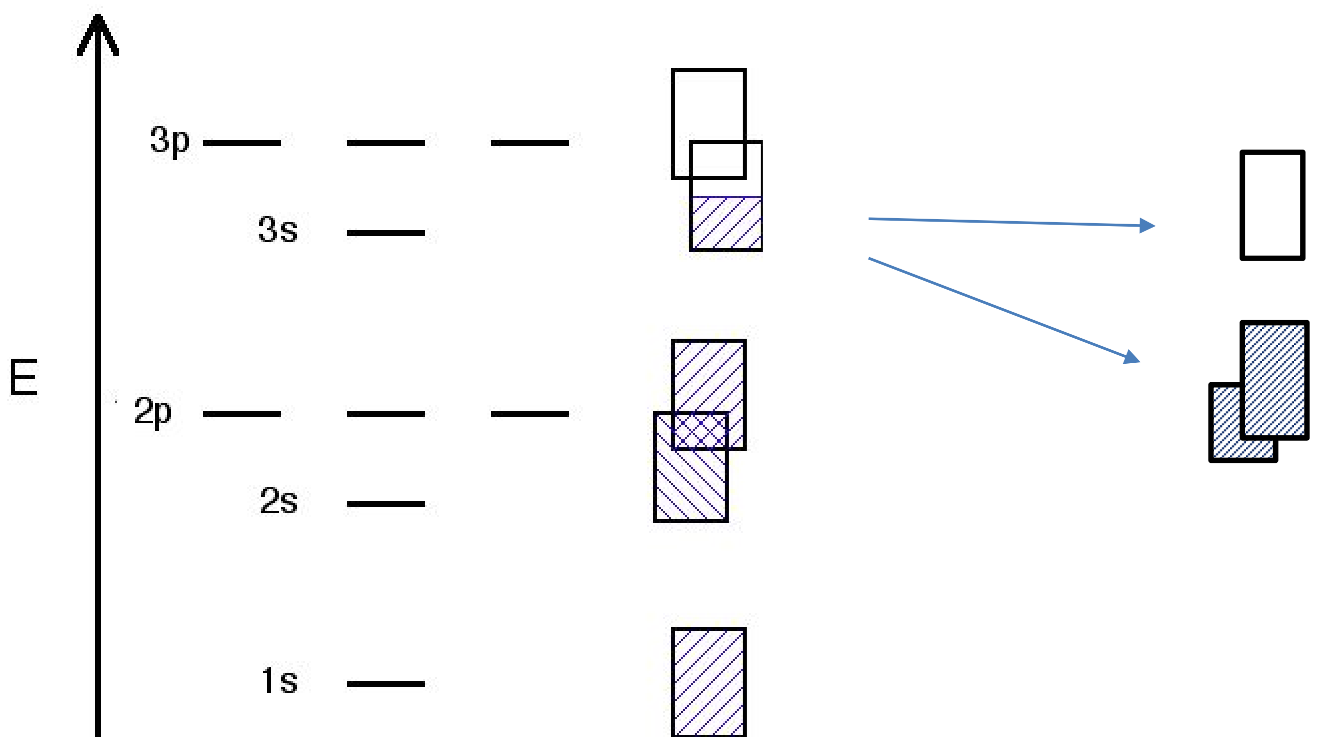


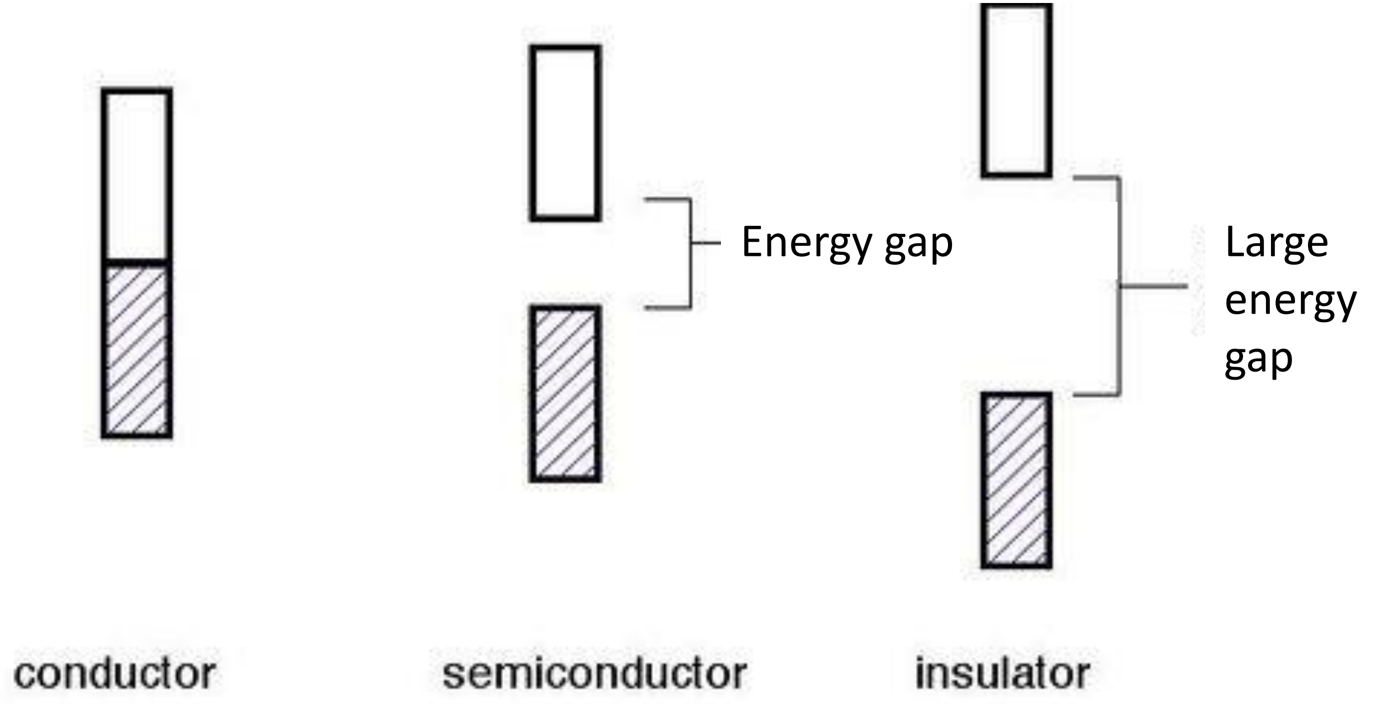
Metal

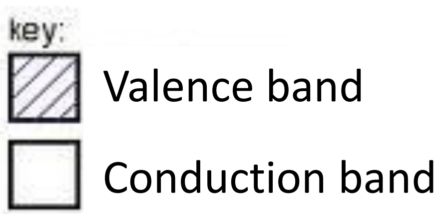
Semiconductor

example: Na

example: Si

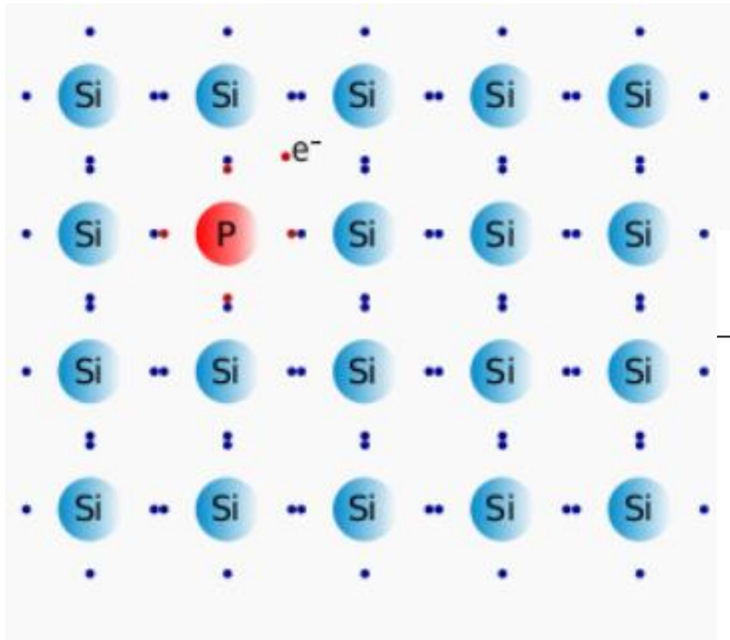


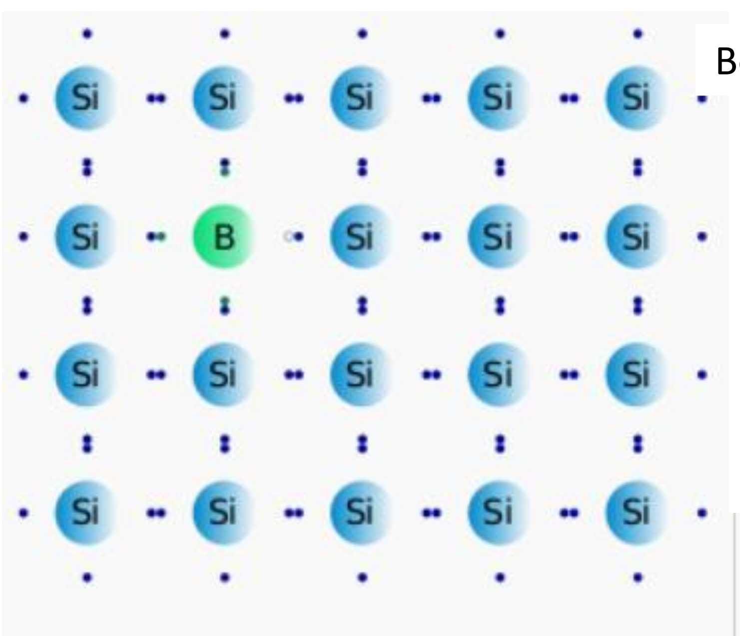


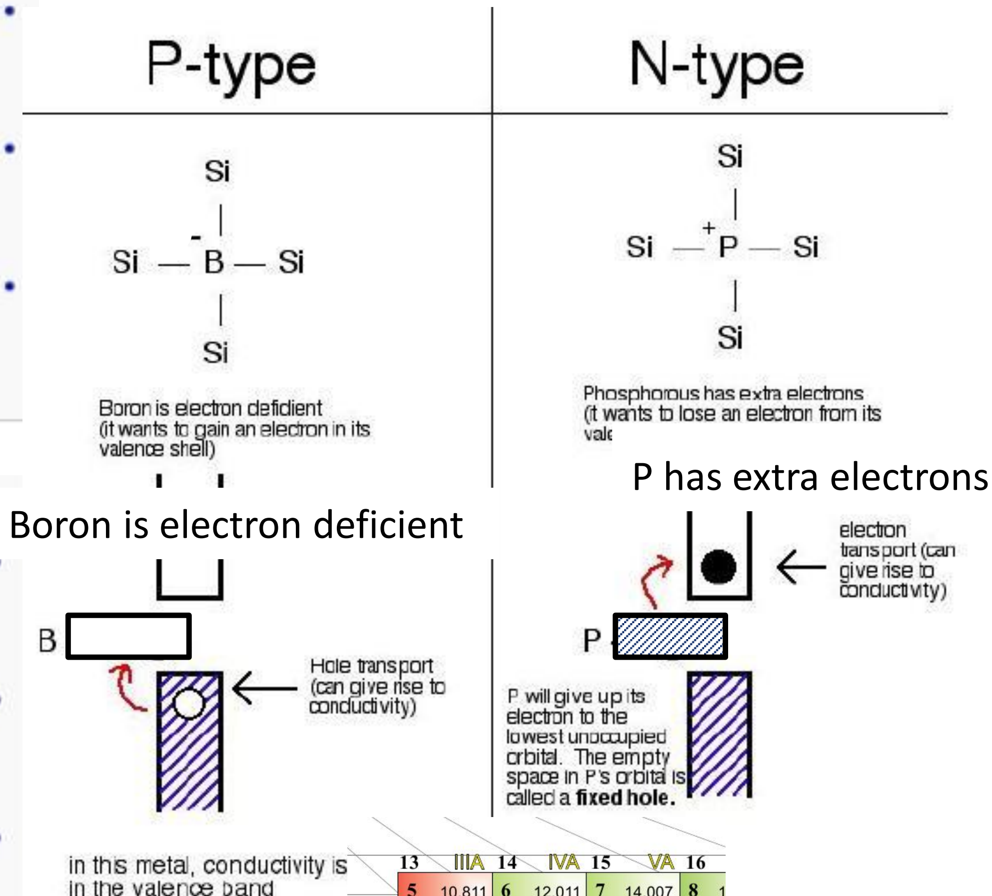


ELEMENT	ΔE(kJ/mol) of energy gap	# of electrons/cm^3 in conduction band	insulator, or conductor?
C (diamond)	524 (big band gap)	10^{-27}	insulator
Si	117 (smaller band gap, but not a full conductor)	10^{9}	semiconductor
Ge	66 (smaller band gap, but still not a full conductor)	10^{13}	semiconductor

Impurity semiconductors

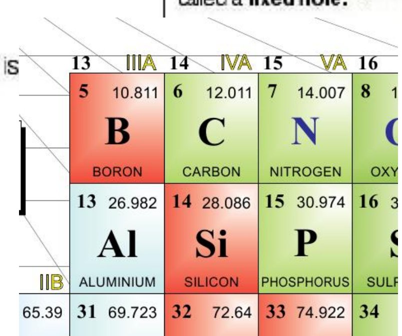




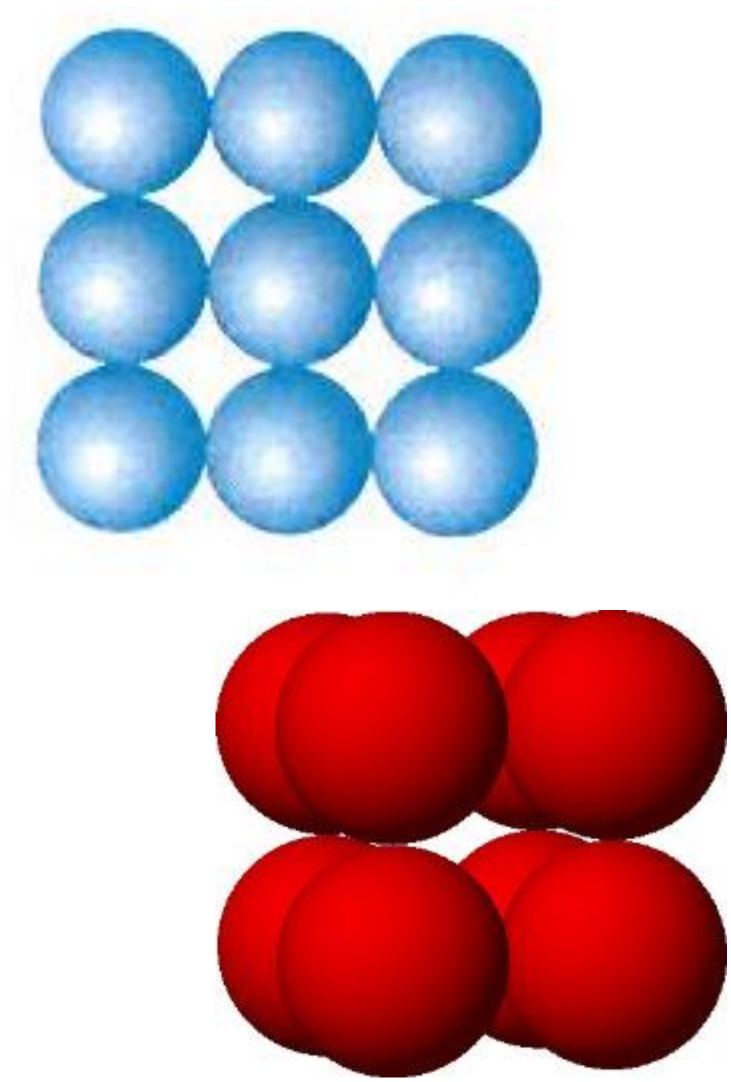


Boron is electron deficient

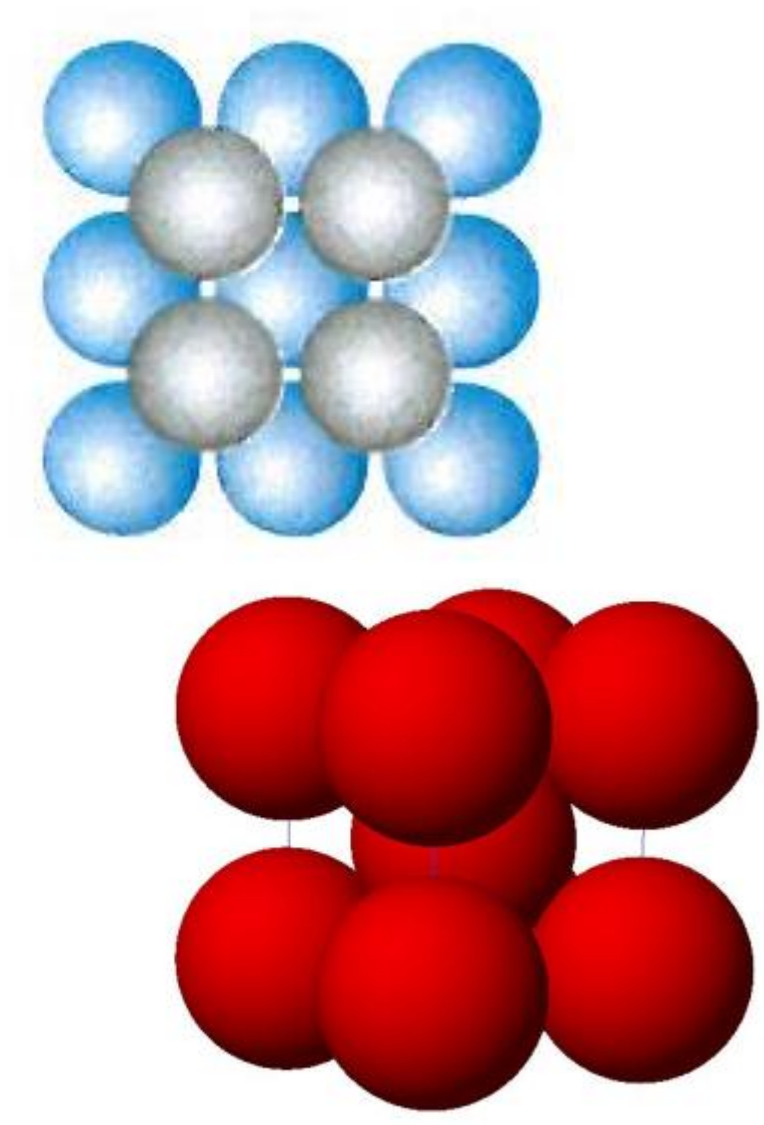
P has extra electrons



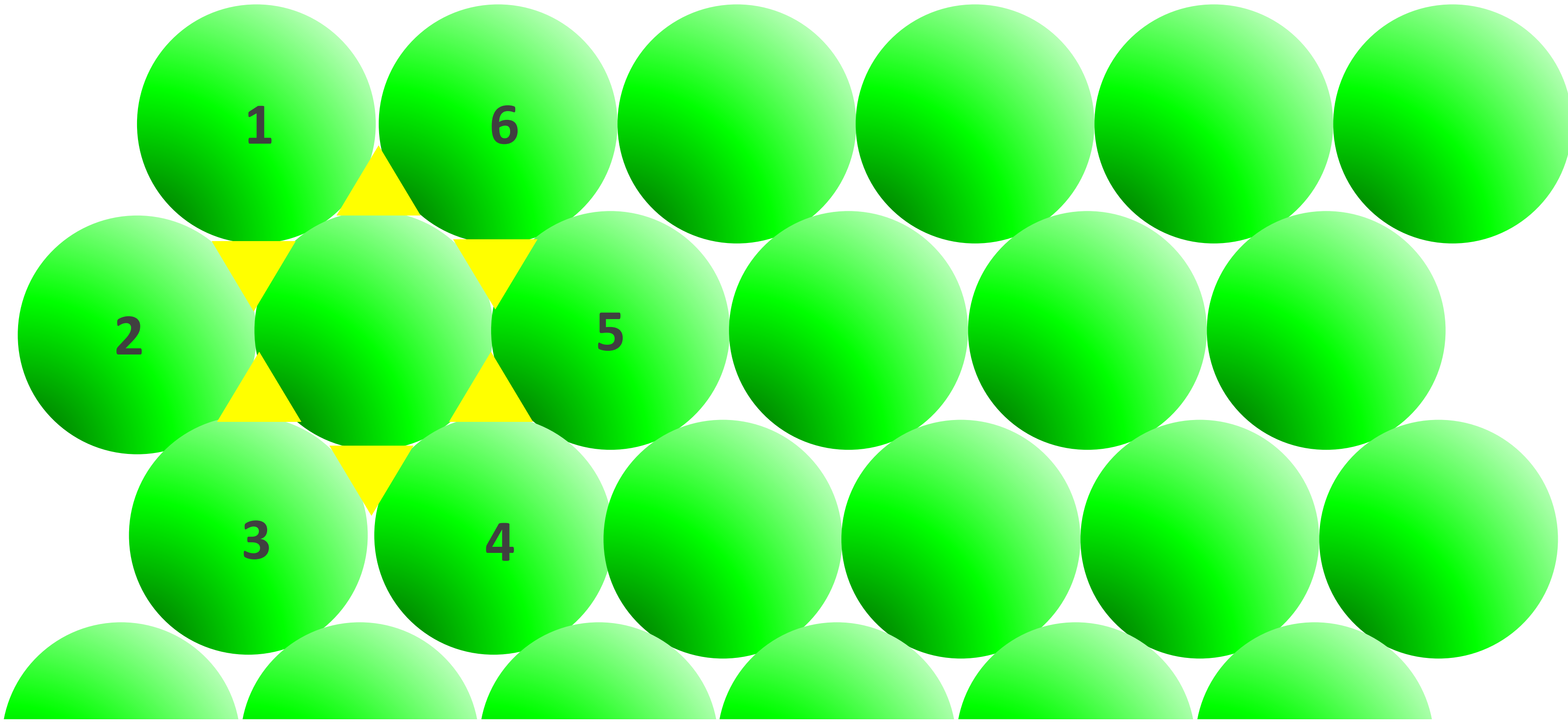
(1) Simple cubic packing



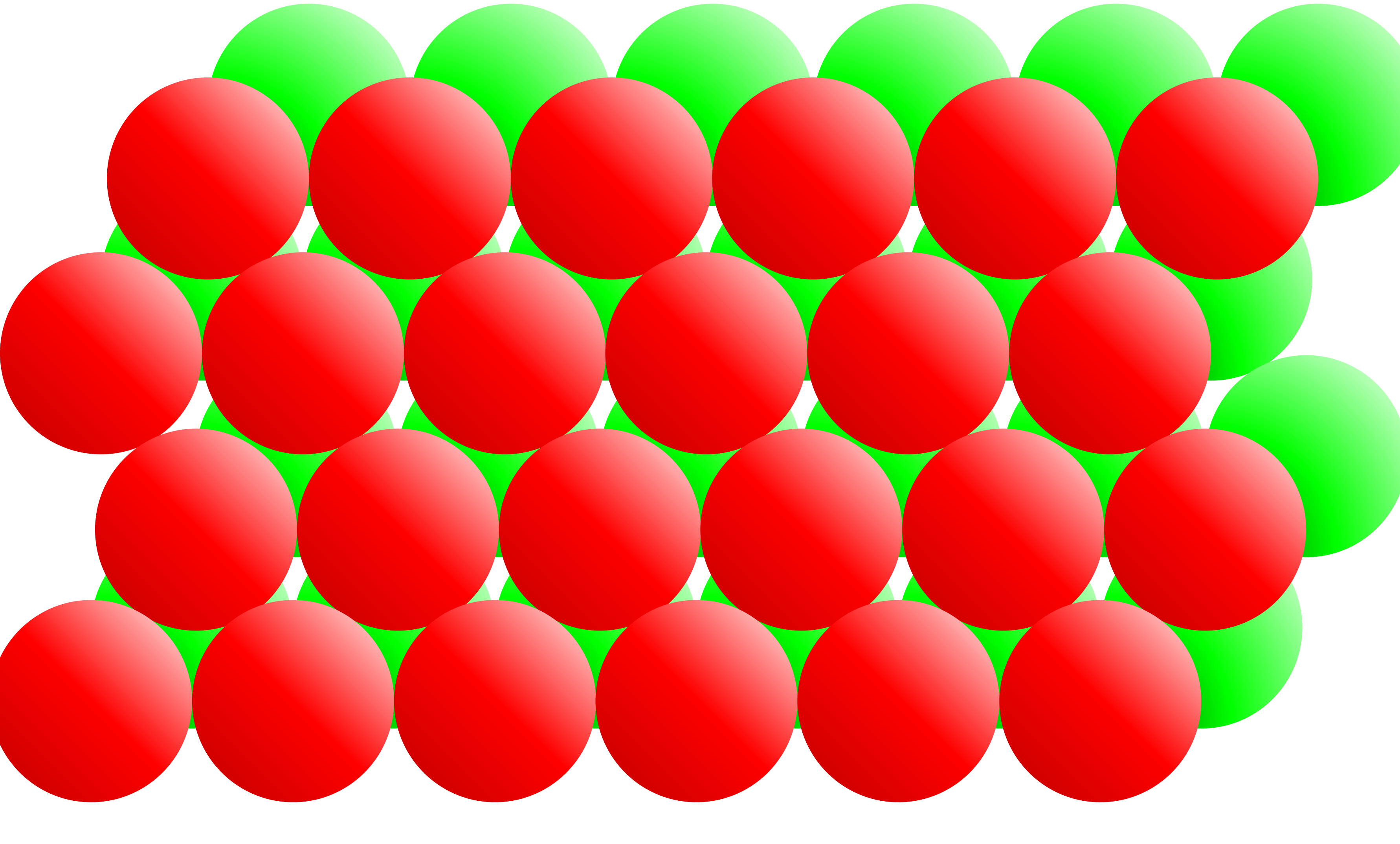
(2) Body-centred cubic packing



(3-4) Close-packed Structure

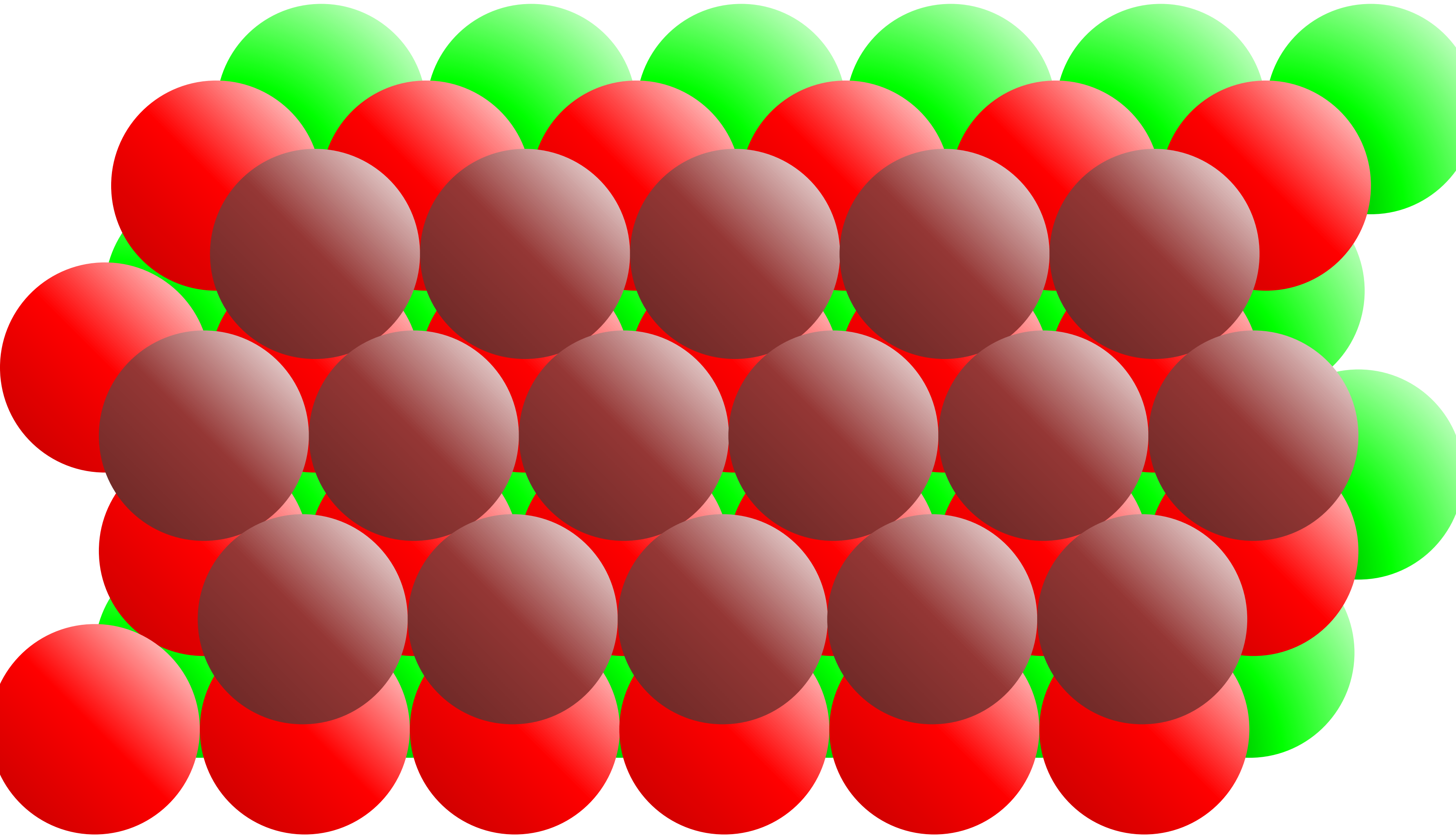


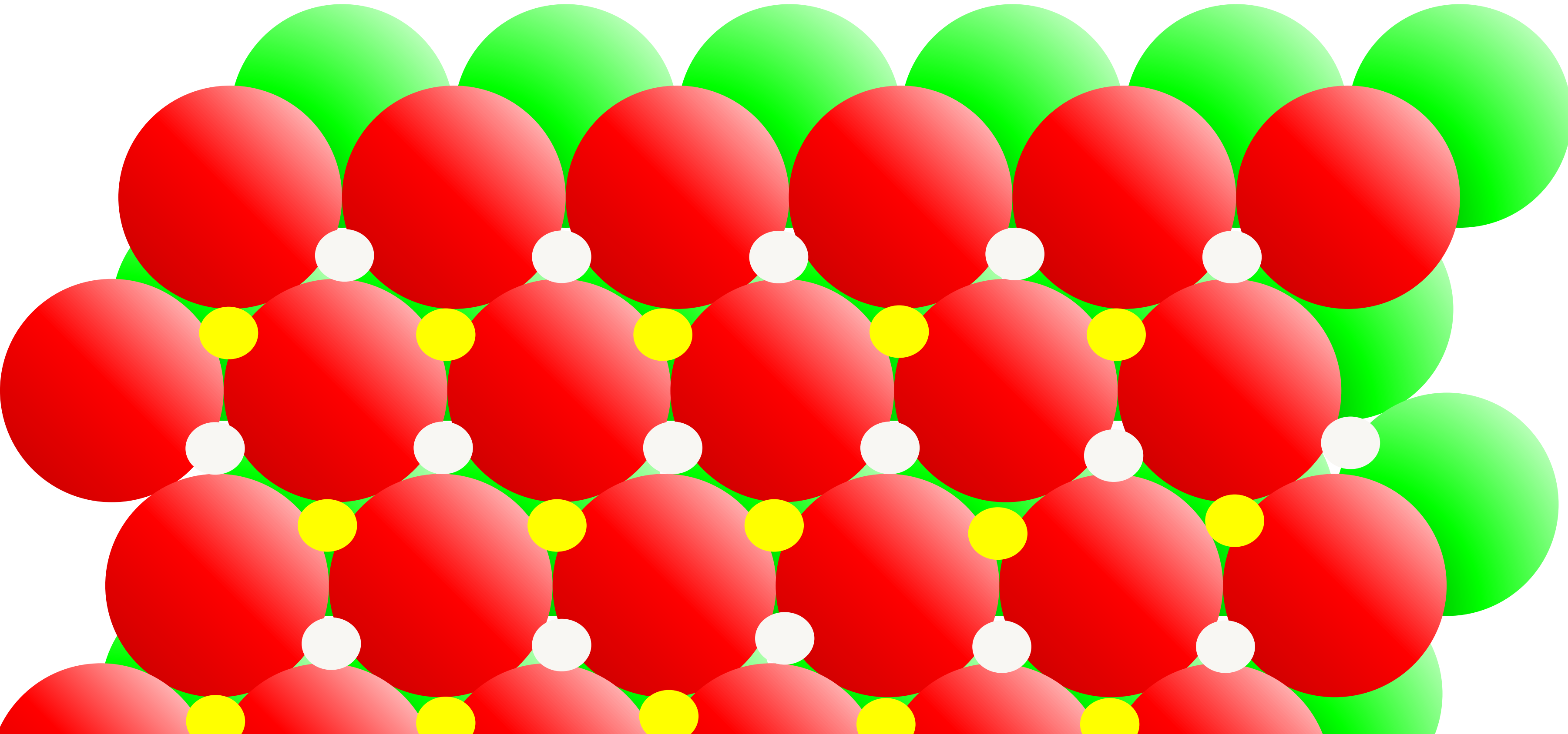
Close packing of spheres in a layer is accomplished when:



ABCABC type

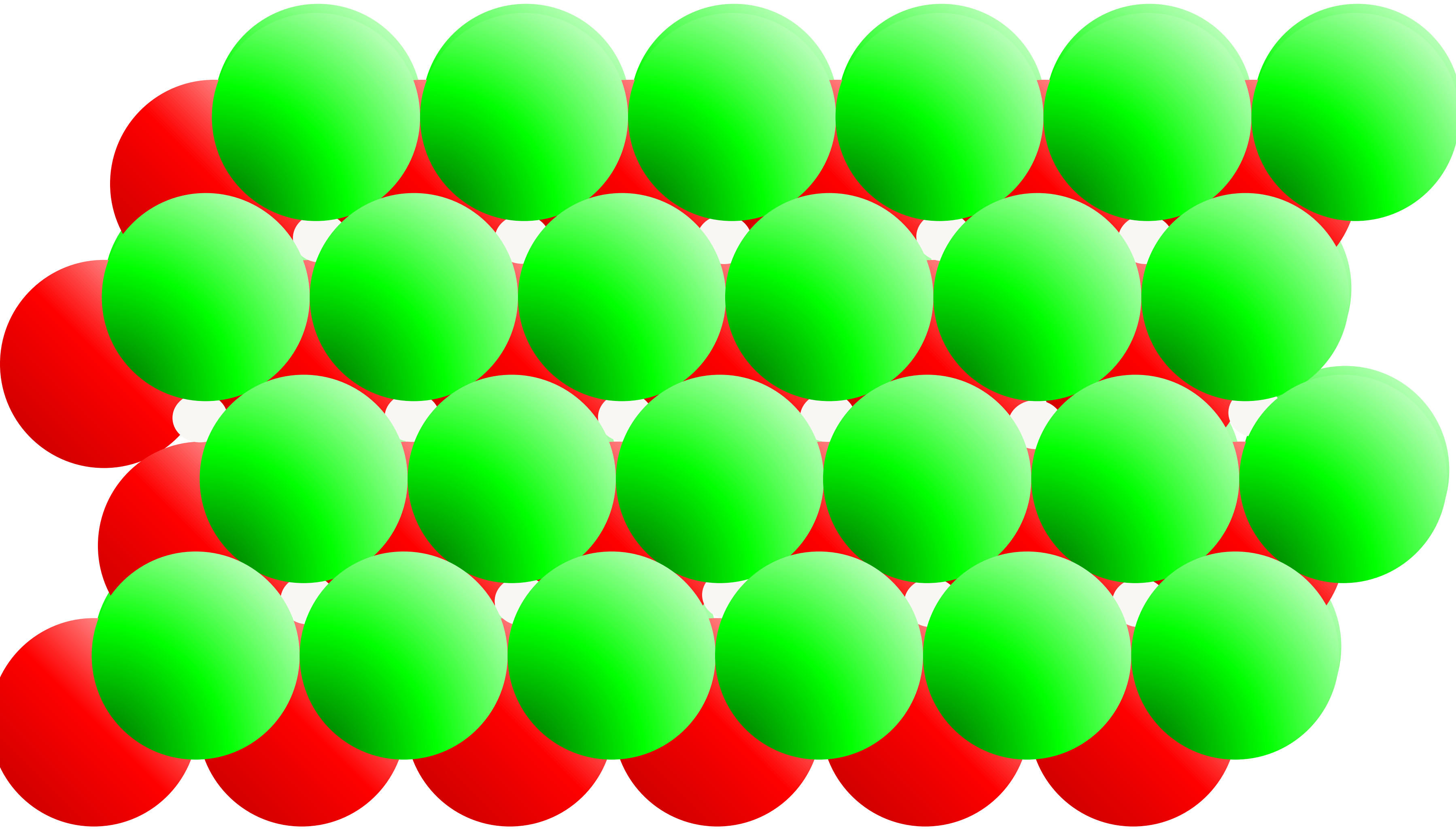
(3) Cubic closest packing (ccp)

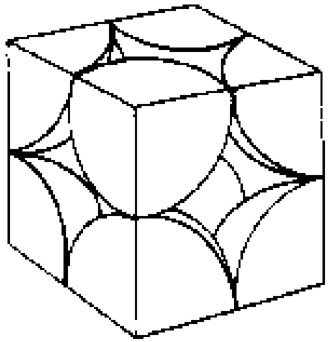
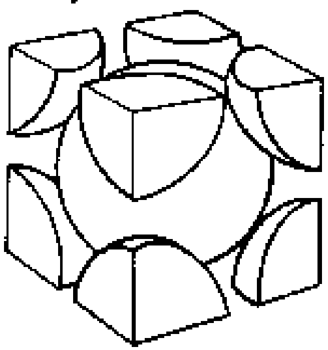




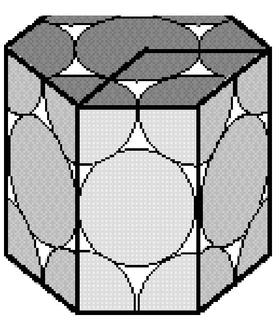
ABABAB type

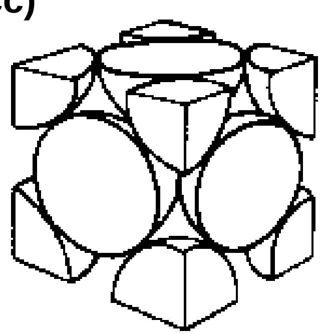
(4) hexagonal closest packing (hcp)

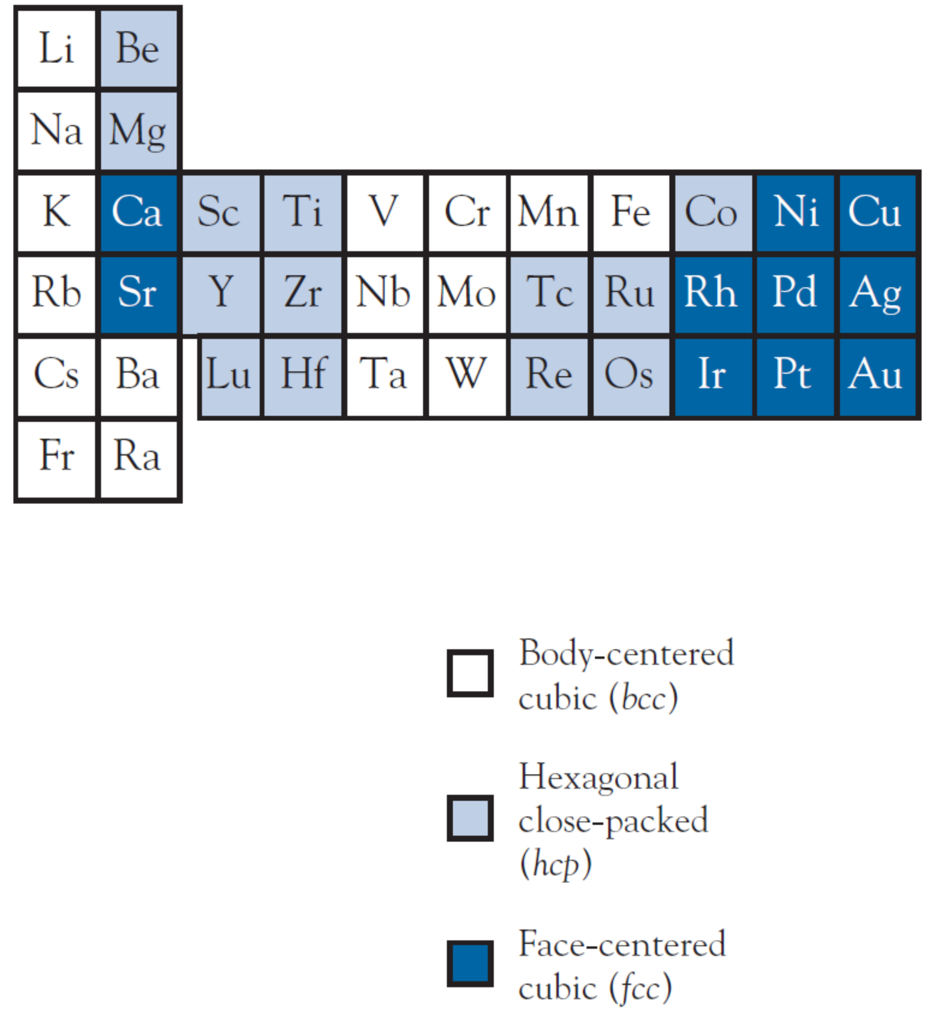


Type of Packing	Packing Efficiency	Coordination Number
Simple cubic (sc) 	52%	6
Body-centered cubic (bcc) 	68%	8
Hexagonal close-packed (hcp)	74%	12
Cubic close-packed or	74%	12

Face centered cubic (ccp or fcc)







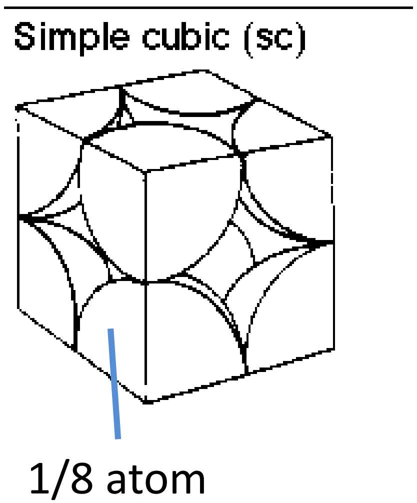
Exercise 1

336 pm = 3.36 Å = 0.336 nm = 3.36 10^{-10} m

The edge length of the unit cell of alpha polonium is 336 pm (picometers).

(a) Determine the radius of a polonium (Po) atom (atomic mass = 209 g/mol).

(b) Determine the density of alpha polonium. Alpha polonium crystallizes in a simple cubic unit cell:



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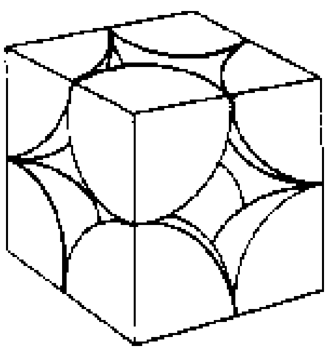
.

Simple cubic (sc)



(a) Two adjacent Po atoms contact each other, so the edge length of this cell is equal to two Po atomic radii: $l = 2r$. Therefore, the radius of Po is 336/2 = 168 pm.

(b) Density is mass/volume.

The density of polonium can be found by determining the density of its unit cell (the mass contained within a unit cell divided by the volume of the unit cell). Since a Po unit cell contains one-eighth of a Po atom at each of its eight corners, a unit cell contains one Po atom.

The mass of a Po unit cell can be found by:

209(g/mol) / $6.022\ 10^{23}$ (atoms/mol) = $3.47\ 10^{-22}$ g/atom = $3.47\ 10^{-22}$ g/unit cell

The volume of a Po unit cell is $(336)^3 = 3.79\ 10^{7}$ pm^3 = $3.79\ 10^{-23}$ cm^3

(Note that the edge length was converted from pm to cm to get the usual volume units for density: g/cm^3).

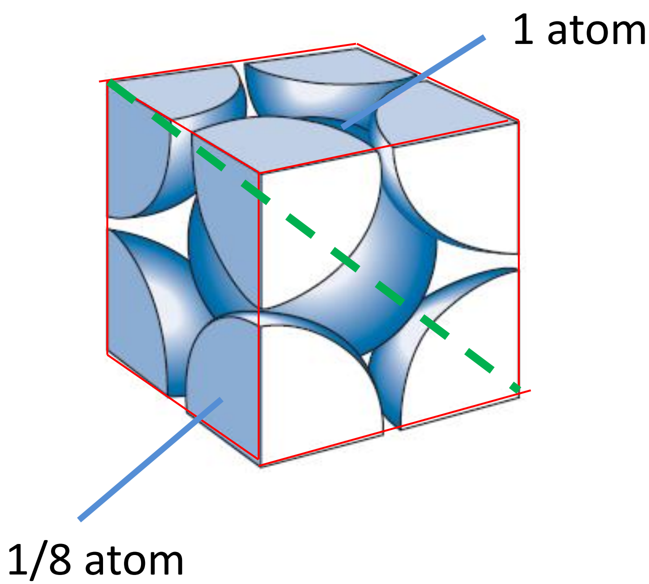
Therefore, the density of Polonium is $3.47\ 10^{-22}$ g / $3.79\ 10^{-23}$ cm^3 =

9.16 g/cm^3

Exercise 2

10 minutes

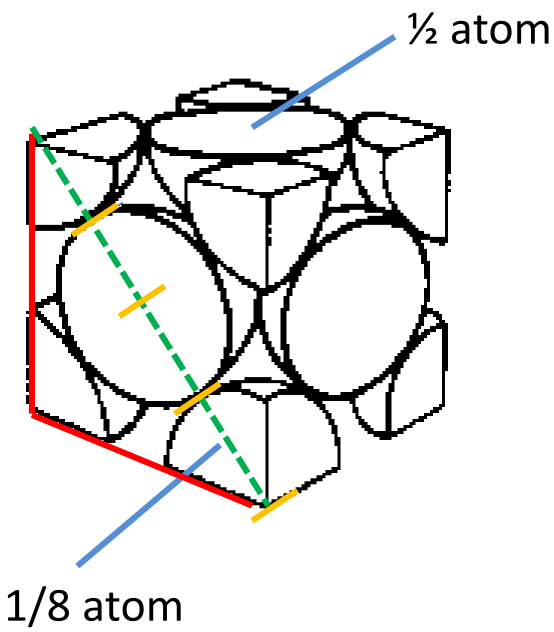
Chromium forms a body-centered cubic lattice in which the edge length of the unit cell is 288 pm. Calculate (a) the metallic radius of a chromium atom and (b) the density of chromium metal.



Exercise 3

10 minutes

The atoms in silver metal are arranged in a face centered cubic unit cell. Calculate the radius of a silver atom if the density of silver is 10.50 g cm^{-3}.



Alloys

Alloy: a combination of two or more solid metals.

Similarly, the metallic bonds of an alloy hold together atoms of different metallic elements.

There are two types of alloys:

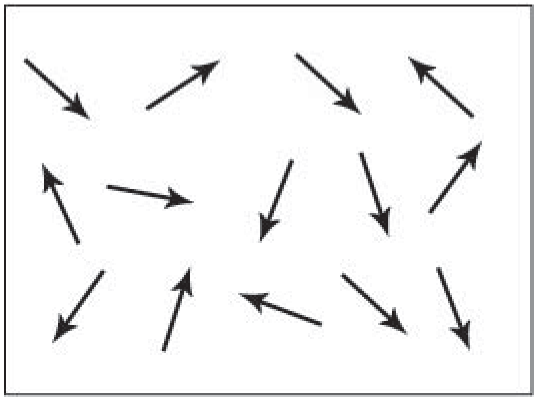
- Solid solutions : metals blend to form a homogeneous mixture.
  requirements: (a) metal atoms should have similar size,
  (b) metals must have the same structure.

Name	Composition (%)	Properties	Uses
Brass	Cu 70–85, Zn 15–30	Harder than pure Cu	Plumbing
Gold, 18-carat	Au 75, Ag 10–20, Cu 5–15	Harder than pure (24-carat) gold	Jewelry
Stainless steel	Fe 65–85, Cr 12–20, Ni 2–15, Mn 1–2, C 0.1–1, Si 0.5–1	Corrosion resistance	Tools, chemical equipment

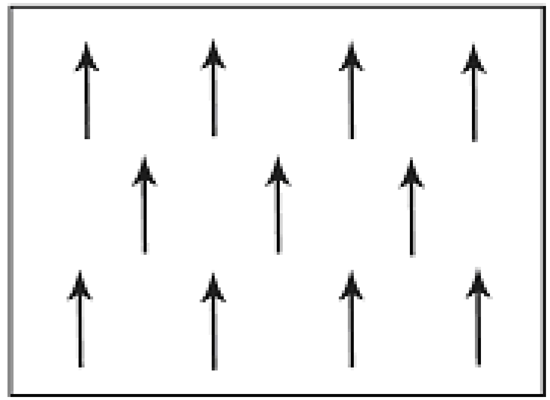
- Intermetallic compounds: formation of precise stoichiometric phases. For example, copper and zinc form three “compounds”: $CuZn$, Cu_5Zn_8, and $CuZn_3$

Magnetic properties of metals

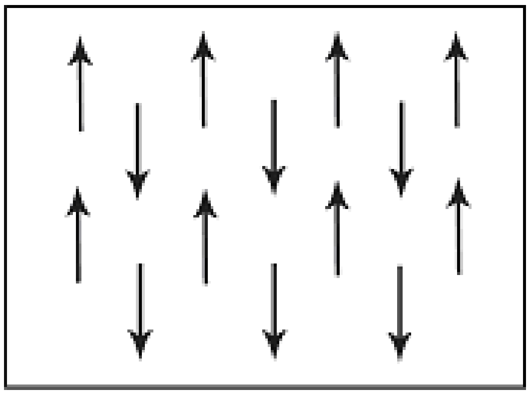
Paramagnetic

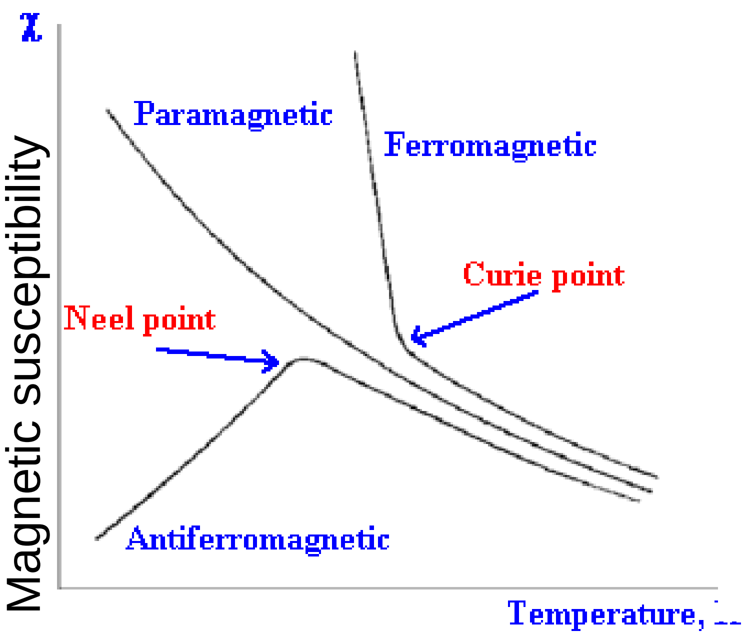


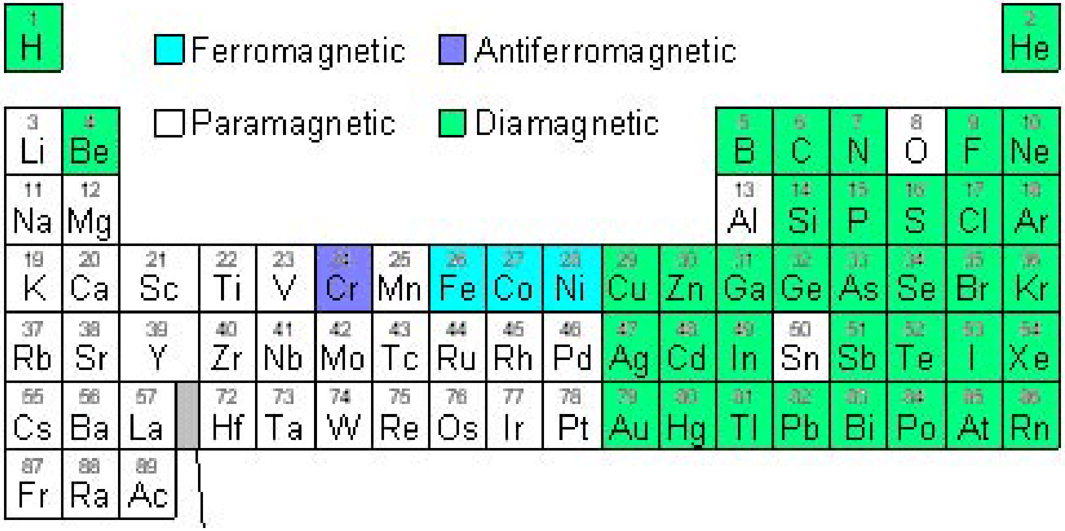
Ferromagnetic



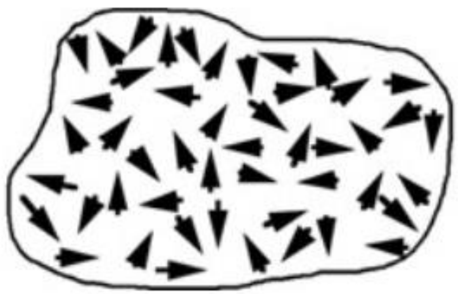
Antiferromagnetic



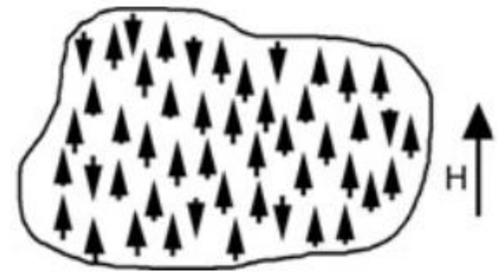




Paramagnetism

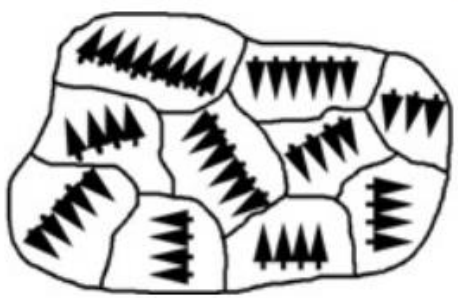


Spins are randomized by thermal energy.

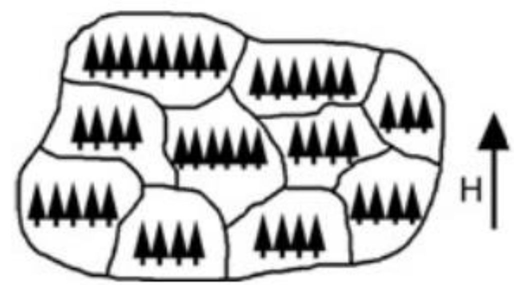


Spins are aligned with or against an applied magnetic field.

Ferromagnetism



Spins are ordered in magnetic domains.



Spins are aligned with an applied magnetic field.

Content:

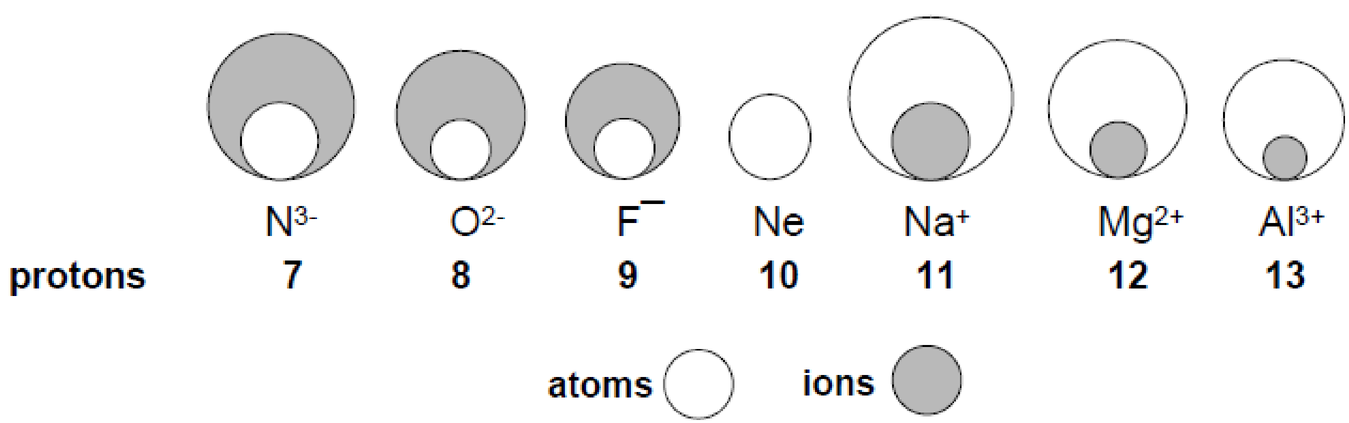
- **Ionic radius;**
- **Ionic and polar bonds;**
- **Interstices and ionic structures;**
- **Exceptions to the packing rules;**
- **Complex ionic solids;**
- **Defects.**

Ionic Crystals

Ionic substances have the following properties:

1) Hard and brittle;
2) High melting points;
3) Molten salts conduct electricity;
4) Dissolve in highly polar solvents as water

the size of ions

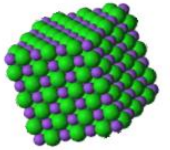


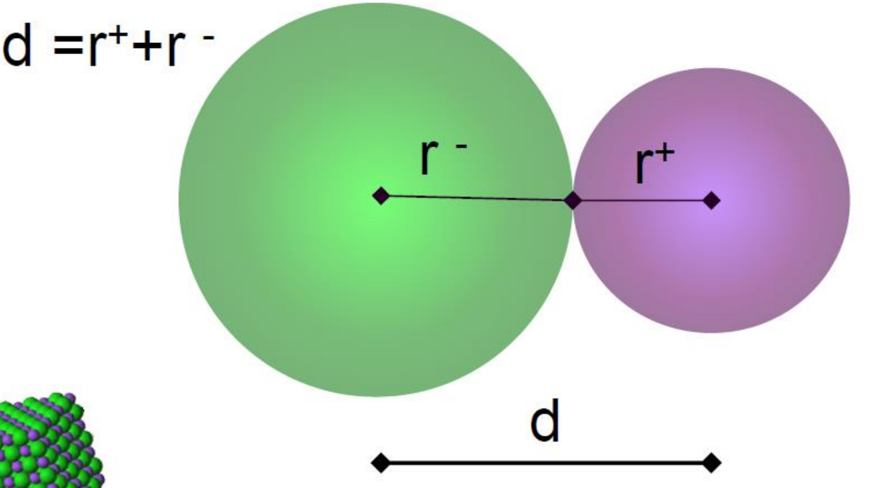
Ionic Radius

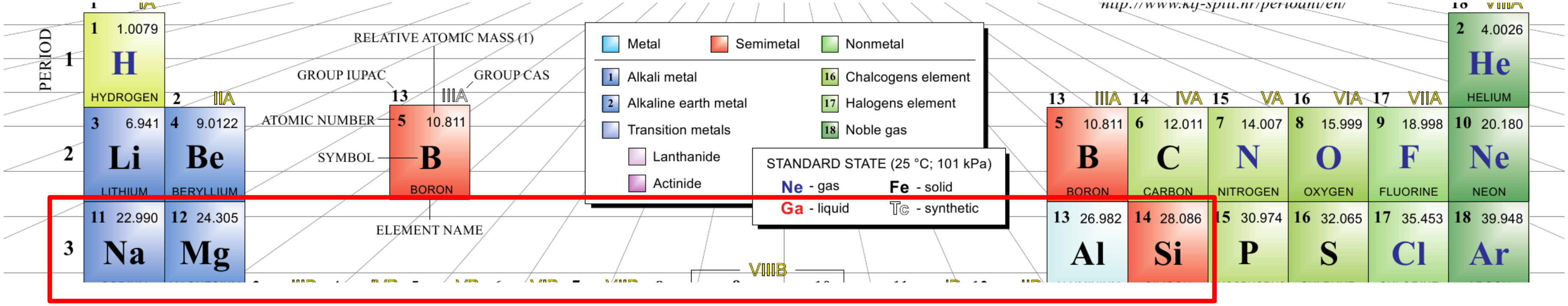
According to Pauling, for two ions with the same noble gas configuration, e.g. Na^+ and F^-, the ratio of the radii should be inversely proportional to the ratio of the "effective" nuclear charges, which can be calculated by Slater's rules.

Exercise 1

Knowing that the interionic distance in NaF is 2.31 Å, calculate the ionic radius of Na^+ and F^-.

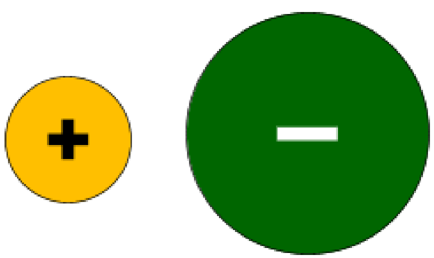


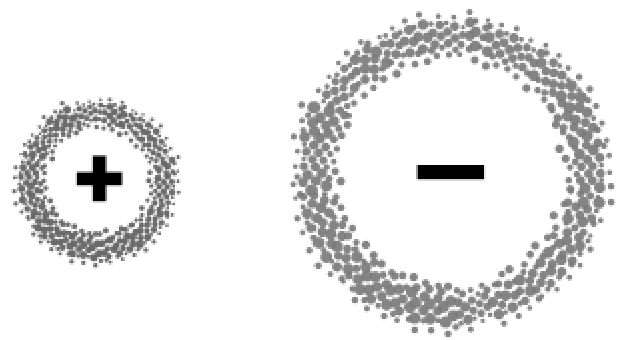




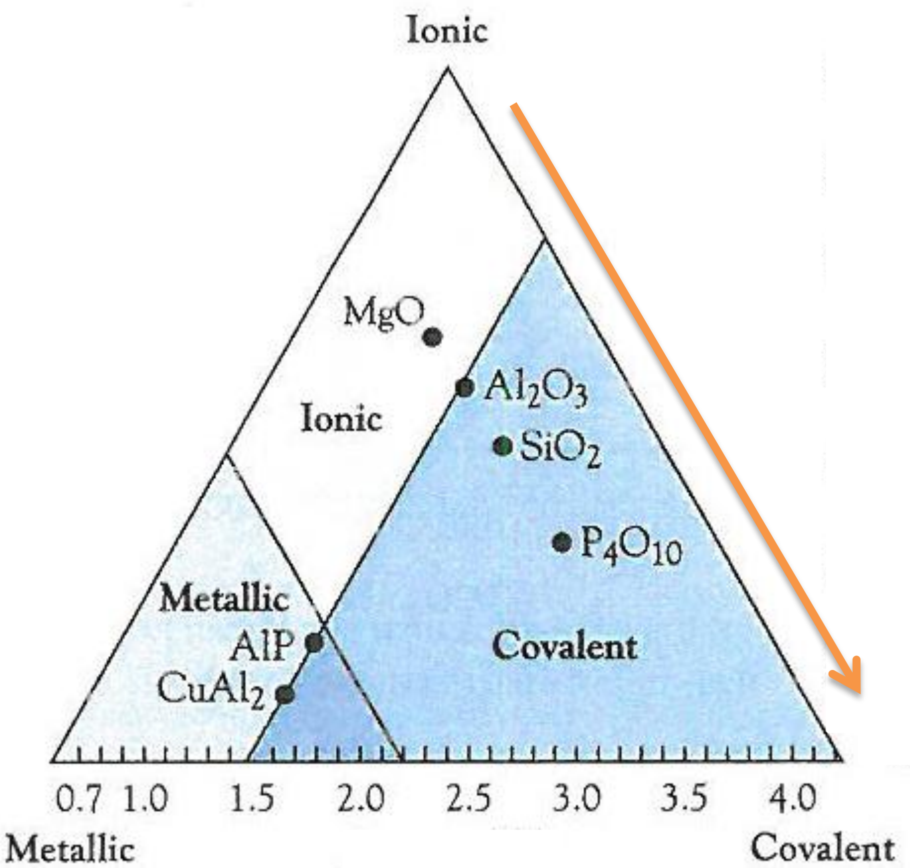
		‘charge’	Melting point /°C
Period 3	NaCl	1+	808
	$MgCl_2$	2+	714
	$AlCl_3$	3+	180
	$SiCl_4$	4+	-70

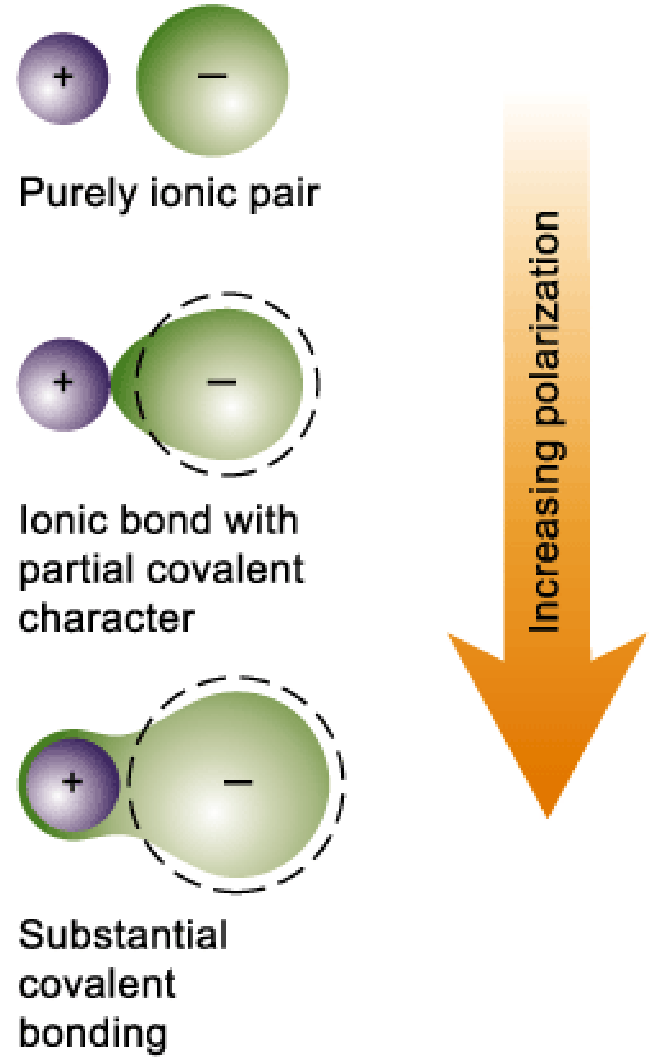
In ideal ionic compounds: we completely separate, spherical ions electron densities are apart from each other;





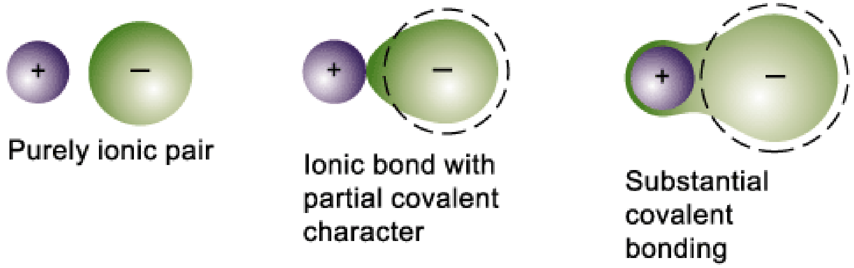
However, if the positive ion has a high charge density it can distort the negative ion by attracting the outer shell electrons to give an area of electron density between the two species ... a bit like a covalent bond





Polarization of an ionic bond means the distortion of the electron cloud of an anion towards a cation by the influence of the electric field of the cation.

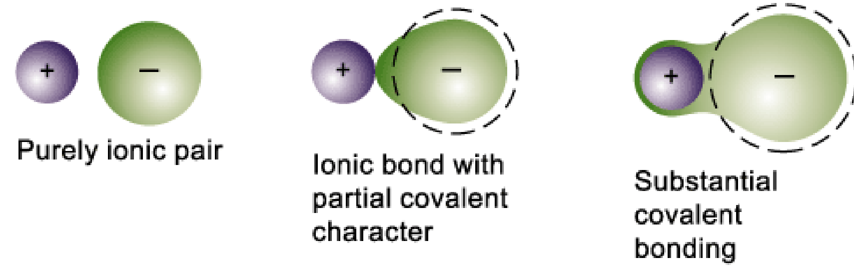
Polarization of an ionic bond results in an ionic bond with covalent character.



Increasing charge

Theoretical metal ion charge	Most predominant bonding type
+1	Ionic, except some lithium compounds
+2	Ionic, except some magnesium and beryllium compounds
+3	Covalent, except some oxides and fluorides of the heavier elements and hydrated cations
+4, +5	Covalent, except some oxides and fluorides of the heavier elements and oxo-cations
+6	Covalent, except some oxides and oxo-cations
+7, +8	Covalent

Compound	Melting point (°C)	Cation charge density ($C \cdot mm^{-3}$)	Bonding classification
MnO	1785	84	Ionic
Mn_2O_7	6	1238	Covalent



Increasing charge
Decreasing cation size

Increasing charge density

Exercise 2

Can charge density explain the different melting temperature of NaCl, $MgCl_2$, $AlCl_3$, and $SiCl_4$?
Calculate the charge density (Coulomb/mm^3) of Na^+ (ionic radius = 0,116 nm), Mg^{2+}(ionic radius = 0,086 nm), Al^{3+} (ionic radius = 0,068 nm), Si^{4+} (ionic radius = 0,041 nm).
Be aware: 1 proton has a charge of 1.60×10^{-19} C (Coulomb)

Chlorides can be used to demonstrate changes in bond type as the positive charge density increases due to higher charge (across Period 3) or larger size (down Group 1)

		Charge density (C/mm^3)	Chloride	Melting point /°C	water solubility
Period 3	Na^+	24	NaCl	808	soluble
	Mg^{2+}	120	$MgCl_2$	714	soluble
	Al^{3+}	365	$AlCl_3$	180	colloidal Al_2O_3
	Si^{4+}	2220	$SiCl_4$	-70	colloidal SiO_2

GREATER POSITIVE CHARGE DENSITY ↓

↑ extent of polarization of ionic bond →

← ↑ extent of polarization of covalent bond

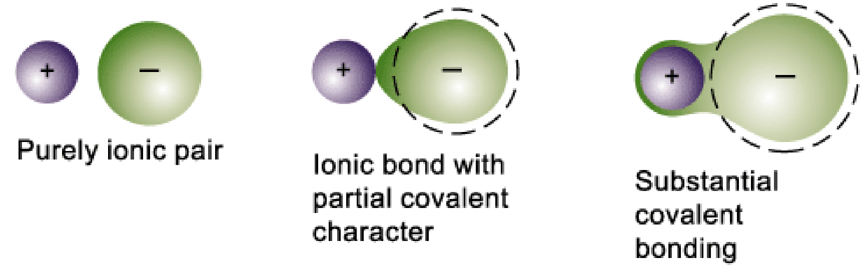
NaCl MgCl$_2$ AlCl$_3$ SiCl$_4$ PCl$_5$ Cl$_2$

Purely Ionic (NaCl)

Ionic with covalent character ($MgCl_2$, $AlCl_3$)

Polar covalent ($SiCl_4$, PCl_5)

Purely covalent (Cl_2)



Increasing charge

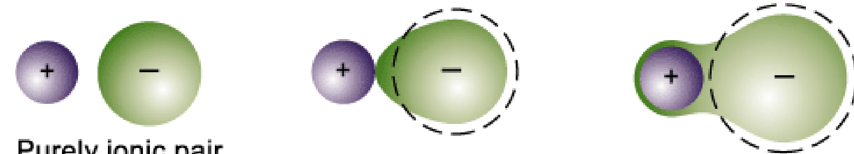
Decreasing cation size

Increasing charge density

Increasing anion size

Ion	Radius (pm)
F^-	119
Cl^-	167
Br^-	182
I^-	206

Compound	Melting point (°C)
KF	857
KCl	772
KBr	735
KI	685



Increasing charge

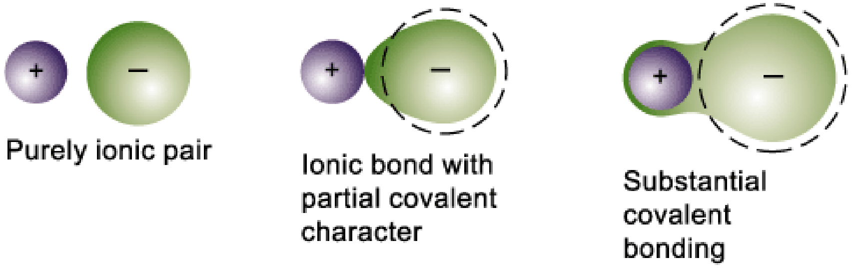
Decreasing cation size

Increasing charge density

Increasing anion size

Incomplete valence shell

Compound	Melting point (°C)	Charge density ($C \cdot mm^{-3}$)	Bonding classification
KCl	770	11	Ionic
AgCl	455	15	Partially covalent



Increasing charge

Decreasing cation size

Increasing charge density

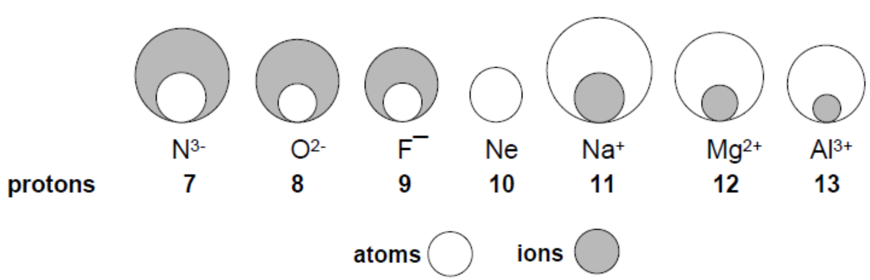
Increasing anion size

Incomplete valence shell

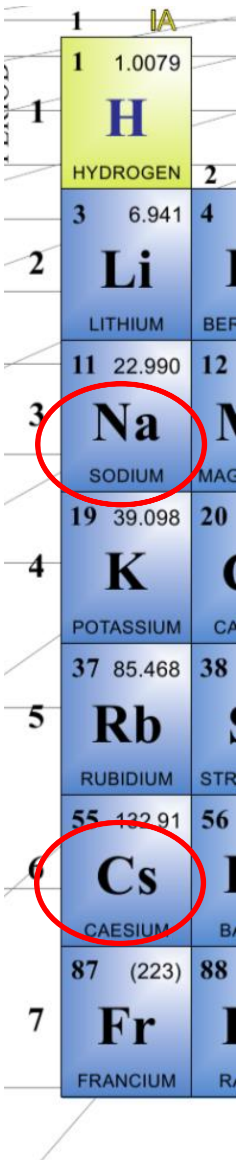
Ionic crystal strucutre

- Ions are assumed to be charged, incompressible, non polarizable spheres;
- Ions try to surround themselves with as many ions of the opposite charge;
- The cation-to-anion ratio must reflect the chemical composition of the compound.

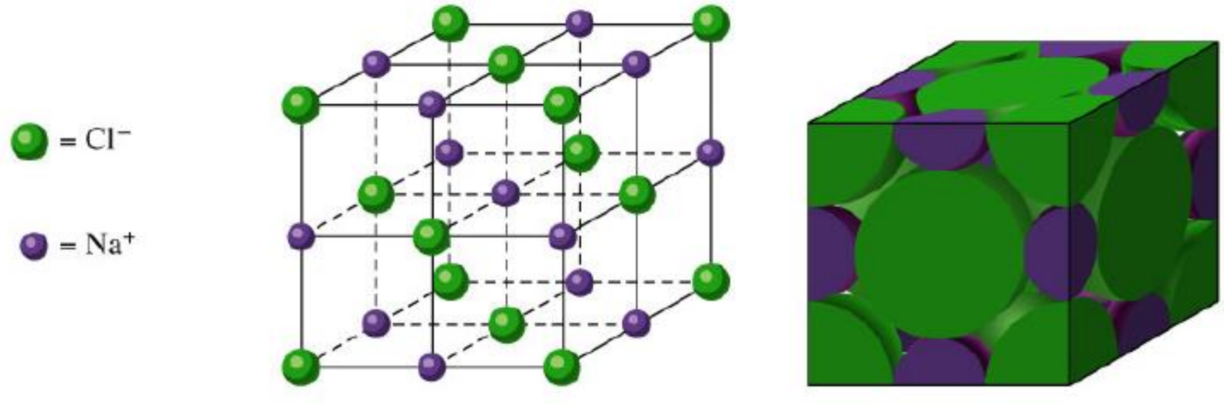
Sodium Chloride (NaCl)



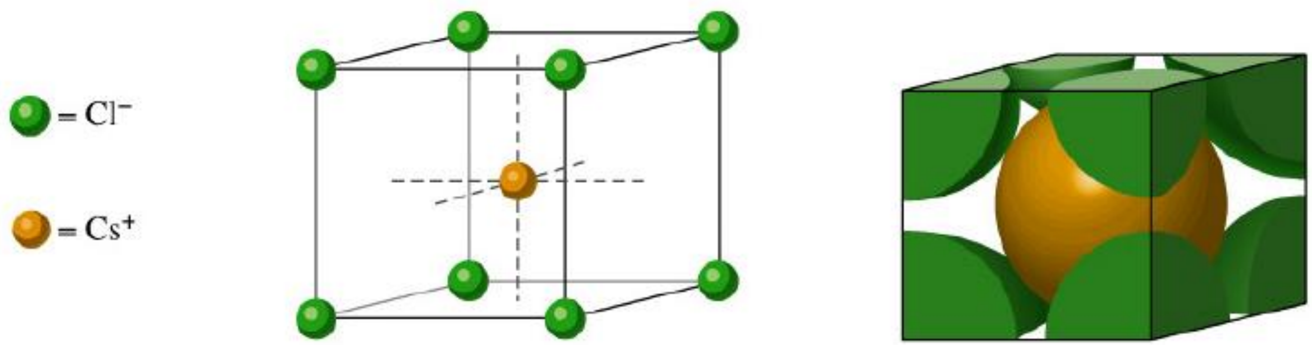
Cation fits interstices between anion.



NaCl



CsCl



Why sodium chloride and cesium chloride have different unit cell structure?

Packing rules for ionic substances

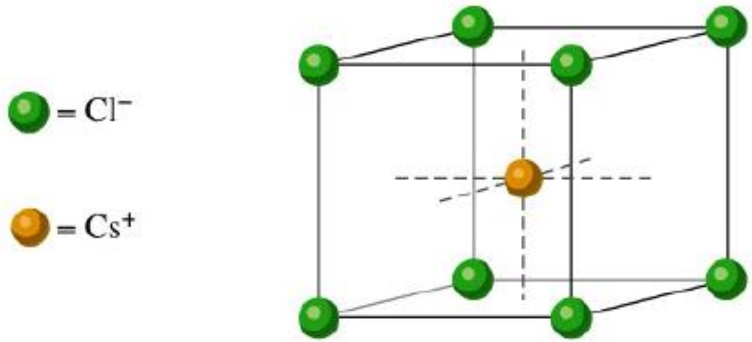
The range of radius ratios corresponding to different ion arrangements

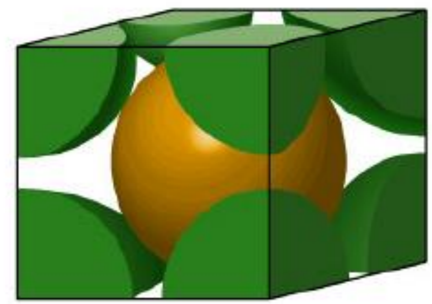
r_+/r_- Values	Coordination number preferred	Name
0.732 to 0.999	8	Cubic
0.414 to 0.732	6	Octahedral
0.225 to 0.414	4	Tetrahedral

The cubic case

Cesium chloride

$$\frac{r_{Cs^+}}{r_{Cl^-}} = 0{,}94 > 0{,}732 \Rightarrow$$ **C. n. = 8**
Cubic sites





Coord. #: Cs^+: 8; Cl^-: 8

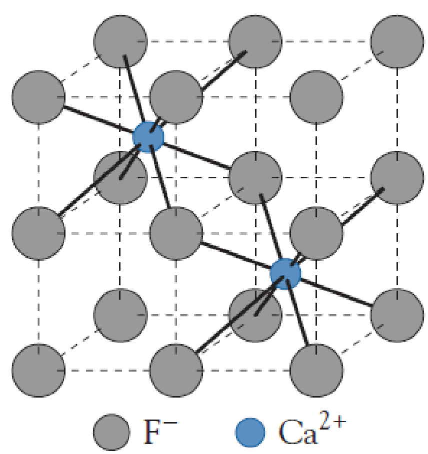
atom/ unit cell

Cs: Cl= 1: 1 ⇒ CsCl

Cl^- at primitive cubic

Cs^+ at Cubic holes

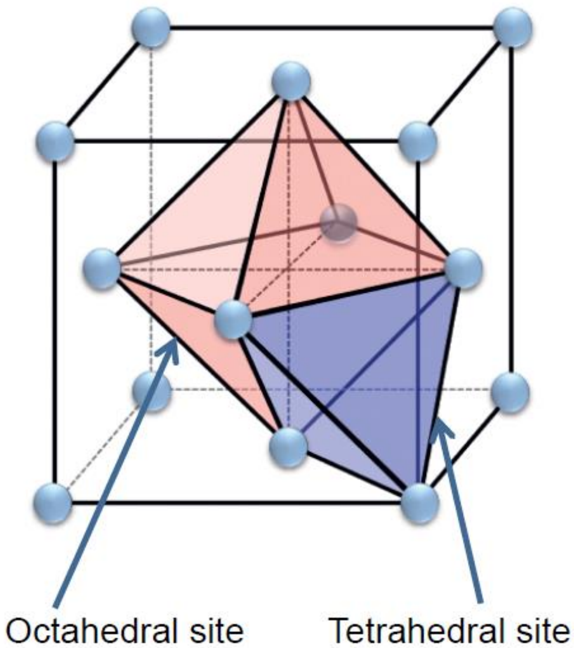
Calcium fluoride

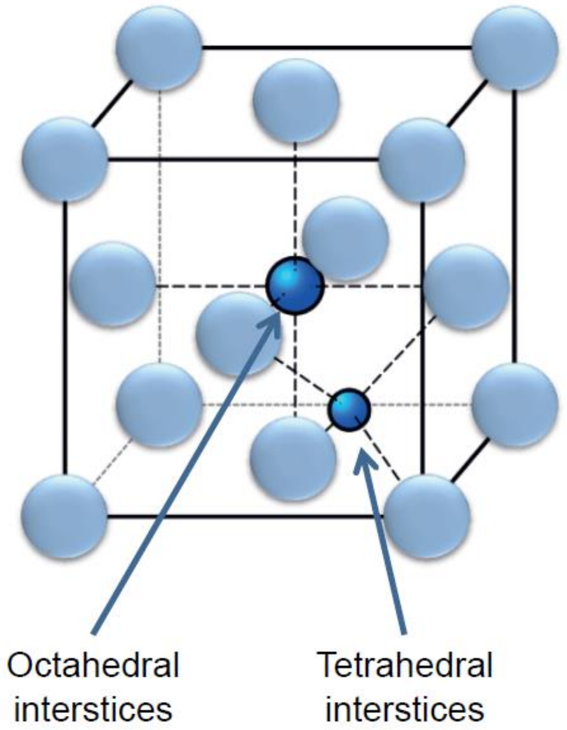


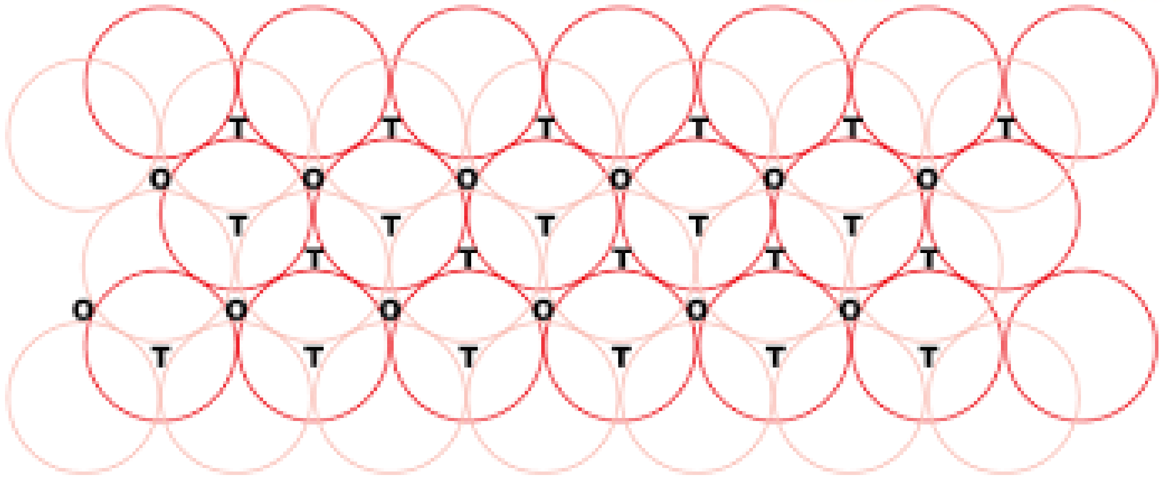
CaF_2

Ca / F = 2 / 1

Face-centered cubic unit cell

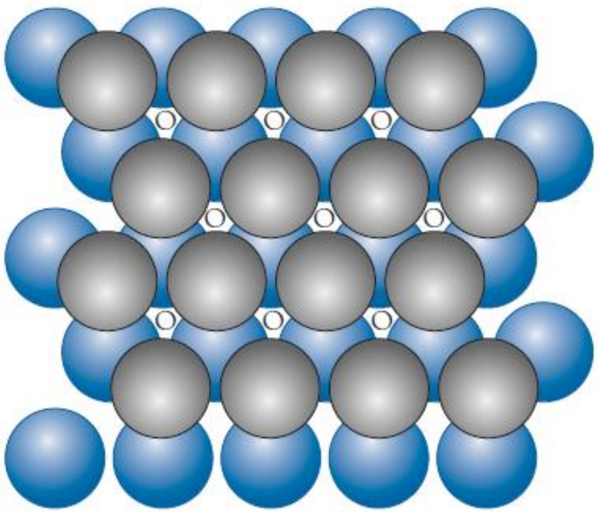




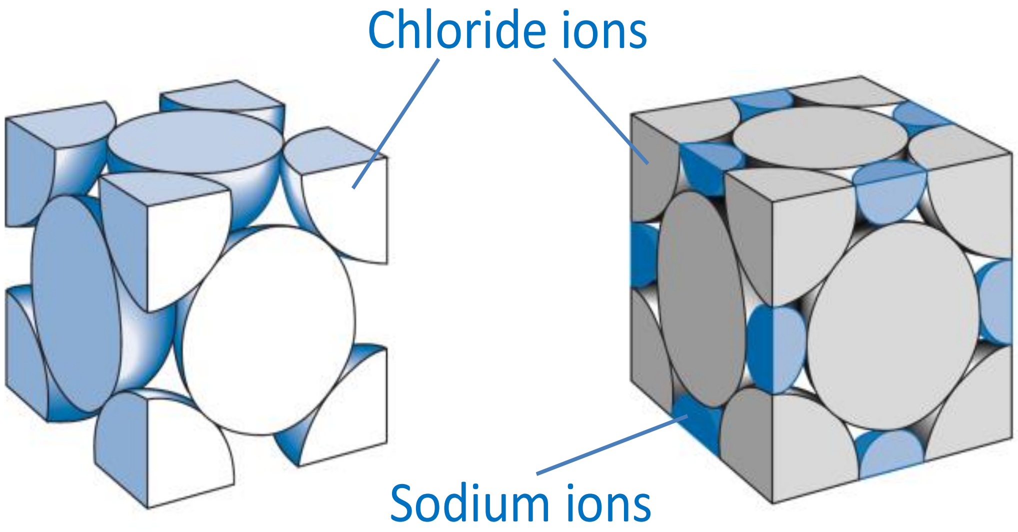


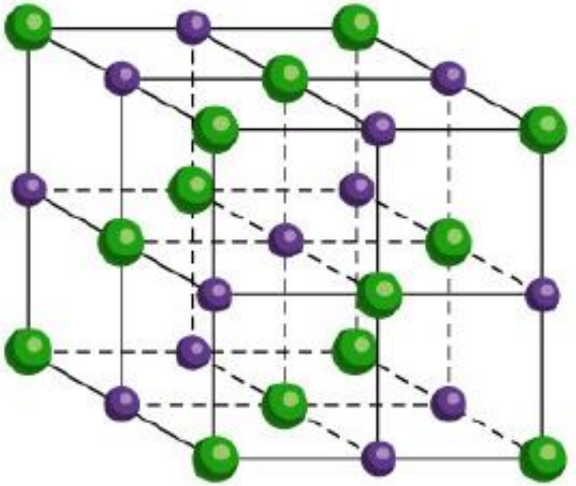
The Octahedra case

Sodium chloride

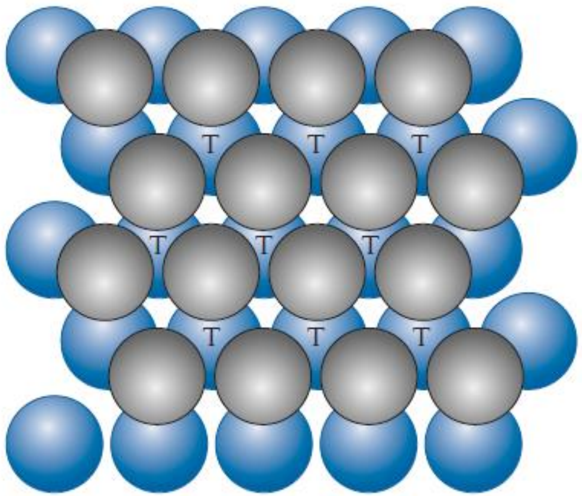


Coord. n. = 6; octahedral sites





The tethrahedral case

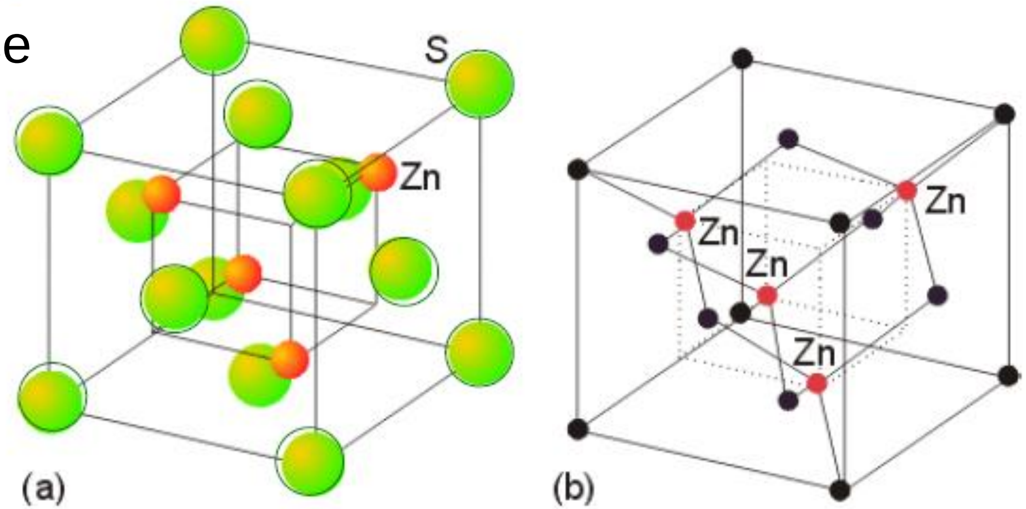


The tetrahedral case

Zinc Blende = ZnS

- S^{2-}: FCC packing
- Zn^{2+}: ½ tetrahedral interstitials
- 4 Zn^{2+} and 4 S^{2-} per unit cell

$$\frac{r_{Zn^{2+}}}{r_{S^{2-}}} = 0.345 \Rightarrow \text{C. n.} = 4$$



Coord. #: Zn^{2+}: 4; S^{2-}: 4

atom/ unit cell

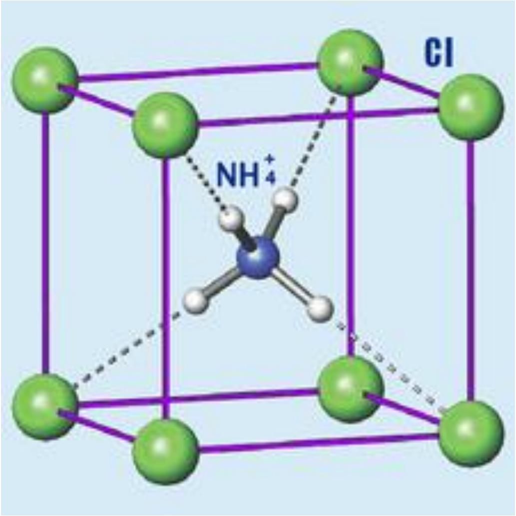
Zn: S = 4: 4 = 1: 1 ⇨ ZnS

S^{2-} at fcc

Zn^{2+} at ½ Td holes

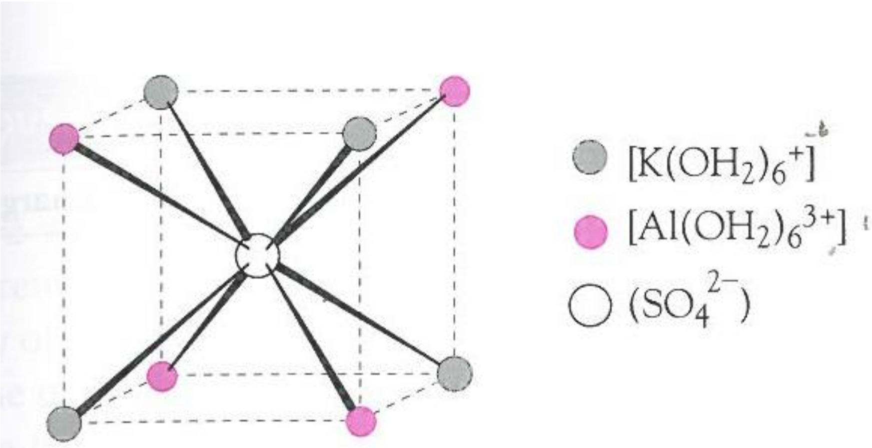
Crystal structures involving polyatomic ions

In general the polyatomic ion is considered to occupy a site as a monoatomic ion



Hydrated salts

The oxygen atoms of the water molecules are attracted towards the positively charge metal ion.
Highly charge cations have higher tendency to for hydrated salts.



Exception to the packing rules

- Partial covalent behavior;
- Low difference in energy between the different packing arrangement;
- Influence of the environment on the ionic radius.

Compound	r_+/r_-	Expected packing	Actual packing
HgS	0.55	NaCl	ZnS
LiI	0.35	ZnS	NaCl
RbCl	0.84	CsCl	NaCl

For some ionic compounds with $r^+/r^- << 0.22$, it will be impossible to achieve good cation-anion contacts.

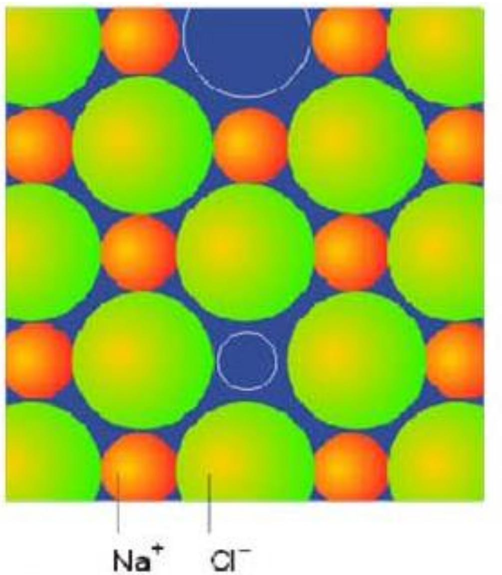
For example, Salts of the small cations Li^+, $Al^{3+}$and Mg^{2+} with large polyatomic anions (ClO_4^-, CO_3^{2-} and NO_3^-) are not stable. The consequences are

1) Transformation of the unstable anhydrous compounds to hydrates. For instance, $Mg(ClO_4)_2$ is a powerful absorbents for water.

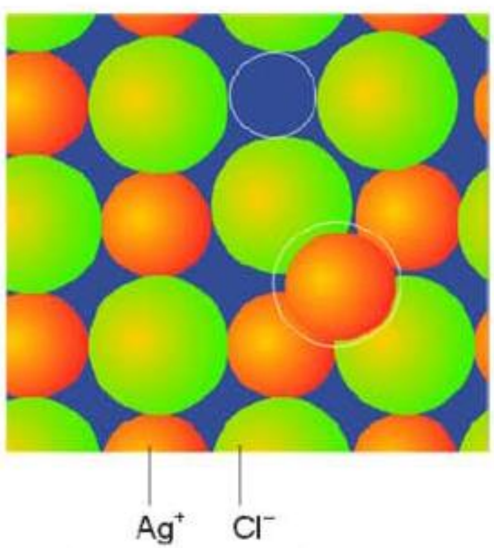
2) Low thermal stability: $Li_2CO_3 \rightarrow Li_2O + CO_2$

3) High solubility: $LiClO_4$ is about 10 times as soluble as $NaClO_4$ which, in turn, is about 1000 times as soluble as $KClO_4$.

Intrinsic defects

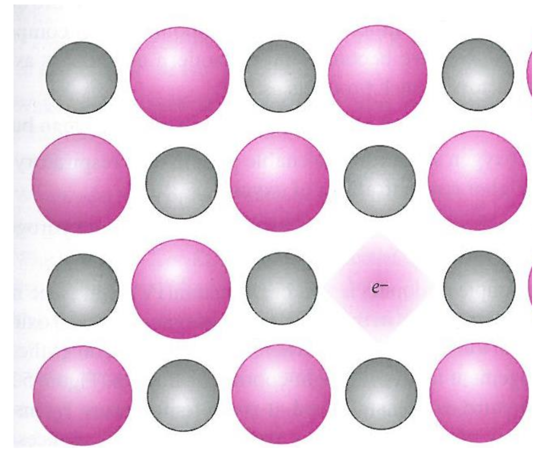


Schottky defects

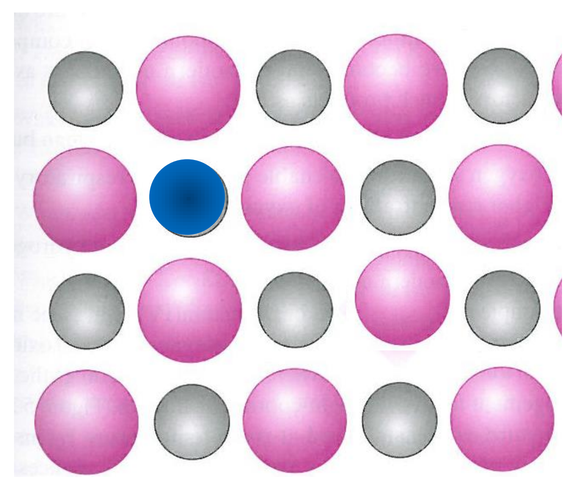


Frenkel Defects

Extrinsic defects

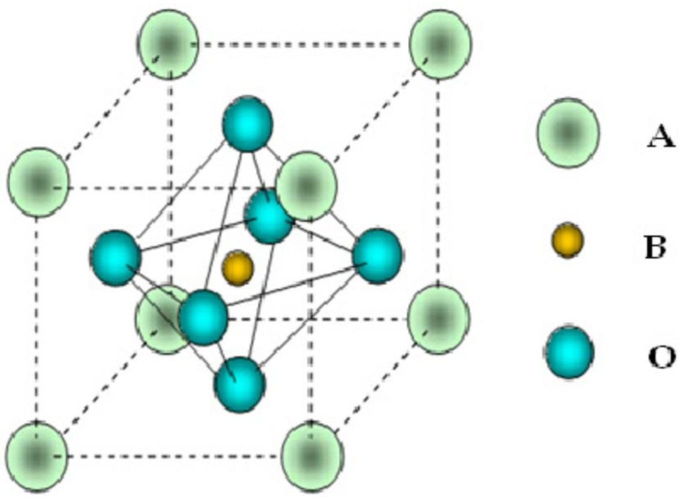


Non stoichiometric compounds

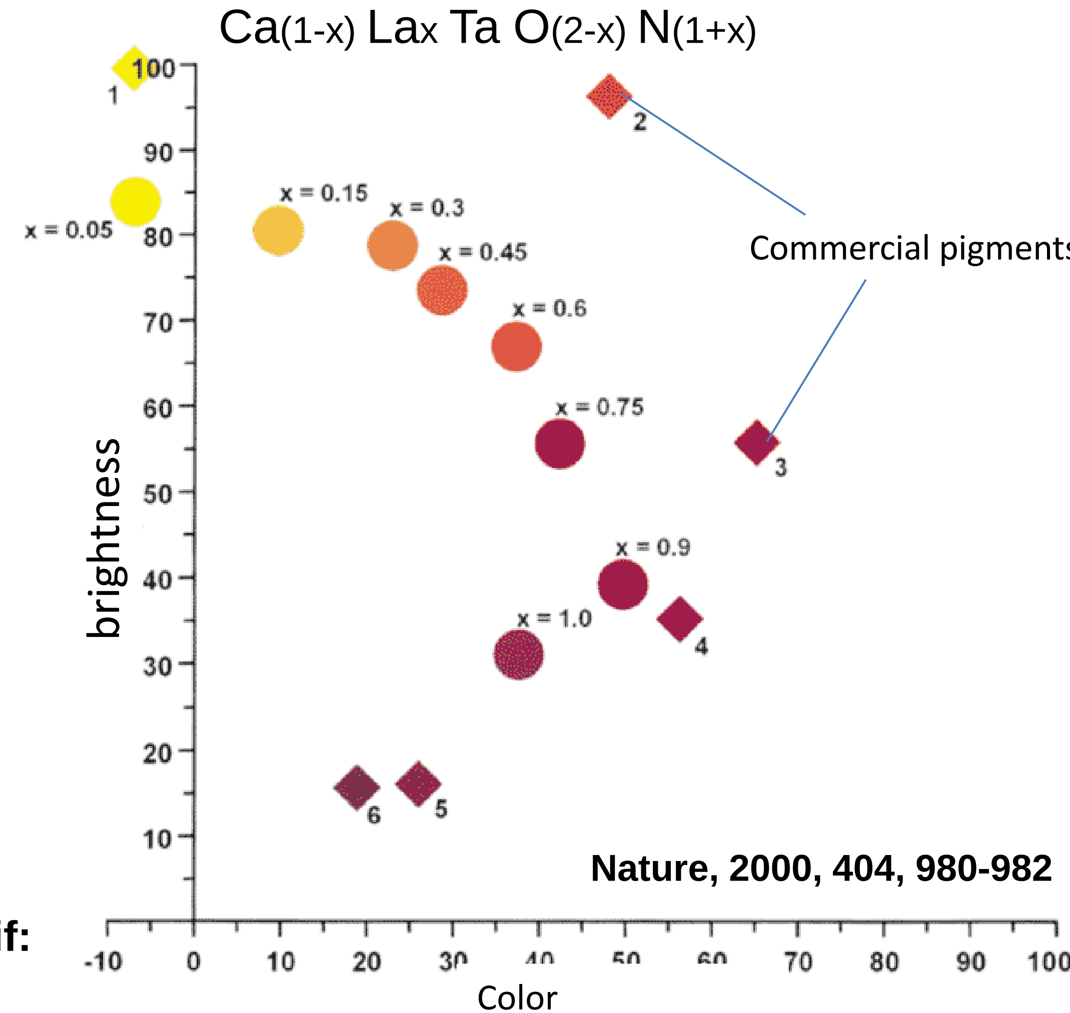


Doping

More complex ionic solids



Perovskite (parent = $CaTiO_3$)
(general ABO_3)



One ion can be substitute in a lattice if:

1) the two ions have identical charge and different radii, but not more than 20%

or

2) the two ions have different charges but the same radii, provided the sum of the charges is unchanged.

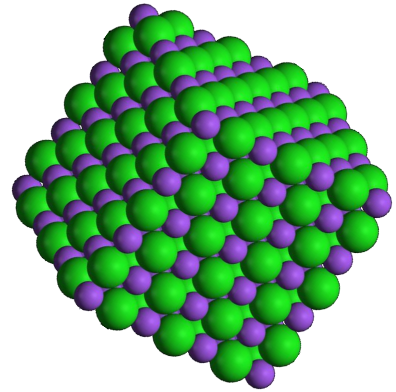
Question 1:

What is the lattice energy for Sodium Chloride?

The lattice energy (enthalpy) is the energy released by the formation of 1 mol of an ionic solid from its constituent gaseous ions.

$$\mathbf{Na^{+}\,(g) + Cl^{-}\,(g) \longrightarrow NaCl\,(s)}$$

Ionic Solids



Inorganic solids: a 3D array of ions

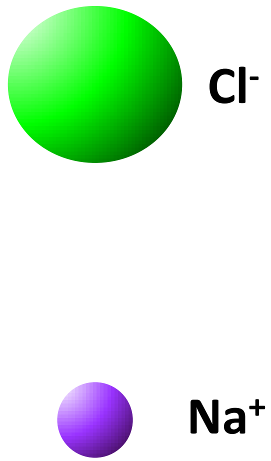
Energy of this array of ions includes:

1) Coulombic attractive and repulsive energies;

2) Additional repulsive energy resulting from repulsion between the overlapping outer electron density of adjacent ions;

3) van der Waals energy and zero point vibrational energy.

Sodium Chloride (NaCl)



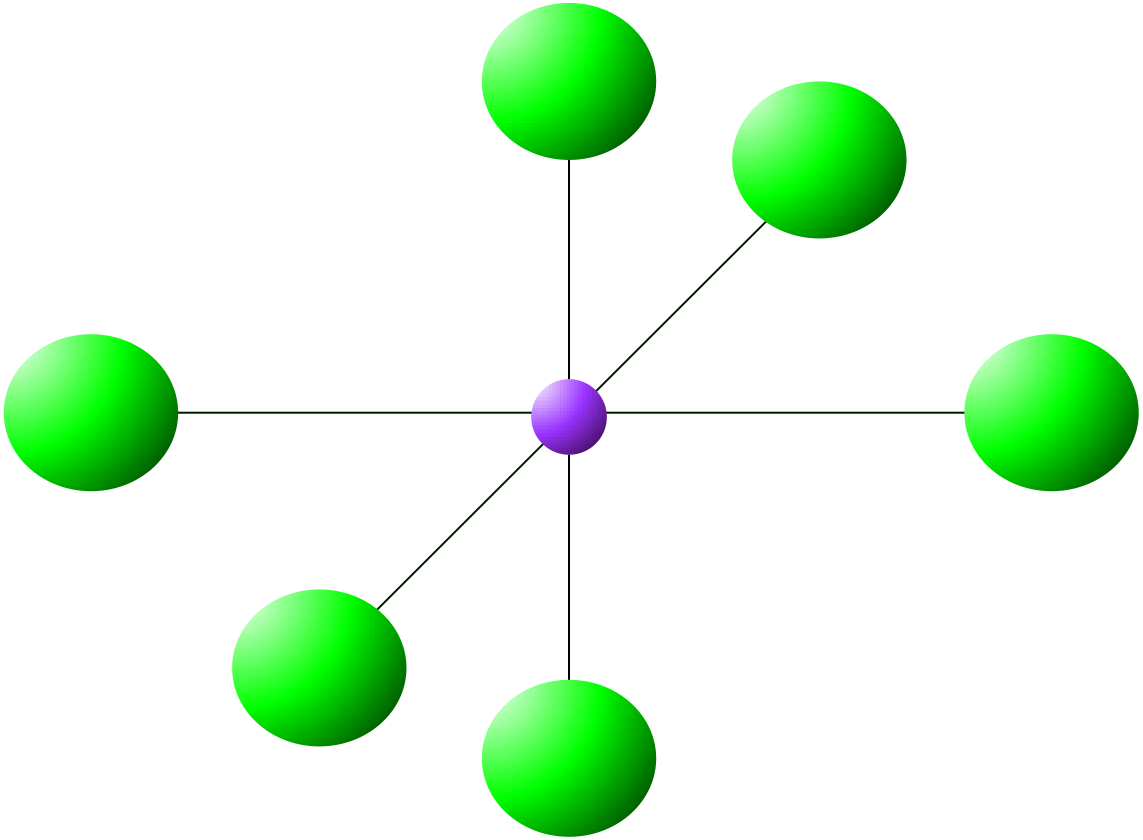
$$E = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r_0}$$

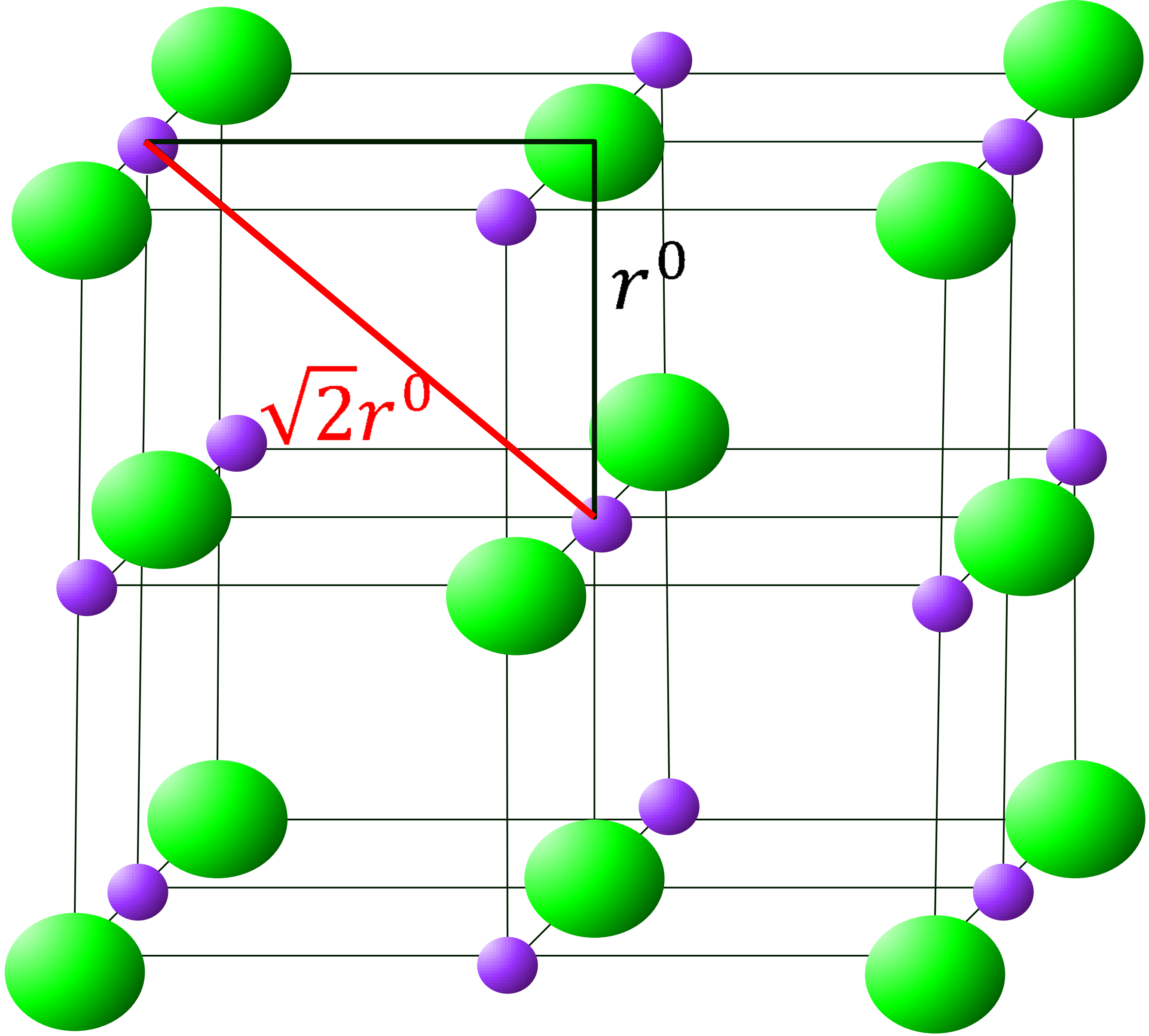
e electron charge in columbs = -1.602×10^{-19} C

ε_0 vacuum dialectric constant = $8.854\ldots \times 10^{-12}\ C^2 \cdot m^{-1} \cdot J^{-1}$

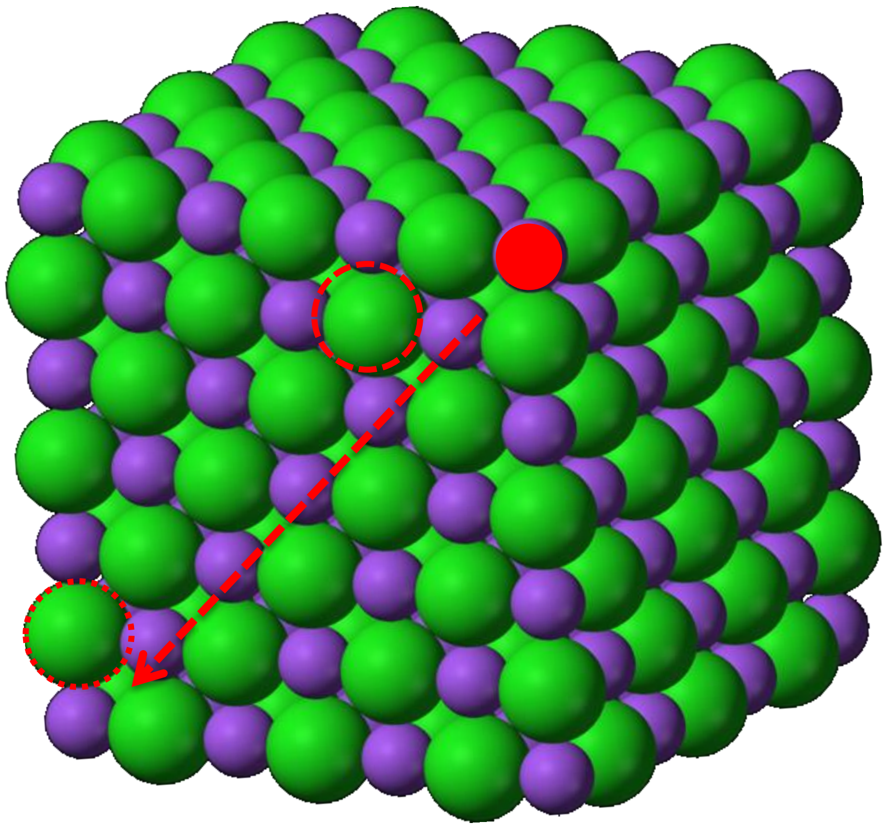
Sodium Chloride (NaCl)

$$E = \frac{1}{4\pi\varepsilon_0} \frac{6e^2}{r_0}$$





$$E = \frac{1}{4\pi\varepsilon_0}\left(-\frac{12e^2}{r_0\sqrt{2}}\right)$$



$$E=\frac{1}{4\pi\varepsilon_0}\left(\frac{6e^2}{r_0}-\frac{12e^2}{r_0\sqrt{2}}+\frac{8e^2}{r_0\sqrt{3}}-\frac{6e^2}{r_0\,2}+\ldots\right)$$

The Lattice Energy of NaCl

$$E = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r_0}\boxed{\left(6 - \frac{12}{\sqrt{2}} + \frac{8}{\sqrt{3}} - \frac{6}{2} + \ldots\right)}$$

$$= \frac{\boxed{1.748}\,e^2}{4\pi\varepsilon_0 r_0} = \frac{A_{NaCl}\,e^2}{4\pi\varepsilon_0 r_0}$$

1.748 is the Madelung constant, A_{NaCl},

for the NaCl structure.

Question 2:

Do crystals of different salts have different Madelung constants?

TABLE 6.3 Madelung constants for common lattice types

Type of lattice	Madelung constant, *A*
Sphalerite, ZnS	1.638
Wurtzite, ZnS	1.641
Sodium chloride, NaCl	1.748
Cesium chloride, CsCl	1.763
Rutile, TiO_2	2.408
Fluorite, CaF_2	2.519

The total electrostatic energy for forming a mole of NaCl from the gaseous ions:

$$U^* = NA_{NaCl}\left(\frac{e^2}{4\pi\varepsilon_0 r_0}\right)$$

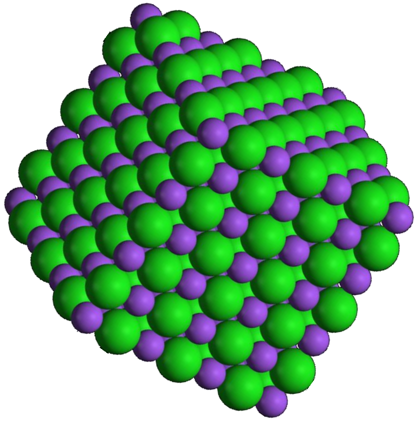
where N is Avogadro's number.

But this energy is not actual energy released in the process:

$$\mathbf{Na^+\ (g) + Cl^-\ (g) \longrightarrow NaCl\ (s)}$$

Why?

Ionic Solids



Inorganic solids: a 3D array of ions

Energy of this array of ions includes:

1) Coulombic attractive and repulsive energies

2) Additional repulsive energy resulting from repulsion between the overlapping outer electron density of adjacent ions

3) Other contributions (e.g.van der Waals energy) are negligible

Finally we obtain the Born-Landé equation

$$U = -\frac{NA_{NaCl}e^2}{4\pi\varepsilon_0 r_0}\left(1 - \frac{1}{n}\right)$$

The value *n* can be estimated for alkali halides by using the average of the following numbers :He 5, Ne 7, Ar 9, Kr 10 and Xe 12.

Electronic configuration of ion	Born exponent, *n*	Examples
He	5	Li^+
Ne	7	$Na^+, Mg^{2+}, O^{2-}, F^-$
Ar	9	$K^+, Ca^{2+}, S^{2-}, Cl^-$
Kr	10	Rb^+, Br^-
Xe	12	Cs^+, I^-

The Born exponent for NaCl can be approximated to 8

$$U = -\frac{(6.02 \times 10^{23}\,\mathrm{mol^{-1}}) \times 1.748 \times 1 \times 1 \times (1.602 \times 10^{-19}\,\mathrm{C})^2}{4 \times 3.142 \times (8.854 \times 10^{-12}\,\mathrm{C^2{\cdot}J^{-1}{\cdot}m^{-1}})(2.83 \times 10^{-10}\,\mathrm{m})}\left(1 - \frac{1}{8}\right)$$

$$= -750 \quad \mathrm{kJ{\cdot}mol^{-1}}$$

Finally we obtain the Born-Landé equation

$$U = -\frac{NA_{NaCl}e^2}{4\pi\varepsilon_0 r_0}\left(1 - \frac{1}{n}\right)$$

The Born coefficient can be estimated from the compressibility of NaCl or can be estimated from the electronic configuration of Na^+ and Cl^-

In this case n = 8.

Then the net energy for the process is

U = -857 + 106 = -750 kJ/mol

Note: the repulsive energy equals to 1/8 of the Coulombic energy.

To make very accurate calculation of crystal energies, sometimes called lattice energy, certain refinements must be introduced:

1. A more accurate, quantum expression for the repulsion energy
2. A correction for van der Waals energy
3. A correction for the "zero-point energy", the vibrational energy present even at 0 K.

Generalisation and Refinement of Lattice Energy Calculations

The general equation is

$$U = -\frac{N\ A\ z^{+}z^{-}e^{2}}{4\pi\varepsilon_0 r_0}\left(1 - \frac{1}{n}\right)$$

for any structure, whose Madelung constant is *A* with ions of charges z^+ and z^-.